

2008 Ford Edge SE

2008 HVAC Climate Control System - General Information and Diagnostics - Edge & MKX

2008 HVAC

Climate Control System - General Information and Diagnostics - Edge & MKX

SPECIFICATIONS

MATERIAL SPECIFICATIONS

Item	Specification	Fill Capacity
A/C Cooling Coil Coating YN-29	-	-
A/C System Flushing Solvent YN-23	-	-
PAG Refrigerant Compressor Oil (R-134a Systems) YN-12-D	WSH-M1C231-B	118 ml (4 fl oz)
R-134a Refrigerant YN-19 (US); CYN-16-P or CYN- 16-R (Canada)	WSH-M17B19-A	0.60kg (21 oz)

GENERAL SPECIFICATIONS

Item	Specification
A/C Compressor	
Type	10S20C
A/C Pressure Relief Valve^a	
Open pressure	3,447-4,137 kPa (500-600 psi)
Close pressure	2758 kPa (400 psi)
Evaporator Temperature Sensor	
ON temperature	3°C (37°F)
OFF temperature	2°C (36°F)
Refrigerant System Dye	
R-134a Leak Detection Dye 164-R6060 or 164-R6081	-

^a Manifold gauge set pressures may vary slightly depending on the distance between the service gauge port valve and the A/C pressure relief valve, the A/C cycling switch or the pressure cutoff switch location.

DESCRIPTION AND OPERATION

CLIMATE CONTROL SYSTEM

Climate Control System

WARNING: Take the following precautions when repairing an air conditioning system containing R-134a:

- Always wear safety goggles.
- Avoid contact with liquid refrigerant R-134a. R-134a vaporizes at approximately -25°C (-13°F) under atmospheric pressure and will freeze skin tissue.
- Never allow refrigerant R-134a gas to escape in quantity in an occupied space. It will displace the oxygen needed to support life.
- Never use a torch in an atmosphere containing R-134a gas. R-134a is non-toxic at all normal conditions, but it decomposes when exposed to high temperatures such as a torch flame. During decomposition it releases irritating and toxic gasses (as described in the MSDS sheet from the manufacturer). Decomposition products are hydrofluoric acid, carbon dioxide and water.

Failure to follow these instructions may result in serious personal injury.

CAUTION: To avoid damaging the vehicle or air conditioning (A/C) components, the following precautions must be observed.

- The A/C refrigerant of all vehicles must be identified and analyzed prior to refrigerant charging. Failure to do so can contaminate the shop bulk refrigerant and other vehicles.
- Do not add R-12 refrigerant to an A/C system that requires the use of R-134a refrigerant. These 2 types of refrigerant must never be mixed. Doing so can damage the A/C system.
- Charge the A/C system with R-134a refrigerant gas while the engine is running only at the low-pressure side, to prevent refrigerant slugging from damaging the A/C compressor.
- Use only R-134a refrigerant. Due to environmental concerns, when the A/C system is drained, the refrigerant must be collected using refrigerant recovery/recycling equipment. Federal, State/Provincial and/or local laws REQUIRE that R-134a be recovered into appropriate recovery equipment and the process be conducted by qualified technicians who have been certified by an approved organization, such as ASE, MACS, HRAI, etc. Use of a recovery machine dedicated to R-134a is necessary to reduce the possibility of oil and refrigerant incompatibility concerns. Refer to the instructions provided by the equipment manufacturer when removing refrigerant from or charging the A/C system.
- Refrigerant R-134a must not be mixed with air for leak testing or used with air for any other purpose above atmospheric pressure. R-134a is combustible when mixed with high concentrations of air and higher pressures.

- A number of manufacturers are producing refrigerant products that are described as direct substitutes for refrigerant R-134a. The use of any unauthorized substitute refrigerant can severely damage the A/C components. If repair is required, use only new or recycled refrigerant R-134a.

CAUTION: To avoid contamination of the air conditioning (A/C) system, the following precautions must be observed.

- Never open or loosen a connection before recovering the refrigerant.
- When loosening a connection, if any residual pressure is evident, allow it to leak out before opening the fitting.
- Evacuate a system that has been opened to install a new component or one that has discharged through leakage before charging.
- Seal open fittings with a cap or plug immediately after disconnecting a component from the system.
- Clean the outside of the fittings thoroughly before disconnecting a component from the system.
- Do not remove the sealing caps from a new component until ready to install.
- Refrigerant oil will absorb moisture from the atmosphere if left uncapped. Do not open an oil container until ready to use and install the cap immediately after using. Store the oil in a clean, moisture-free container.
- Install a new O-ring seal before connecting an open fitting. Coat the fitting and O-ring seal with PAG oil before connecting.
- When installing a refrigerant line, avoid sharp bends. Position the line away from the exhaust or any sharp edges that can chafe the line.
- Tighten threaded fittings only to specifications. The steel and aluminum fittings used in the refrigerant system will not tolerate overtightening.
- When disconnecting a fitting, use a wrench on both halves of the fitting to prevent twisting of the refrigerant lines or tubes.
- Do not open a refrigerant system or uncap a new component unless it is as close as possible to room temperature. This will prevent condensation from forming inside a component that is cooler than the surrounding air.

The electronic manual temperature control (EMTC) system heats or cools the vehicle depending on the HVAC module - EMTC control panel selection.

- The HVAC module - EMTC control panel selections determine heating or cooling, air distribution and

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enables blower motor operation.

- The temperature control setting determines the air temperature.
- The blower motor switch varies the blower motor speed.
- During A/C operation the system also reduces the relative humidity of the air.

The dual automatic temperature control (DATC) system maintains the selected vehicle interior temperature by heating and/or cooling the air.

- During A/C operation the system also reduces the relative humidity of the air.
- The driver may override the automatic mode of operation.
- The temperature control setting determines the air temperature.
- The blower motor control override buttons vary the blower motor speed.
- The driver side and passenger side temperature settings can be individually controlled.

The system consists of the following components:

- A/C compressor
- A/C compressor clutch assembly
- A/C condenser core with A/C receiver/drier
- A/C evaporator core
- Connecting refrigerant lines
- Thermostatic expansion valve (TXV)
- A/C pressure transducer
- Heater core and evaporator core housing
- HVAC module - EMTC or HVAC module - DATC
- Blower switch (EMTC systems)
- Temperature blend door actuators
- Floor/defrost/panel door actuator
- Air inlet door actuator
- Blower motor resistor or speed control module
- Blower motor
- Ambient temperature sensor
- In-vehicle temperature sensor
- Solar radiation sensor
- Evaporator temperature sensor

External Temperature Display

The external temperature is displayed when the EXT button is pressed. While the actual external temperature can vary continuously, the value that is displayed will update at a specific rate depending on whether or not the

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engine is "hot" and whether or not the vehicle is moving.

When the external temperature is rising, the display will update slowly. Updates (when the actual external temperature is higher than currently displayed value) will be limited to 0.6°C (1°F) every 20 seconds while the vehicle is moving at 40 kph (25 mph) or greater for more than 90 seconds or 0.6°C (1°F) every 20 minutes if the vehicle is not moving at this speed. This is to prevent the heat from the engine compartment from affecting the accuracy of the display.

When the external temperature is dropping, the display will update quickly.

Updates (when the actual external temperature is lower than currently displayed value) will only be limited to 0.6°C (1°F) every 2 seconds (regardless of vehicle speed). Consequently, the display will essentially follow the drop experienced by the external temperature thermistor.

Climate Control System Operation

Electronic Manual Temperature Control (EMTC) System

The electronic manual temperature control (EMTC) system operation is determined by the settings on the HVAC module - EMTC. The EMTC system is composed of the HVAC module - EMTC, blower motor resistor, mode door actuator, air inlet door actuator and temperature blend door actuator to control the various functions.

The blower motor switch:

- sets the blower motor speed.
- directs the blower motor path to ground through the blower motor resistor to allow blower motor operation in LO, MED LO and MED HI.
- directs the blower motor path directly to ground, bypassing the blower motor resistor to allow blower motor operation in HI.

The A/C request button:

- can command the A/C compressor ON (when the function selector switch is in PANEL, FLOOR/PANEL, FLOOR/DEFROST and FLOOR) when the A/C request switch is pressed. Indicator illuminates when A/C request switch is toggled ON.
- is inoperative in MAX A/C, OFF and DEFROST.
- indicator illuminates when the function selector switch is in MAX A/C and cannot be toggled.
- indicator does not illuminate in OFF and cannot be toggled.
- indicator does not illuminate in DEFROST but can be toggled ON or OFF. The A/C compressor will operate regardless of indicator status if the outside air temperature is above 6°C (43°F).
- is serviced only with the climate control module.

The function selector knob:

- selects airflow direction.

The air recirculation button:

- selects either recirculated or outside air source.
- recirculates air in any mode except DEFROST, if requested.
- is disabled in MAX A/C or OFF mode where recirculated air only is used.
- is disabled in DEFROST mode where outside air only is used.
- is serviced only with the climate control module.

The temperature selector knob:

- selects full warm and full cool as well as blended outlet temperatures.

The HVAC module - EMTC also includes:

- a rear window DEFROST button. Refer to **GLASS, FRAMES & MECHANISMS** article.

Dual Automatic Temperature Control (DATC) System

The dual automatic temperature control (DATC) system operation is determined by the settings on the HVAC module - DATC. The DATC system automatically maintains a selected temperature for vehicle interior comfort. The DATC system is composed of the HVAC module - DATC, blower motor speed control, mode door actuator, air inlet door actuator, temperature blend door actuator, in-vehicle temperature sensor, ambient temperature sensor, evaporator temperature sensor and solar radiation sensor to control the various functions.

The HVAC module - DATC will display outside air temperature when the EXT button is pressed.

When the HVAC module - DATC is set to AUTO:

- the air distribution direction, blower motor speed and A/C compressor operation are automatically controlled based on the temperature(s) selected.

The HVAC module - DATC manual override settings:

- allow the air distribution direction to be manually selected.
- allow the blower motor speed to be manually selected.
- allow RECIRCULATION mode to be manually selected in all modes except DEFROST. The A/C request can be cancelled by pressing the A/C request switch, turning the indicator OFF.
- allow A/C compressor operation to be manually selected except in DEFROST.

The HVAC module - DATC also includes:

- a rear window DEFROST button. Refer to **GLASS, FRAMES & MECHANISMS** article.
- heated/cooled seat controls. Refer to **SEATING** article.

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Max A/C

When MAX A/C is selected:

- the air inlet door actuator closes off outside air and admits only recirculated air.
- the recirc button is disabled and the indicator is illuminated.
- the floor/defrost/panel door directs airflow to the instrument panel A/C registers.
- blended air temperature is available.
- the A/C request button is illuminated and will be disabled.
- the blower motor is ON.

PANEL

When PANEL is selected:

- the recirc request button is enabled. If the recirc request button is selected (indicator ON), the air inlet door actuator closes off outside air from entering the passenger compartment. If the recirc request button is not selected (indicator OFF), the air inlet door actuator admits only outside air into the passenger compartment.
- the floor/defrost/panel door directs airflow to the instrument panel A/C registers.
- blended air temperature is available.
- the A/C request button is enabled. The A/C compressor will operate and the indicator will illuminate if the A/C request button is selected.
- the blower motor is ON.

PANEL/FLOOR

When PANEL/FLOOR is selected:

- the recirc request button is enabled. If the recirc request button is selected (indicator ON), the air inlet door actuator closes off outside air from entering the passenger compartment. If the recirc request button is not selected (indicator OFF), the air inlet door actuator admits only outside air into the passenger compartment.
- the floor/defrost/panel door directs airflow to the floor duct and the instrument panel A/C registers. A small amount of airflow from the side window demisters and defrost duct will be present.
- blended air temperature is available.
- the A/C request button is enabled. The A/C compressor will operate and the indicator will illuminate if the A/C request button is selected.
- the blower motor is ON.

OFF

When OFF is selected:

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- the recirc request button is disabled.
- the air inlet door actuator closes off outside air and admits only recirculated air.
- the A/C request button is disabled.
- the blower motor is OFF.

FLOOR

When FLOOR is selected:

- the recirc request button is enabled. If the recirc request button is selected (indicator ON), the air inlet door actuator closes off outside air from entering the passenger compartment. If the recirc request button is not selected (indicator OFF), the air inlet door actuator admits only outside air into the passenger compartment.
- the floor/defrost/panel door directs airflow to the floor duct. A small amount of airflow from the defroster duct and side window demisters will be present.
- blended air temperature is available.
- the A/C request button is enabled. The A/C compressor will operate and the indicator will illuminate if the A/C request button is selected.
- the blower motor is ON.

FLOOR/DEFROST

When FLOOR/DEFROST is selected:

- the recirc request button is enabled. If the recirc request button is selected (indicator ON), the air inlet door actuator closes off outside air from entering the passenger compartment. If the recirc request button is not selected (indicator OFF), the air inlet door actuator admits only outside air into the passenger compartment.
- the floor/defrost/panel door directs airflow to the floor duct, the defroster duct and the side window demisters.
- blended air temperature is available.
- the A/C request button will toggle the indicator on and off. To reduce fogging, the A/C compressor will operate automatically, regardless of indicator status.
- the blower motor is ON.

DEFROST

When DEFROST is selected:

- the air inlet door actuator opens, admitting only outside air into the passenger compartment.
- the floor/defrost/panel door directs airflow to the defroster duct and side window demisters. A small amount of airflow from the floor duct will be present.
- blended air temperature is available.

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- to reduce fogging, the A/C compressor will operate automatically, regardless of indicator status.
- the blower motor is ON.

System Airflow Description - Dual Automatic Temperature Control (DATC) System

AUTO

When AUTO is selected:

- the temperature control setting is manually set to the desired setting.
- the air inlet door actuator is automatically controlled by the HVAC module - DATC, based on the temperature setting.
- the mode door actuators are automatically controlled by the HVAC module - DATC based on the temperature setting.
- the A/C compressor is automatically controlled by the HVAC module - DATC based on the temperature setting.
- the blower motor is ON. The blower motor speed is automatically controlled by the HVAC module - DATC based on the temperature setting, but can be manually overridden.

OFF

When OFF is selected:

- the air inlet door actuator closes off outside air and admits only recirculated air.
- the A/C request button is disabled.
- the blower motor is OFF.

PANEL

When PANEL is selected:

- the recirc request button is enabled. If the recirc request button is selected (indicator ON), the air inlet door actuator closes off outside air from entering the passenger compartment. If the recirc request button is not selected (indicator OFF), the air inlet door actuator admits only outside air into the passenger compartment.
- the floor/defrost/panel door directs airflow to the instrument panel A/C registers.
- blended air temperature is available. Only when A/C compressor operation has been selected by pressing the A/C request button (indicator ON), can the airflow temperature be cooled below the outside air temperature.
- the A/C request button is enabled. The A/C compressor will operate and the indicator will illuminate if the A/C request button is selected.
- the blower motor is ON.

PANEL/FLOOR

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When PANEL/FLOOR is selected:

- the recirc request button is enabled. If the recirc request button is selected (indicator ON), the air inlet door actuator closes off outside air from entering the passenger compartment. If the recirc request button is not selected (indicator OFF), the air inlet door actuator admits only outside air into the passenger compartment.
- the floor/defrost/panel door directs airflow to the floor duct and the instrument panel A/C registers. A small amount of airflow from the side window demisters and defrost duct will be present.
- blended air temperature is available. Only when A/C compressor operation has been selected by pressing the A/C request button (indicator ON), can the airflow temperature be cooled below the outside air temperature.
- the A/C request button is enabled. The A/C compressor will operate and the indicator will illuminate if the A/C request button is selected.
- the blower motor is ON.

FLOOR

When FLOOR is selected:

- the recirc request button is enabled. If the recirc request button is selected (indicator ON), the air inlet door actuator closes off outside air from entering the passenger compartment. If the recirc request button is not selected (indicator OFF), the air inlet door actuator admits only outside air into the passenger compartment.
- the floor/defrost/panel door directs airflow to the floor duct. A small amount of airflow from the defroster duct and side window demisters will be present.
- blended air temperature is available. Only when A/C compressor operation has been selected by pressing the A/C request button (indicator ON), can the airflow temperature be cooled below the outside air temperature.
- the A/C request button is enabled. The A/C compressor will operate and the indicator will illuminate if the A/C request button is selected.
- the blower motor is ON.

FLOOR/DEFROST

When FLOOR/DEFROST is selected:

- the recirc request button is enabled. If the recirc request button is selected (indicator ON), the air inlet door actuator closes off outside air from entering the passenger compartment. If the recirc request button is not selected (indicator OFF), the air inlet door actuator admits only outside air into the passenger compartment.
- the floor/defrost/panel door directs airflow to the floor duct, the defroster duct and the side window demisters.
- blended air temperature is available.
- the A/C request button is enabled. The A/C compressor will operate automatically and the indicator will

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illuminate if the A/C request button is selected.

- the blower motor is ON.

DEFROST

When DEFROST is selected:

- the recirc request button and indicator is disabled. Regardless of button and indicator status, the air inlet door actuator admits only outside air into the passenger compartment.
- the floor/defrost/panel door directs airflow to the defroster duct and side window demisters. A small amount of airflow from the floor duct will be present.
- blended air temperature is available.
- to reduce fogging, the A/C compressor will operate automatically, regardless of indicator status.
- the blower motor is ON.

DIAGNOSTIC TESTS

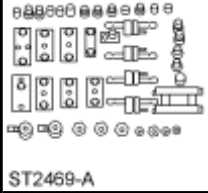
CLIMATE CONTROL SYSTEM

Special Tools

Illustration	Tool Name	Tool Number
 ST2574-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool	
 ST1137-A	Fluke 77 III Automotive Meter	105-R0056 or equivalent
 ST2570-A	R-134a Manifold Gauge Set	176-R032A or equivalent
 ST2826-A	A/C Flush Adapter Kit	219-00074 or equivalent

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A/C Flush and Purge Fitting Kit

219-00024 or equivalent



Pressure Test Kit

014-R1072 or equivalent

Principles of Operation

There are 4 main principles involved with the basic theory of operation:

- Heat transfer
- Latent heat of vaporization
- Relative humidity
- Effects of pressure

Heat Transfer

If 2 substances of different temperature are placed near each other, the heat in the warmer substance will transfer to the colder substance.

Latent Heat of Vaporization

When a liquid boils (converts to gas) it absorbs heat without raising the temperature of the resulting gas. When the gas condenses (converts back to a liquid), it gives off heat without lowering the temperature of the resulting liquid.

Relative Humidity

The amount of moisture (water vapor content) that the air can hold is directly related to the air temperature. The more heat there is in the air, the more moisture the air can hold. The lower the moisture content in the air, the more comfortable it is. Removing the moisture from the air lowers its relative humidity and improves personal comfort.

Effects of Pressure on Boiling or Condensation

As the pressure is increased on a liquid, the temperature at which the liquid boils (converts to gas) also increases. Conversely, when the pressure on a liquid is reduced, its boiling point is also reduced. When in the gas state, an increase in pressure causes an increase in temperature, while a decrease in pressure will decrease the temperature of the gas.

Compressor Anti-Slugging Strategy (CASS)

Liquid refrigerant may accumulate in the A/C compressor under certain conditions. To alleviate damage to the A/C compressor, the compressor anti-slugging strategy (CASS) is utilized.

CASS is initiated only under specific conditions:

- Ignition is OFF for more than 8 hours
- Ambient temperature is above -4°C (25°F)
- Battery voltage is above 8.5 volts during engine cranking

When these conditions are present, the PCM will activate the A/C control relay prior to cranking of the engine. The A/C control relay engages the A/C compressor for approximately 4-15 A/C compressor revolutions or a maximum of 2 seconds (depending upon vehicle application), allowing the liquid refrigerant to be pushed from the A/C compressor. CASS is initiated by the PCM regardless of the HVAC - EMTC or the HVAC - DATC system settings.

The Refrigerant Cycle

During stabilized conditions (A/C system shutdown), the refrigerant pressures are equal throughout the system. When the A/C compressor is in operation it increases pressure on the refrigerant vapor, raising its temperature. The high-pressure and high-temperature vapor is then released into the top of the A/C condenser core.

The A/C condenser, being close to ambient temperature, causes the refrigerant vapor to condense into a liquid when heat is removed from the refrigerant by ambient air passing over the fins and tubing. The now liquid refrigerant, still at high pressure, exits from the bottom of the A/C condenser and enters the inlet side of the A/C receiver/drier. The receiver/drier is designed to remove moisture from the refrigerant.

The outlet of the receiver/drier is connected to the thermostatic expansion valve (TXV). The TXV provides the orifice which is the restriction in the refrigerant system and separates the high and low pressure sides of the A/C system. As the liquid refrigerant passes across this restriction, its pressure and boiling point are reduced.

The liquid refrigerant is now at its lowest pressure and temperature. As it passes through the A/C evaporator, it absorbs heat from the airflow passing over the plate/fin sections of the A/C evaporator. This addition of heat causes the refrigerant to boil (convert to gas). The now cooler air can no longer support the same humidity level of the warmer air and this excess moisture condenses on the exterior of the evaporator coils and fins and drains outside the vehicle.

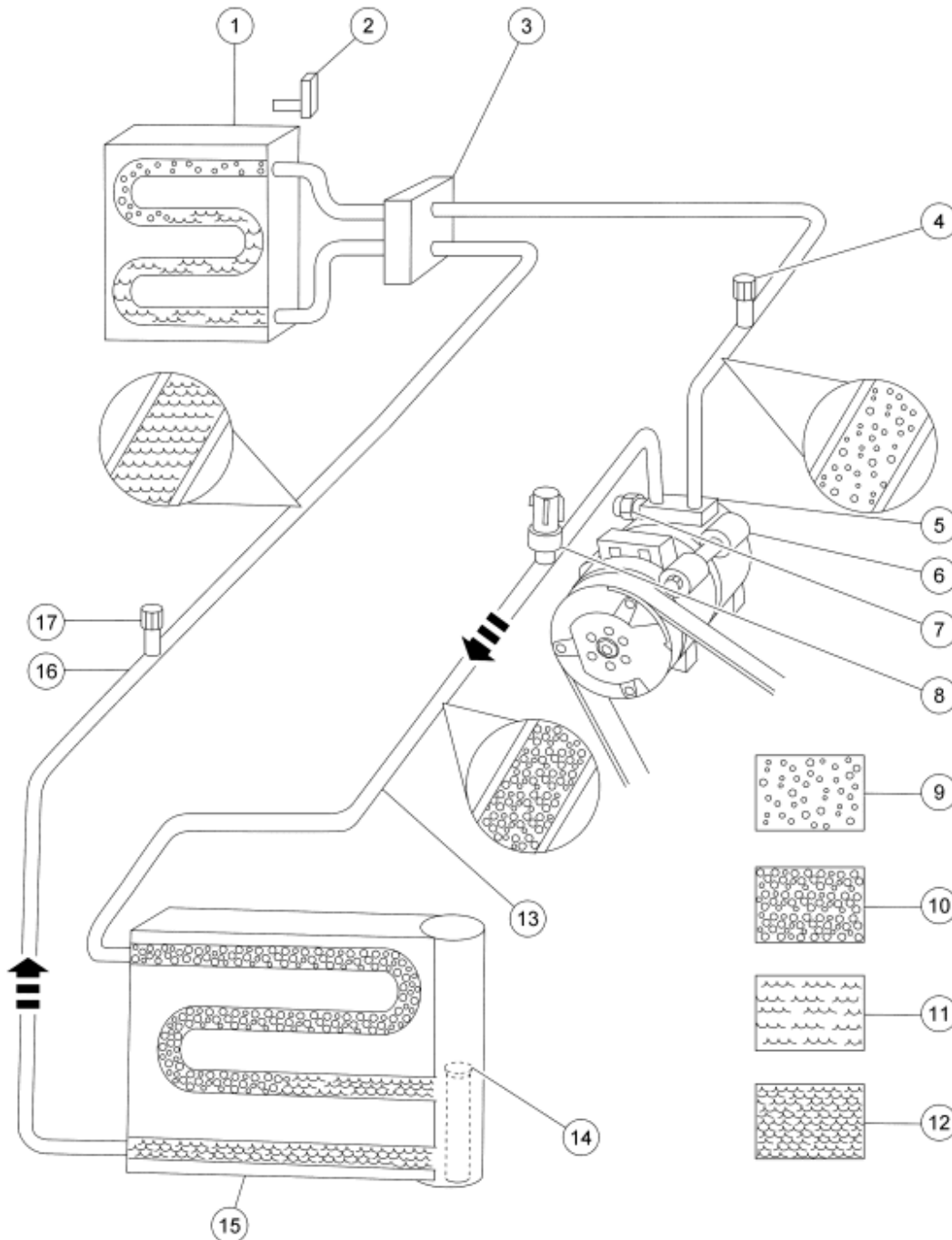
The refrigerant cycle is now repeated with the A/C compressor again increasing the pressure and temperature of the refrigerant.

A thermistor which monitors the temperature of the air that has passed through the evaporator core controls A/C clutch cycling. If the temperature of the evaporator core discharge air is low enough to cause the condensed water vapor to freeze, the A/C clutch is disengaged by the vehicle PCM.

The high-side line pressure is also monitored so that A/C compressor operation will be interrupted if the system pressure becomes too high or is determined to be too low (low charge condition).

The A/C compressor thermal protection switch will interrupt compressor operation if the compressor housing exceeds temperature limits.

The A/C compressor relief valve will open and vent refrigerant to relieve unusually high system pressure.



N0062913

Fig. 1: Thermostatic Expansion Valve (TXV) Type Refrigerant System
 Courtesy of FORD MOTOR CO.

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Item	Description
1	A/C evaporator core
2	A/C evaporator core temperature sensor
3	Thermostatic expansion valve (TXV)
4	A/C charge valve port (low side)
5	A/C suction line
6	A/C compressor
7	A/C pressure relief valve
8	A/C pressure transducer
9	Low-pressure vapor
10	High-pressure vapor
11	Low-pressure liquid
12	High-pressure liquid
13	Compressor discharge line
14	A/C desiccant cartridge
15	A/C condenser core with A/C receiver/drier
16	Condenser-to-evaporator line
17	A/C charge valve port (high side)

Inspection and Verification

1. Verify the customer's concern by operating the climate control system to duplicate the condition.
2. Inspect to determine if one of the following mechanical or electrical concerns apply:

VISUAL INSPECTION CHART

Mechanical	Electrical
• Loose, missing or damaged A/C compressor drive belt • Loose or disconnected A/C clutch • Broken or binding door/actuator • Broken or leaking refrigerant lines • Obstructed in-vehicle temperature sensor • Disconnected in-vehicle temperature aspirator hose	• Smart junction box (SJB) fuse(s): ○ 15 (10A) ○ 37 (10A) ○ 45 (5A) ○ 66 (40A) • Battery junction box (BJB) fuse 35 (10A) • Blower motor inoperative • A/C compressor inoperative • Circuitry open/shorted • Disconnected electrical connectors

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- Cooling fan inoperative

3. As pinpoint tests and measurements are being carried out, be sure to inspect for any disconnected, loose-fitting or incorrectly installed component, module and in-line electrical connectors and pins.
4. If the inspection reveals obvious concerns that can be readily identified, repair as necessary.

NOTE: **Make sure to use the latest scan tool software release.**

5. If the cause is not visually evident, connect the scan tool to the data link connector (DLC).

NOTE: **The vehicle communication module (VCM) LED prove out confirms power and ground from the DLC are provided to the VCM.**

6. If the scan tool does not communicate with the VCM:
 - check the VCM connection to the vehicle.
 - check the scan tool connection to the VCM.
 - refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no communication with the scan tool.
7. If the scan tool does not communicate with the vehicle:
 - verify the ignition key is in the ON position.
 - verify the scan tool operation with a known good vehicle.
 - refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM, HVAC module - dual automatic temperature control (DATC).
8. Carry out the network test.
 - If the scan tool responds with no communication from one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.
 - If the network test passes, retrieve and record continuous memory DTCs.
9. Clear the continuous DTCs and carry out the self-test diagnostics for the HVAC module - DATC.

NOTE: **Some PCM DTCs may inhibit A/C operation. If any PCM DTCs are retrieved, diagnose those first. Refer to Introduction - Gasoline Engines article.**

10. If the HVAC module - DATC DTCs retrieved are related to the concern, go to the **HVAC MODULE - DUAL AUTOMATIC TEMPERATURE CONTROL DTC CHART**. If the PCM DTCs retrieved are related to the concern, go to the **Introduction - Gasoline Engines** article. For all other DTCs, refer to **MULTIFUNCTION ELECTRONIC MODULES** article.
11. If no DTCs related to the concern are retrieved, go to **Symptom Chart - Climate Control System**.

HVAC Module - Dual Automatic Temperature Control (DATC) - Diagnostic Methods

The dual automatic temperature control (DATC) system must be diagnosed by first retrieving any DTCs, if present.

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- An on-demand (hard fault) DTC indicates that the fault is currently present. An on-demand DTC suggests a wiring fault, disconnected connector or component failure.
- A continuous (intermittent) DTC alone (corresponding on-demand DTC is not present) indicates that the fault is an intermittent condition and may not be currently present. A continuous only DTC suggests a poor wiring connection, loose pin or terminal or intermittent component failure.

On-demand (hard fault) or continuous (intermittent fault) DTCs can be retrieved using a scan tool. If using a scan tool, refer to the scan tool operating manual.

On-demand DTCs can also be retrieved by carrying out the Dual Automatic Temperature Control Module On-Demand Self Test. To retrieve and/or clear continuous DTCs, carry out the HVAC Module - Dual Automatic Temperature Control (DATC) - Retrieve Continuous DTCs procedure. Always carry out the HVAC Module - Dual Automatic Temperature Control (DATC) On-Demand Self Test before retrieving continuous DTCs.

If no DTCs are present, go to **Symptom Chart - Climate Control System** or Go to **Symptom Chart - Noise, Vibration and Harshness (NVH)** for the appropriate diagnostic action.

HVAC Module - Dual Automatic Temperature Control (DATC) On-Demand Self Test

The HVAC Module - Dual Automatic Temperature Control (DATC) On-Demand Self Test will retrieve on-demand (hard fault) DTCs only, it will not retrieve continuous DTCs. Continuous DTCs can be cleared when exiting the HVAC Module - Dual Automatic Temperature Control (DATC) On-Demand Self Test. Make sure to retrieve continuous DTCs by carrying out the HVAC Module - Dual Automatic Temperature Control (DATC) - Retrieve Continuous DTCs procedure before clearing any continuous DTCs.

- The HVAC Module - Dual Automatic Temperature Control (DATC) On-Demand Self Test will not detect concerns associated with data link messages like engine coolant temperature or vehicle speed signals. A scan tool must be used to retrieve these concerns.
- The HVAC Module - Dual Automatic Temperature Control (DATC) On-Demand Self Test will detect concerns in the system control functions and will display on-demand (hard fault) DTCs for concerns that are present during the self-test. The vehicle interior temperature should be between 4°C-38°C (40°F-100°F) when carrying out the self-test. If the temperatures are not within the specified ranges, false DTCs may be displayed.
- The self-test can be initiated after cycling the ignition switch from OFF to ON. Normal operation of the climate control system stops when the self-test is activated.
- To enter the self-test, press the OFF and DEFROST buttons simultaneously and release, then press the AUTO button within 2 seconds. The display will show a flashing "----" for a maximum of 20 seconds, after which "8888" will be displayed if there are no on-demand DTCs present. If DTCs are present, the HVAC module - DATC will display the on-demand DTCs. Record all DTCs displayed.
- If any DTCs appear during the self-test, carry out the diagnostic procedure. Refer to the **HVAC MODULE - DUAL AUTOMATIC TEMPERATURE CONTROL DTC CHART** and follow the ACTION for each DTC given.
- If a condition exists, but no DTCs appear during the self-test, go to **Symptom Chart - Climate Control System** condition: The DATC System is Inoperative, Intermittent or Incorrect Operation.
- Always exit the self-test before powering the system down (system turned OFF).

- Continuous DTCs will be deleted after 80 ignition switch ON cycles after the intermittent fault occurs.

HVAC Module - Dual Automatic Temperature Control (DATC) - Retrieve Continuous DTCs

The HVAC module - DATC will retrieve only continuous (intermittent) DTCs when carrying out this procedure.

- Retrieval of continuous DTCs can be initiated after cycling the ignition switch from OFF to ON. Normal operation of the climate control system stops when retrieving continuous DTCs.
- To retrieve continuous DTCs, press the OFF and DEFROST buttons simultaneously and release, then press the RECIRC button within 2 seconds. The display will show a flashing "----" for a maximum of 20 seconds, after which "8888" will be displayed if there are no continuous DTCs present. If DTCs are present, the HVAC module - DATC will display the continuous DTCs present. Record all DTCs displayed.
- If any DTCs appear, carry out the diagnostic procedure. Refer to the **HVAC MODULE - DUAL AUTOMATIC TEMPERATURE CONTROL DTC CHART** and follow the ACTION for each DTC given.
- If a condition exists but no DTCs appear, go to **Symptom Chart - Climate Control System** condition: The Dual Automatic Temperature Control (DATC) System is Inoperative, Intermittent or Incorrect Operation.
- To exit and **retain** all continuous DTCs, press any button **except** DEFROST. The HVAC module - DATC will exit the retrieved continuous DTCs mode and retain all continuous DTCs.
- To exit and **clear** all continuous DTCs, press the DEFROST button. The HVAC module - DATC will exit the retrieve continuous DTCs mode and all continuous DTCs will be cleared.
- Always exit the procedure before powering the system down (system turned OFF).
- Continuous DTCs will be deleted after 80 ignition switch ON cycles after the intermittent fault occurs.

HVAC Module Cold Boot Process

The purpose of the cold boot process is to allow the HVAC module to reinitialize. To carry out the cold boot process, follow the steps below.

NOTE: HVAC module - DATC only.

1. Clear the DTCs.
2. Turn the ignition switch to the OFF position.
3. Disconnect the HVAC module electrical connectors.
4. Inspect module connectors for:
 - corrosion.
 - pushed-out connectors.
 - incorrectly seated connector.
5. Wait one minute.

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6. Connect the HVAC module electrical connectors.
7. Turn the ignition switch to the ON position.
8. Select any position except OFF on the HVAC module.

The HVAC module will now initialize.

HVAC Module - Dual Automatic Temperature Control (DATC) DTC Chart

NOTE: Some PCM DTCs may inhibit A/C operation. If any PCM DTCs are retrieved, diagnose those first. Refer to the Introduction - Gasoline Engines article.

HVAC MODULE - DUAL AUTOMATIC TEMPERATURE CONTROL DTC CHART

DTC	Description	Action to Take
B1251	Air Temperature Internal Sensor Circuit Open	Go to Pinpoint Test A .
B1253	Air Temperature Internal Sensor Circuit Short to Ground	Go to Pinpoint Test A .
B1255	Air Temperature External Sensor Circuit Open	Go to Pinpoint Test B .
B1257	Air Temperature External Sensor Circuit Short to Ground	Go to Pinpoint Test B .
B1317	Battery Voltage High	CLEAR the DTCs. REPEAT the self-test. If the DTC returns and the DTC is not present in other modules, MEASURE the voltage between ground and: HVAC module - dual automatic temperature control (DATC) C228a-12, circuit CBP37 (WH) and HVAC module - DATC C228a-13, circuit SBP15 (WH/RD). • If the voltage is less than 16 volts, INSTALL a new HVAC module - DATC. REFER to <u>CLIMATE CONTROL</u> article. • If the voltage is greater than 16 volts, INSTALL a new HVAC module - DATC. REFER to <u>CHARGING SYSTEM - GENERAL INFORMATION</u> article. If the DTC returns and the DTC is present in other modules, REFER to <u>CHARGING SYSTEM - GENERAL INFORMATION</u> article.
B1318	Battery Voltage Low	Go to Pinpoint Test G .
B1342	ECU is Faulted	CLEAR the DTCs. REPEAT the self-test. If the DTC returns, INSTALL a new HVAC module - DATC. REFER to <u>CLIMATE CONTROL</u> article.

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B1947	Climate Control A/C Post Evaporator Sensor Circuit Short to Ground	Go to Pinpoint Test D .
B2014	Climate Control A/C Post Evaporator Sensor Open Circuit	Go to Pinpoint Test D .
B226C	Left Blend Door Motor Stall	Go to Pinpoint Test F .
B226D	Right Blend Door Motor Stall	Go to Pinpoint Test F .
B226E	Mode Blend Door Motor Stall	Go to Pinpoint Test F .
B2426	Passenger Solar Radiation Sensor Circuit Open	Go to Pinpoint Test C .
B2427	Passenger Solar Radiation Sensor Circuit Short to Ground	Go to Pinpoint Test C .
B2795	Driver Solar Radiation Sensor Circuit Short to Ground	Go to Pinpoint Test C .
B2796	Driver Solar Radiation Sensor Circuit Open	Go to Pinpoint Test C .
B295A	Driver Blend Door Potentiometer Feedback Circuit Open	Go to Pinpoint Test F .
B295B	Driver Blend Door Potentiometer Feedback Circuit Short to Ground	Go to Pinpoint Test F .
B295C	Passenger Blend Door Potentiometer Feedback Circuit Open	Go to Pinpoint Test F .
B295D	Passenger Blend Door Potentiometer Feedback Circuit Short to Ground	Go to Pinpoint Test F .
B295E	Mode Door Potentiometer Feedback Circuit Open	Go to Pinpoint Test F .
B295F	Mode Door Potentiometer Feedback Circuit Short to Ground	Go to Pinpoint Test F .
U0073	Control Module Communication Bus A Off	CLEAR the DTCs. REPEAT the self-test. If the DTC returns, REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article to diagnose the network concern.
U0140	Lost Communication With Body Control Module (GEM)	CLEAR The DTCs. REPEAT the self-test. If the DTC returns, REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article to diagnose the lost communication with the smart junction box (SJB).
U0155	Lost Communication With Instrument Panel Cluster Control Module	CLEAR The DTCs. REPEAT the self-test. If the DTC returns, REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article to diagnose the lost communication with the instrument panel cluster.

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U0246	Lost Communication With Seat Control Module (DCSM)	CLEAR The DTCs. REPEAT the self-test. If the DTC returns, REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article to diagnose the lost communication with the heated seat module.
U2050	No Application Present	REFLASH the HVAC module - DATC. REFER to <u>MODULE CONFIGURATION</u> article. If the DTC returns, INSTALL a new HVAC module - DATC. REFER to <u>CLIMATE CONTROL</u> article.
U2051	One or More Calibration Files Missing/Corrupt	REFLASH the HVAC module - DATC. REFER to <u>MODULE CONFIGURATION</u> article. If the DTC returns, INSTALL a new HVAC module - DATC. REFER to <u>CLIMATE CONTROL</u> article.

Symptom Chart - Climate Control System

Symptom Chart - Climate Control System

Condition	Possible Sources	Action
No communication with the HVAC module - dual automatic temperature control (DATC)	- Fuse(s) - Circuitry open - HVAC module - DATC communication network	REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article.
Air inlet door is inoperative	- Circuitry open/shorted - HVAC module - DATC - HVAC module - electronic manual temperature control (EMTC) - Air inlet door actuator/linkage	Go to Pinpoint Test E .
Incorrect/erratic direction of airflow from outlets	- Circuitry short/open - Door actuator - Defrost/panel/floor door actuator/linkage - LH or RH blend door actuator/linkage - HVAC module - EMTC or HVAC module - DATC	Go to Pinpoint Test F .
The temperature control is inoperative/does not operate correctly	- Circuitry open/shorted - Door actuator - LH or RH blend door actuator/linkage	Go to Pinpoint Test F .

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	HVAC module - EMTC or HVAC module - DATC	
The DATC system is inoperative, intermittent or incorrect operation	- Fuse(s) - Circuitry open	Go to Pinpoint Test H .
Insufficient, erratic or no heat	- Low engine coolant level - Engine overheating - Plugged or partially plugged heater core - Temperature blend door is binding or stuck - Temperature blend door actuator - Heater hose is kinked or binding	Go to Pinpoint Test I .
The air conditioning (A/C) is inoperative/does not operate correctly	- Open fuse - Circuitry short/open - PCM - HVAC module - EMTC or HVAC module - DATC - A/C compressor - A/C control relay - A/C evaporator temperature sensor - A/C pressure transducer - Battery junction box (BJB)	Go to Pinpoint Test J .
The air conditioning (A/C) is always on	- Circuitry shorted - PCM - HVAC module - EMTC or HVAC module - DATC - A/C compressor - A/C control relay - A/C evaporator temperature sensor - A/C pressure transducer	Go to Pinpoint Test K .
The blower motor is	- Fuse(s) - Circuitry open - A/C blower motor switch	

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inoperative - EMTC	- Blower motor relay - A/C blower motor - HVAC module - EMTC	Go to Pinpoint Test L .
The blower motor does not operate correctly - EMTC	- Circuitry open/shorted - A/C blower motor resistor - A/C blower motor switch	Go to Pinpoint Test M .
The blower motor is inoperative - DATC	- Fuse(s) - Circuitry open/shorted - Blower motor relay - A/C blower motor - HVAC module - DATC - Blower motor speed control	Go to Pinpoint Test N .
The blower motor does not operate correctly - DATC	- Circuitry open/shorted - Blower motor speed control - HVAC module - DATC - A/C blower motor relay	Go to Pinpoint Test O .
The dual temperature control is inoperative/does not operate correctly	- Temperature blend door/actuator broken/binding - HVAC module - DATC - Low A/C refrigerant charge	CHECK that the A/C system pressure is above 290 kPa (42 psi). If the pressure is below 290 kPa (42 psi), REFER to <u>Fluorescent Dye Leak Detection</u>. CHECK the temperature blend doors/actuators for a binding or broken condition. If no condition is found, INSTALL a new HVAC module - DATC.
The temperature set point does not repeat after turning the ignition switch OFF	- Open fuse - Circuitry short/open - HVAC module - DATC	CHECK circuit SBP15 (WH/RD) for a short or open and repair as necessary. If OK, INSTALL a new HVAC module - DATC.
The temperature display will not switch between Celsius and Fahrenheit	- HVAC module - DATC - Instrument cluster module	- If the instrument cluster does not switch between English and Metric, REFER to <u>INSTRUMENT CLUSTER & PANEL ILLUMINATION</u> article. - If only the HVAC module - DATC does not switch between

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		English and Metric, INSTALL a new HVAC module - DATC.
Inaccurate external temperature display	- HVAC module - DATC - Ambient temperature sensor	REFER to Description and Operation of the External Temperature Display. If the external temperature display is not operating as described and no DTC is present, CARRY OUT the ambient temperature sensor component test in this service information. If the sensor tests OK, CARRY OUT the HVAC Module Cold Boot Process. If the condition returns, INSTALL a new HVAC module - DATC. REFER to <u>CLIMATE CONTROL</u> article. TEST the system for normal operation.

Symptom Chart - Noise, Vibration and Harshness (NVH)

NOTE: Noise, vibration and harshness (NVH) symptoms will be identified using the diagnostic tools that are available. For a list of these tools, an explanation of their uses and a glossary of common terms, refer to **NOISE, VIBRATION & HARSHNESS** article. Since it is possible any one of multiple systems may be the cause of a symptom, it may be necessary to use a process of elimination type of diagnostic approach to pinpoint the responsible system. If this is not the causal system for the symptom, refer back to **NOISE, VIBRATION AND HARSHNESS** article for the next likely system and continue diagnosis.

Symptom Chart - Noise, Vibration and Harshness (NVH)

Condition	Possible Sources	Action
Noisy A/C compressor clutch	A/C compressor clutch air gap	INSTALL a new A/C compressor. REFER to <u>CLIMATE CONTROL</u> article. TEST the system for normal operation.

Pinpoint Tests

Pinpoint Test A: DTC B1251 or B1253 - Air Temperature Internal Sensor Circuit Open or Short to Ground

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Automatic Climate Control System for schematic and connector information.

Normal Operation

Under normal operation, the in-vehicle air temperature sensor receives a ground from the HVAC module - dual automatic temperature control (DATC) through circuit RH111 (GY/BU). A 5-volt reference voltage is supplied to the in-vehicle air temperature sensor from the HVAC module - DATC through circuit VH414 (GN/BU).

- DTC B1251 Air Temperature Internal Sensor Circuit Open - The module senses no voltage drop on the sensor reference voltage circuit, indicating an open circuit.
- DTC B1253 Air Temperature Internal Sensor Circuit Short to Ground - The module senses excessive voltage drop on the sensor reference voltage circuit, indicating a short directly to ground.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- In-vehicle air temperature sensor
- HVAC module - DATC

PINPOINT TEST A: DTC B1251 OR B1253 - AIR TEMPERATURE INTERNAL SENSOR CIRCUIT OPEN OR SHORT TO GROUND

A1 CHECK THE SENSOR RESISTANCE

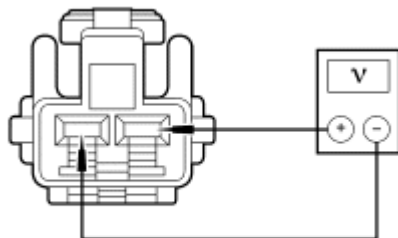
- Key in OFF position.
- Disconnect: In-Vehicle Temperature Sensor C233
- Carry out the In-Vehicle Temperature Sensor component test in this service information.
- **Does the in-vehicle temperature sensor test OK?**

YES : Go to A2.

NO : INSTALL a new in-car temperature sensor. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

A2 CHECK THE HVAC MODULE - DATC SENSOR OUTPUT VOLTAGE

- Key in ON position.
- Press the AUTO button.



N0061460

Fig. 2: Measuring Voltage Between In-Vehicle Temperature Sensor C233-2 & C233-1, Circuit VH414 (GN/BU) & RH111 (GY/BU)

Courtesy of FORD MOTOR CO.

- Measure the voltage between in-vehicle temperature sensor C233-2, circuit VH414 (GN/BU), harness side and C233-1, circuit RH111 (GY/BU), harness side.

- **Is the voltage between 4.7 and 5.1 volts?**

YES : INSTALL a new in-vehicle sensor. CLEAR the DTCs. REPEAT the self-test. If code returns, go to A7.

NO : If diagnosing DTC B1251, go to A3.

If diagnosing DTC B1253, go to A5.

A3 CHECK CIRCUIT VH414 (GN/BU) FOR AN OPEN

- Key in OFF position.
- Disconnect: HVAC Module - DATC C228a

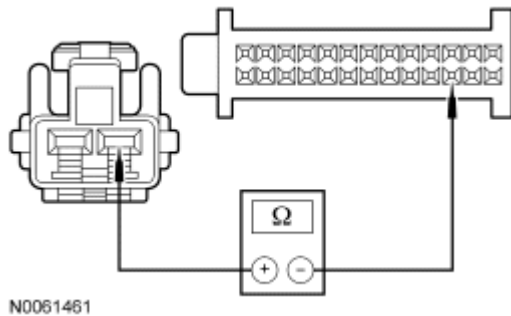


Fig. 3: Measuring Resistance Between EATC Module C228A-3, Circuit VH414 (GN/BU) & In-Vehicle Temperature Sensor C233-2

Courtesy of FORD MOTOR CO.

- Measure the resistance between HVAC module - DATC C228a-3, circuit VH414 (GN/BU), harness side and in-vehicle temperature sensor C233-2, circuit VH414 (GN/BU), harness side.

- **Is the resistance less than 5 ohms?**

YES : Go to A4.

NO : REPAIR circuit VH414 (GN/BU) for an open. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

A4 CHECK CIRCUIT RH111 (GY/BU) FOR AN OPEN

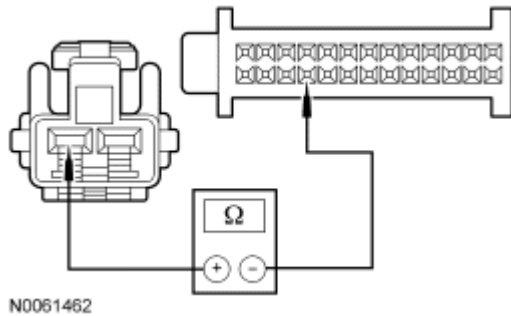


Fig. 4: Measuring Resistance Between EATC Module C228A-10, Circuit RH111 (GY/BU) & In-Vehicle Temperature Sensor C233-1
Courtesy of FORD MOTOR CO.

- Measure the resistance between HVAC module - DATC C228a-10, circuit RH111 (GY/BU), harness side and in-vehicle temperature sensor C233-1, circuit RH111 (GY/BU), harness side.

- **Is the resistance less than 5 ohms?**

YES : Go to A7.

NO : REPAIR circuit RH111 (GY/BU) for an open. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

A5 CHECK CIRCUIT VH414 (GN/BU) FOR A SHORT TO CIRCUIT RH111 (GY/BU)

- Key in OFF position.
- Disconnect: HVAC Module - DATC C228a

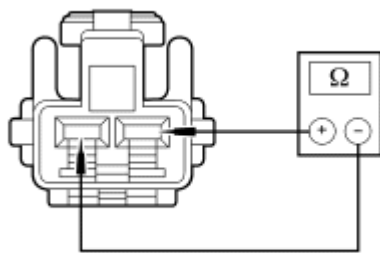


Fig. 5: Measuring Resistance Between In-Vehicle Temperature Sensor C233-2 & C233-1, Circuit VH414 (GN/BU) & RH111 (GY/BU)
Courtesy of FORD MOTOR CO.

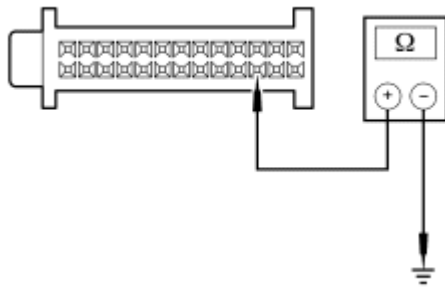
- Measure the resistance between in-vehicle temperature sensor C233-2, circuit VH414 (GN/BU), harness side and C233-1, circuit RH111 (GY/BU), harness side.

- **Is the resistance greater than 10,000 ohms?**

YES : Go to A6.

NO : REPAIR circuit VH414 (GN/BU) for a short to circuit RH111 (GY/BU). CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

A6 CHECK CIRCUIT VH414 (GN/BU) FOR A SHORT TO GROUND



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Fig. 6: Measuring Resistance Between EATC Module C228A-3, Circuit VH414 (GN/BU) & Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between HVAC module - DATC C228a-3, circuit VH414 (GN/BU), harness side and ground.
- **Is the resistance greater than 10,000 ohms?**
YES : Go to A7.
NO : REPAIR circuit VH414 (GN/BU) for a short to ground. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

A7 CHECK THE MODULE CONNECTION

- Carry out the HVAC Module Cold Boot Process.
- Carry out the HVAC Module - Dual Automatic Temperature Control (DATC) On-Demand Self Test.
- Operate the system.
- **Does the concern return?**
YES : INSTALL a new HVAC module - DATC. REFER to **CLIMATE CONTROL** article. TEST the system for normal operation.
NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test B: DTC B1255 or B1257 - Air Temperature External Sensor Circuit Open or Short to Ground

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Automatic Climate Control System for schematic and connector information.

Normal Operation

Under normal operation, the ambient air temperature sensor receives a ground from the HVAC module - dual automatic temperature control (DATC) through circuit RH111 (GY/BU). A 5-volt reference voltage is supplied to the ambient air temperature sensor from the HVAC module - DATC through circuit VH407 (YE/GN).

- DTC B1255 Air Temperature External Sensor Circuit Open - The module senses no voltage drop on the

sensor reference voltage circuit, indicating an open circuit.

- DTC B1257 Air Temperature External Sensor Circuit Short To Ground - The module senses excessive voltage drop on the sensor reference voltage circuit, indicating a short directly to ground.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Ambient air temperature sensor
- HVAC module - DATC

PINPOINT TEST B: DTC B1255 OR B1257 - AIR TEMPERATURE EXTERNAL SENSOR CIRCUIT OPEN OR SHORT TO GROUND

B1 CHECK THE SENSOR RESISTANCE

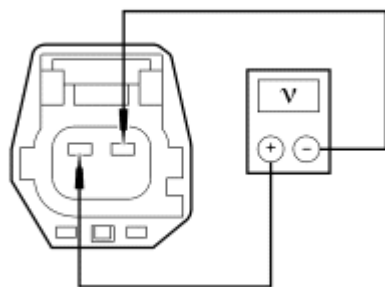
- Key in OFF position.
- Disconnect: Ambient Temperature Sensor C198
- Carry out the Ambient Temperature Sensor component test in this service information.
- **Does the ambient temperature sensor test OK?**

YES : Go to B2.

NO : INSTALL a new ambient temperature sensor. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

B2 CHECK THE HVAC - DATC SENSOR OUTPUT VOLTAGE

- Key in ON position.
- Press the AUTO button.



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Fig. 7: Measuring Voltage Between Ambient Temperature Sensor C198-2 & C198-1, Circuit VH407 (YE/GN) & RH111 (GY/BU)

Courtesy of FORD MOTOR CO.

- Measure the voltage between ambient temperature sensor C198-2, circuit VH407 (YE/GN), harness side and C198-1, circuit RH111 (GY/BU), harness side.
- **Is the voltage between 4.7 and 5.1 volts?**

YES : INSTALL a new ambient sensor. CLEAR the DTCs. REPEAT the self-test. If code returns,

go to B7.

NO : If diagnosing DTC B1255, go to B3.

If diagnosing DTC B1257, go to B5.

B3 CHECK CIRCUIT VH407 (YE/GN) FOR AN OPEN

- Key in OFF position.
- Disconnect: HVAC Module - DATC C228a

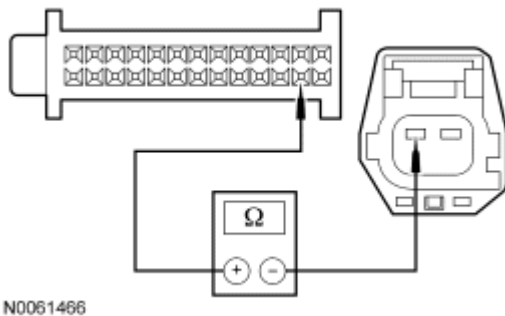


Fig. 8: Measuring Resistance Between EATC Module C228A-2, Circuit VH407 (YE/GN) & Ambient Temperature Sensor C198-2
Courtesy of FORD MOTOR CO.

- Measure the resistance between HVAC module - DATC C228a-2, circuit VH407 (YE/GN), harness side and ambient temperature sensor C198-2, circuit VH407 (YE/GN), harness side.
- **Is the resistance less than 5 ohms?**
YES : Go to B4.
NO : REPAIR circuit VH407 (YE/GN) for an open. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

B4 CHECK CIRCUIT RH111 (GY/BU) FOR AN OPEN

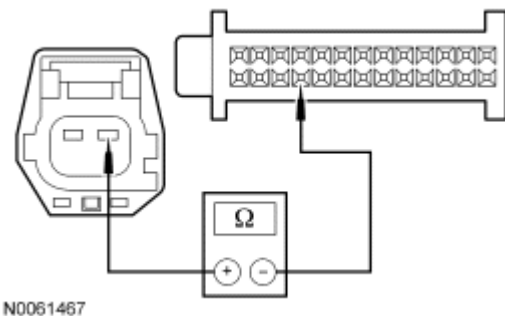


Fig. 9: Measuring Resistance Between EATC Module C228A-10, Circuit RH111 (GY/BU) & Ambient Temperature Sensor C198-1
Courtesy of FORD MOTOR CO.

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- Measure the resistance between HVAC module - DATC C228a-10, circuit RH111 (GY/BU), harness side and ambient temperature sensor C198-1, circuit RH111 (GY/BU), harness side.
- **Is the resistance less than 5 ohms?**
YES : Go to B7.
NO : REPAIR circuit RH111 (GY/BU) for an open. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

B5 CHECK CIRCUIT VH407 (YE/GN) FOR A SHORT TO CIRCUIT RH111 (GY/BU)

- Key in OFF position.
- Disconnect: HVAC Module - DATC C228a

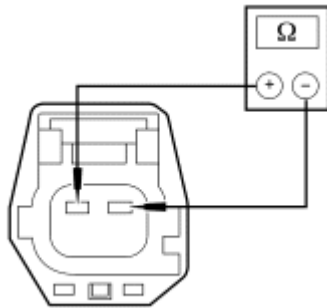


Fig. 10: Measuring Resistance Between Ambient Temperature Sensor C198-2 & C198-1, Circuit VH407 (YE/GN) & RH111 (GY/BU)
Courtesy of FORD MOTOR CO.

- Measure the resistance between ambient temperature sensor C198-2, circuit VH407 (YE/GN), harness side and C198-1, circuit RH111 (GY/BU), harness side.
- **Is the resistance greater than 10,000 ohms?**
YES : Go to B6.
NO : REPAIR circuit VH407 (YE/GN) for a short to circuit RH111 (GY/BU). CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

B6 CHECK CIRCUIT VH407 (YE/GN) FOR A SHORT TO GROUND

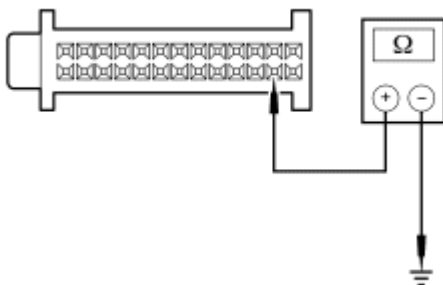


Fig. 11: Measuring Resistance Between EATC Module C228A-2, Circuit VH407 (YE/GN) & Ground

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Courtesy of FORD MOTOR CO.

- Measure the resistance between HVAC module - DATC C228a-2, circuit VH407 (YE/GN), harness side and ground.

- **Is the resistance greater than 10,000 ohms?**

YES : Go to B7.

NO : REPAIR circuit VH407 (YE/GN) for a short to ground. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

B7 CHECK THE MODULE CONNECTION

- Carry out the HVAC Module Cold Boot Process.
- Carry out the HVAC Module - Dual Automatic Temperature Control (DATC) On-Demand Self Test.
- Operate the system.

- **Does the concern return?**

YES : INSTALL a new HVAC module - DATC. REFER to **CLIMATE CONTROL** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test C: DTC B2426, B2427, B2795 or B2796 - Solar Radiation Sensor Circuit Open or Short to Ground

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Automatic Climate Control System for schematic and connector information.

Normal Operation

Under normal operation, the solar radiation sensor receives a ground through circuit GD115 (BK/GY). A 5-volt reference voltage is supplied to the solar radiation sensor from the HVAC module - dual automatic temperature control (DATC) through circuit VH417 (YE/OG) (LH) and circuit VH416 (VT/GY) (RH).

DTC	Description	Fault Trigger Conditions
B2426	Passenger Solar Radiation Sensor Circuit Open	The module senses no voltage drop on the sensor reference voltage circuit, indicating an open circuit.
B2427	Passenger Solar Radiation Sensor Circuit Short to Ground	The module senses excessive voltage drop on the sensor reference voltage circuit, indicating a short directly to ground.
B2795	Driver Solar Radiation Sensor Circuit Short to Ground	The module senses no voltage drop on the sensor reference voltage circuit, indicating an open circuit.
		The module senses excessive voltage

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B2796

Driver Solar Radiation Sensor Circuit
Open

drop on the sensor reference voltage circuit, indicating a short directly to ground.

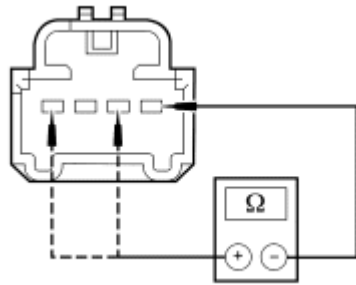
This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Solar radiation sensor
- HVAC module - DATC

PINPOINT TEST C: DTC B2426, DTC B2427, DTC B2795 OR DTC B2796 - A/C SOLAR RADIATION SENSOR OPEN CIRCUIT OR SHORT TO GROUND

C1 CHECK THE SOLAR RADIATION SENSOR RESISTANCE

- Key in OFF position.
- Disconnect: Solar Radiation Sensor C286



N0041877

Fig. 12: Checking Sensor Resistance
Courtesy of FORD MOTOR CO.

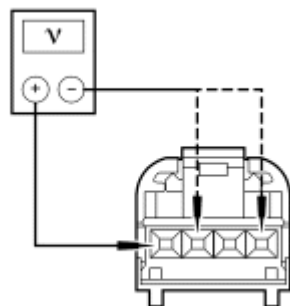
- Measure the resistance between the solar radiation sensor terminals 1 and 4 and terminals 3 and 4.
- **Is continuity present and the resistance greater than 0 ohm?**

YES : Go to C2.

NO : INSTALL a new solar radiation sensor. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

C2 CHECK THE SOLAR RADIATION SENSOR REFERENCE VOLTAGE

- Key in ON position.
- Press the AUTO button.



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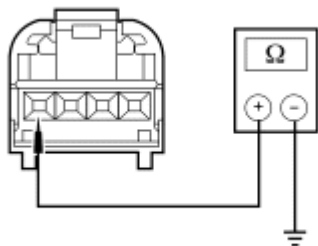
Fig. 13: Checking Circuit For Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between solar radiation sensor C286-4, circuit GD115 (BK/GY), harness side and:
 - **DTC B2426 or B2427** , C286-1, circuit VH416 (VT/GY), harness side.
 - **DTC B2795 or B2796** , C286-3, circuit VH417 (YE/OG), harness side.
- **Are the voltages between 4.7 and 5.1 volts?**
YES : Go to C6.
NO : If diagnosing DTC B2426 or B2796, go to C3.

If diagnosing DTC B2427 or B2795, go to C5.

C3 CHECK CIRCUIT GD115 (BK/GY) FOR AN OPEN

- Key in OFF position.



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Fig. 14: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between solar radiation sensor C286-4, circuit GD115 (BK/GY), harness side and ground.
- **Is the resistance less than 5 ohms?**
YES : Go to C4.
NO : REPAIR circuit GD115 (BK/GY) for an open. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

C4 CHECK CIRCUIT VH417 (YE/OG) OR VH416 (VT/GY) FOR AN OPEN

- Disconnect: HVAC Module - DATC C228a

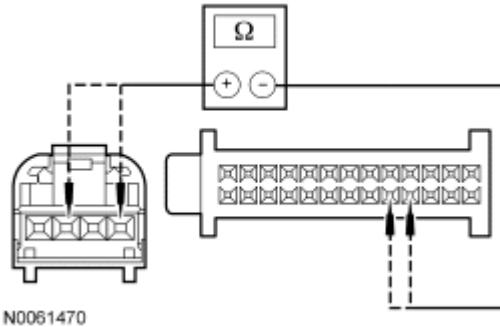


Fig. 15: Measuring Resistance Between EATC Module C228A-4 & C228A-5, Circuit VH416 (VT/GY) & VH417 (YE/OG)
 Courtesy of FORD MOTOR CO.

- Measure the resistance between:
 - **DTC B2426** , HVAC module - DATC C228a-4, circuit VH416 (VT/GY), harness side and solar radiation sensor C286-1, circuit VH416 (VT/GY), harness side.
 - **DTC B2796** , HVAC module - DATC C228a-5, circuit VH417 (YE/OG), harness side and solar radiation sensor C286-3, circuit VH417 (YE/OG), harness side.

- **Are the resistances less than 5 ohms?**

YES : Go to C6.

NO : REPAIR circuit VH417 (YE/OG) or VH416 (VT/GY) for an open. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

C5 CHECK CIRCUIT VH417 (YE/OG) OR VH416 (VT/GY) FOR A SHORT TO GROUND

- Key in OFF position.
- Disconnect: HVAC Module - DATC C228a

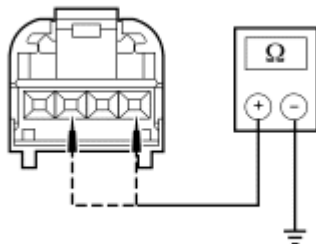


Fig. 16: Checking Circuits For Short To Ground
 Courtesy of FORD MOTOR CO.

- Measure the resistance between ground and:

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- **DTC B2427** , solar radiation sensor C286-1, circuit VH416 (VT/GY), harness side.
- **DTC B2795** , solar radiation sensor C286-3, circuit VH417 (YE/OG), harness side.
- **Are the resistances greater than 10,000 ohms?**

YES : Go to C6.

NO : REPAIR circuit VH417 (YE/OG) or VH416 (VT/GY) for a short to ground. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

C6 CHECK THE MODULE CONNECTION

- Carry out the HVAC Module Cold Boot Process.
- Carry out the HVAC Module - Dual Automatic Temperature Control (DATC) On-Demand Self Test.
- Operate the system.
- **Does the concern return?**

YES : INSTALL a new HVAC module - DATC. REFER to **CLIMATE CONTROL** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test D: DTC B1947 or B2014 - Climate Control A/C Post Evaporator Sensor Circuit Open or Short to Ground

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Automatic Climate Control System for schematic and connector information.

Normal Operation

Under normal operation, the evaporator temperature sensor receives a ground from the HVAC module - dual automatic temperature control (DATC) through circuit RH111 (GY/BU)/(BK). A 5-volt reference voltage is supplied to the evaporator discharge air temperature sensor from the HVAC module - DATC through circuit VH406 (VT/BN)/(LG/BK).

- DTC B1947 Climate Control A/C Post Evaporator Sensor Circuit Short To Ground - The module senses excessive voltage drop on the sensor reference voltage circuit, indicating a short directly to ground.
- DTC B2014 Climate Control A/C Post Evaporator Sensor Open Circuit - The module senses no voltage drop on the sensor reference voltage circuit, indicating an open circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Evaporator temperature sensor
- HVAC module - electronic manual temperature control (EMTC) or HVAC module - DATC

PINPOINT TEST D: DTC B1947 OR DTC B2014 - CLIMATE CONTROL A/C POST EVAPORATOR

SENSOR CIRCUIT OPEN OR SHORT TO GROUND

NOTE: The evaporator temperature sensor cannot be installed independently of the heater core and evaporator core housing.

D1 CHECK THE SENSOR RESISTANCE

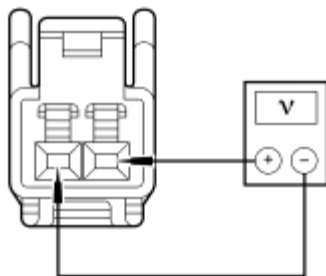
- Key in OFF position.
- Disconnect: Evaporator Temperature Sensor C296
- Carry out the evaporator temperature sensor component test in this service information.
- **Is the resistance within the specified values for the temperatures?**

YES : Go to D2.

NO : INSTALL a new heater core and evaporator core housing. REFER to **CLIMATE CONTROL** article. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

D2 CHECK THE HVAC MODULE - DATC SENSOR OUTPUT VOLTAGE

- Key in ON position.
- Press the AUTO button.



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Fig. 17: Measuring Voltage Between Evaporator Temperature Sensor C296-2, Circuit (LG/BK) & C296-1, Circuit (BK)

Courtesy of FORD MOTOR CO.

- Measure the voltage between evaporator temperature sensor C296-2, circuit (LG/BK), harness side and C296-1, circuit (BK), harness side.
- **Is the voltage between 4.7 and 5.1 volts?**

YES : Go to D7.

NO : If diagnosing DTC B2826, go to D3.

If diagnosing DTC B2827, go to D5.

D3 CHECK CIRCUIT VH406 (VT/BN)/(LG/BK) FOR AN OPEN

- Key in OFF position.
- Disconnect: HVAC Module - DATC C228a

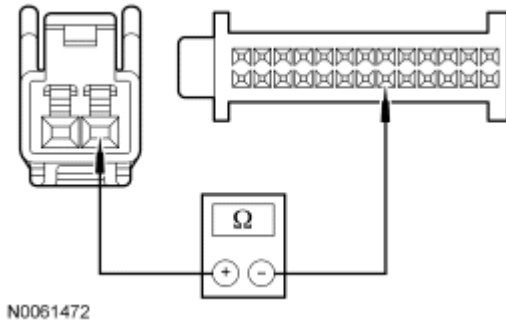


Fig. 18: Measuring Resistance Between EATC Module C228A-6 & C296-2, Circuit VH406 (VT/BN) & (LG/BK)
 Courtesy of FORD MOTOR CO.

- Measure the resistance between HVAC module - DATC C228a-6, circuit VH406 (VT/BN), harness side and evaporator temperature sensor C296-2, circuit (LG/BK), harness side.

- **Is the resistance less than 5 ohms?**

YES : Go to D4.

NO : REPAIR circuit VH406 (VT/BN)/(LG/BK) for an open. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

D4 CHECK CIRCUIT RH111 (GY/BU)/(BK) FOR AN OPEN

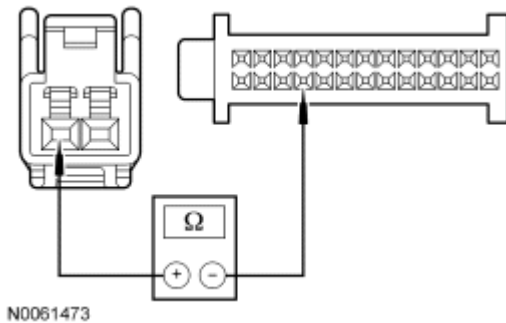


Fig. 19: Measuring Resistance Between EATC Module C228A-10 & C296-1, Circuit RH111 (GY/BU) & Circuit (BK)
 Courtesy of FORD MOTOR CO.

- Measure the resistance between HVAC module - DATC C228a-10, circuit RH111 (GY/BU), harness side and evaporator temperature sensor C296-1, circuit (BK), harness side.

- **Is the resistance less than 5 ohms?**

YES : Go to D7.

NO : REPAIR circuit RH111 (GY/BU)/(BK) for an open. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

D5 CHECK CIRCUIT VH406 (VT/BN)/(LG/BK) FOR A SHORT TO CIRCUIT RH111 (GY/BU)/(BK)

- Key in OFF position.

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- Disconnect: HVAC Module - DATC C228a

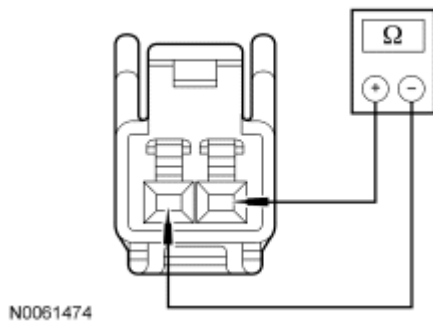


Fig. 20: Measuring Resistance Between Evaporator Temperature Sensor C296-2 & C296-1, Circuit (LG/BK) & Circuit (BK)
Courtesy of FORD MOTOR CO.

- Measure the resistance between evaporator temperature sensor C296-2, circuit (LG/BK), harness side and C296-1, circuit (BK), harness side.

- **Is the resistance greater than 10,000 ohms?**

YES : Go to D6.

NO : REPAIR circuit VH406 (VT/BN)/(LG/BK) for a short to circuit RH111 (GY/BU)/(BK).
CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

D6 CHECK CIRCUIT VH406 (VT/BN)/(LG/BK) FOR A SHORT TO GROUND

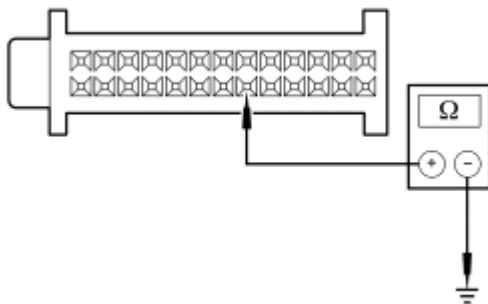


Fig. 21: Measuring Resistance Between EATC Module C228A-6, Circuit VH406 (VT/BN) & Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between HVAC module - DATC C228a-6, circuit VH406 (VT/BN), harness side and ground.

- **Is the resistance greater than 10,000 ohms?**

YES : Go to D7.

NO : REPAIR circuit VH406 (VT/BN)/(LG/BK) for a short to ground. CLEAR the DTCs.
REPEAT the self-test. TEST the system for normal operation.

D7 CHECK THE MODULE CONNECTION

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- Carry out the HVAC Module Cold Boot Process.
- Carry out the HVAC Module - Dual Automatic Temperature Control (DATC) On-Demand Self Test.
- Operate the system.
- **Does the concern return?**

YES : INSTALL a new HVAC module - DATC. REFER to **CLIMATE CONTROL** article. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation. If the concern returns, INSTALL a new heater core and evaporator core housing. REFER to **CLIMATE CONTROL** article. CLEAR the DTCs. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test E: Air Inlet Door is Inoperative

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Manual Climate Control System for schematic and connector information.

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Automatic Climate Control System for schematic and connector information.

Normal Operation

Under normal operation, the HVAC module - electronic manual temperature control (EMTC) or HVAC module - dual automatic temperature control (DATC) supplies ground to the air inlet door actuator motor through circuit CH208 (GN/OG)/(GN) to rotate the air inlet door actuator 180 degrees to the outside air position, or circuit CH207 (BU/GY)/(BN) to rotate the air inlet door actuator 180 degrees to the recirculated air position. Ignition voltage is supplied to the air inlet door actuator through circuit CBP37 (WH)/(GY).

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Air inlet door actuator
- HVAC module - DATC
- HVAC module - EMTC
- Stuck or damaged linkage or door

PINPOINT TEST E: AIR INLET DOOR IS INOPERATIVE

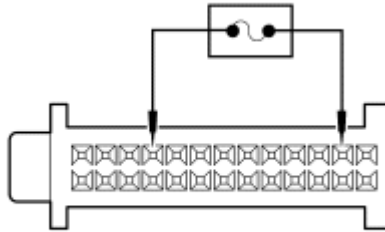
E1 CHECK THE DOOR ACTUATOR OPERATION

- Key in OFF position.
- Disconnect: HVAC Module - DATC C228a or HVAC Module - EMTC C2357a
- Remove the door actuator and disengage the actuator driveshaft from the actuator door.

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- Mark the door actuator driveshaft position.
- Key in ON position.

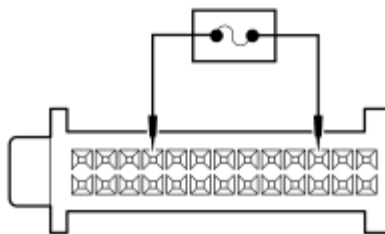


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Fig. 22: Connecting Fused Jumper Wire Between EATC C228A-23 & C2357A-15, Circuit GD115 (BK/GY) & CH207 (BU/GY)
Courtesy of FORD MOTOR CO.

NOTE: This step positions the actuator for the following steps.

- Connect a fused jumper wire between HVAC module - DATC C228a-23 or HVAC module - EMTC C2357a-23, circuit GD115 (BK/GY), harness side and HVAC module - DATC C228a-15 or HVAC module - EMTC C2357a-15, circuit CH207 (BU/GY), harness side.
- Remove the fused jumper wire.



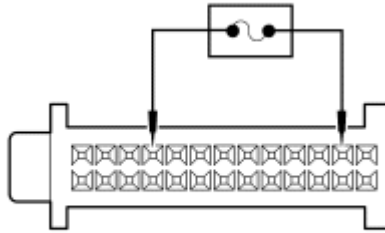
N0061477

Fig. 23: Connecting Fused Jumper Wire Between EATC C228A-23 & C2357A-16, Circuit GD115 (BK/GY) & CH208 (GN/OG)
Courtesy of FORD MOTOR CO.

- Connect a second fused jumper wire between: HVAC module - DATC C228a-23 or HVAC module - EMTC C2357a-23, circuit GD115 (BK/GY), harness side and HVAC module - DATC C228a-16 or HVAC module - EMTC C2357a-16, circuit CH208 (GN/OG), harness side.
- Remove the fused jumper wire.

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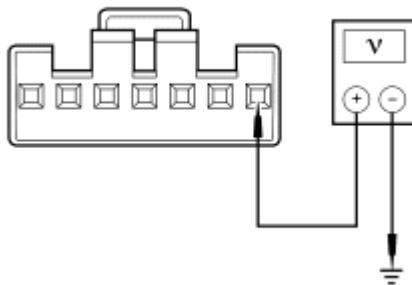
N0061476

Fig. 24: Connecting Fused Jumper Wire Between EATC C228A-23 & C2357A-15, Circuit GD115 (BK/GY) & CH207 (BU/GY)
Courtesy of FORD MOTOR CO.

- Connect a fused jumper wire between HVAC module - DATC C228a-23 or HVAC module - EMTC C2357a-23, circuit GD115 (BK/GY), harness side and HVAC module - DATC C228a-15 or HVAC module - EMTC C2357a-15, circuit CH207 (BU/GY), harness side.
- **Does the actuator motor rotate 180 degrees in the previous steps (360 degrees total)?**
YES : INSPECT for binding or broken linkage. If no condition is found, go to E7.
NO : Go to E2.

E2 CHECK THE ACTUATOR MOTOR FOR VOLTAGE

- Key in OFF position.
- Disconnect: Air Inlet Door Actuator C2363
- Key in ON position.



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Fig. 25: Measuring Voltage Between Ground & Air Inlet Door Actuator C2363-7, Circuit (GY)
Courtesy of FORD MOTOR CO.

- Measure the voltage between ground and air inlet door actuator C2363-7, circuit (GY), harness side.
- **Is any voltage present?**
YES : REPAIR circuit CBP37 (WH)/(GY) for an open. TEST the system for normal operation.
NO : Go to E3.

E3 CHECK THE ACTUATOR MOTOR DRIVE CIRCUITS FOR AN OPEN

- Key in OFF position.

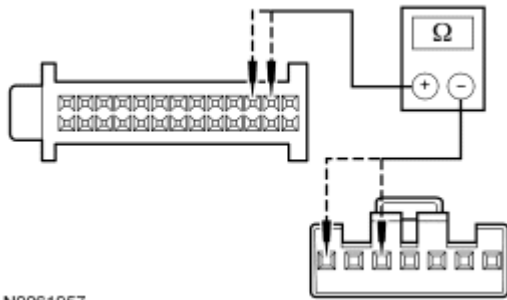


Fig. 26: Measuring Resistance Between EATC C228A-15 & C228A-16, Circuit CH207 (BU/GY) & CH208 (GN/OG)
 Courtesy of FORD MOTOR CO.

- Measure the resistance between:
 - HVAC module - DATC C228a-15 or HVAC module - EMTC C2357a-15, circuit CH207 (BU/GY), harness side and air inlet door actuator C2363-1, circuit (BN), harness side.
 - HVAC module - DATC C228a-16 or HVAC module - EMTC C2357a-16, circuit CH208 (GN/OG), harness side and air inlet door actuator C2363-3, circuit (GN), harness side.

- **Are the resistances less than 5 ohms?**

YES : Go to E4.

NO : REPAIR circuit CH207 (BU/GY)/(BN) or CH208 (GN/OG)/(GN) for an open. TEST the system for normal operation.

E4 CHECK THE ACTUATOR MOTOR DRIVE CIRCUITS FOR A SHORT TO GROUND

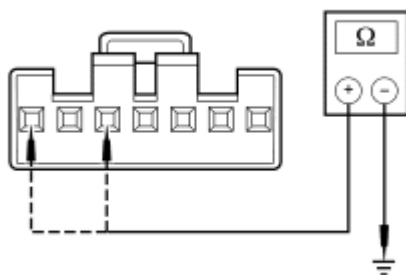


Fig. 27: Measuring Resistance Between Ground And Air Inlet Door Actuator C2363-1 & C2363-3, Circuit (BN)
 Courtesy of FORD MOTOR CO.

- Measure the resistance between ground and:
 - air inlet door actuator C2363-1, circuit (BN), harness side.
 - air inlet door actuator C2363-3, circuit (GN), harness side.

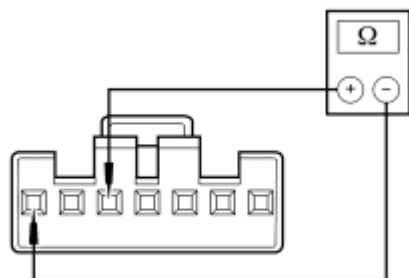
- **Are the resistances greater than 10,000 ohms?**

YES : Go to E5.

NO : REPAIR circuit CH207 (BU/GY)/(BN) or CH208 (GN/OG)/(GN) for a short to ground.

TEST the system for normal operation.

E5 CHECK THE ACTUATOR MOTOR DRIVE CIRCUITS FOR A SHORT TOGETHER



N0061481

Fig. 28: Measuring Resistance Between Air Inlet Door Actuator C2363-1 & C2363-3, Circuit (BN)

Courtesy of FORD MOTOR CO.

- Measure the resistance between air inlet door actuator C2363-1, circuit (BN), harness side and air inlet door actuator C2363-3, circuit (GN), harness side.

- **Is the resistance greater than 10,000 ohms?**

YES : Go to E6.

NO : REPAIR circuits CH207 (BU/GY)/(BN) and CH208 (GN/OG)/(GN) for a short together.

TEST the system for normal operation.

E6 CHECK THE ACTUATOR CONNECTION

NOTE: DATC systems only.

- Clear the DTCs.
- Disconnect the actuator connector.
- Check for:
 - corrosion.
 - pushed-out connectors.
 - incorrectly seated connector.
- Connect and correctly seat the actuator connector.
- Carry out the HVAC Module Cold Boot Process.

NOTE: DATC systems only.

- Carry out the HVAC Module - Dual Automatic Temperature Control (DATC) On-Demand Self

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Test.

- Operate the system.
- **Does the concern return?**

YES : INSTALL a new actuator. REFER to **CLIMATE CONTROL** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs.

E7 CHECK THE MODULE CONNECTION

- Carry out the HVAC Module Cold Boot Process.

NOTE: **DATC systems only.**

- Carry out the HVAC Module - Dual Automatic Temperature Control (DATC) On-Demand Self Test.
- Operate the system.
- **Does the concern return?**

YES : INSTALL a new HVAC module - DATC or HVAC module - EMTC. REFER to **CLIMATE CONTROL** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test F: Incorrect/Erratic Direction of Airflow From Outlet(s)

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Manual Climate Control System for schematic and connector information.

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Automatic Climate Control System for schematic and connector information.

Normal Operation

Under normal operation, to rotate the blend door actuator, the HVAC module - electronic manual temperature control (EMTC) or HVAC module - dual automatic temperature control (DATC) supplies voltage to the actuator motors through the door actuator clockwise circuits, and supplies ground through the door actuator counterclockwise circuits. To reverse the actuator rotation, the HVAC module - EMTC or HVAC module - DATC reverses the voltage and ground circuits.

The actuator feedback resistors are supplied a ground from the HVAC module - EMTC or HVAC module - DATC by the door actuator return circuits and a 5-volt reference voltage on the door actuator reference circuits. The HVAC module - EMTC or HVAC module - DATC reads the voltage on the blend door actuator feedback circuits to determine the door actuator position, by the position of the actuator feedback resistor wiper arm.

Door Actuator Clockwise Circuits

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- (HVAC module - EMTC) Blend - CH233 (VT/BN)/(OG/BK)
- (HVAC module - DATC) LH Blend - CH238 (YE/OG)/(RD/YE)
- (HVAC module - DATC) RH Blend - CH213 (BN/GN)/(VT)
- Defrost/Panel/Floor - CH228 (YE/GN)/(GY/BK)

Door Actuator Counterclockwise Circuits

- (HVAC module - EMTC) Blend - CH234 (YE/GN)/(VT)
- (HVAC module - DATC) LH Blend - CH239 (BU/WH)/(WH)
- (HVAC module - DATC) RH Blend - CH212 (BU/OG)/(OG/BK)
- Defrost/Panel/Floor - CH229 (WH/BU)/(BU)

Door Actuator Return Circuits

- (HVAC module - EMTC) Blend - RH111 (GY/BU)/(BK)
- (HVAC module - DATC) LH Blend - RH111 (GY/BU)/(BK)
- (HVAC module - DATC) RH Blend - RH111 (GY/BU)/(BK)
- Defrost/Panel/Floor - RH111 (GY/BU)/(BK)

Door Actuator Reference Circuits

- (HVAC module - EMTC) Blend - LH111 (BN/WH)/(OG)
- (HVAC module - DATC) LH Blend - LH111 (BN/WH)/(OG)
- (HVAC module - DATC) RH Blend - LH111 (BN/WH)/(OG)
- Defrost/Panel/Floor - LH111 (BN/WH)/(OG)

Door Actuator Feedback Circuits

- (HVAC module - EMTC) Blend - VH439 (GY/VT)/(RD)
- (HVAC module - DATC) LH Blend - VH440 (BU/BN)/(YE)
- (HVAC module - DATC) RH Blend - VH441 (WH/BN)/(RD)
- Defrost/Panel/Floor - VH436 (YE/VT)/(BU/RD)

DTC	Description	Fault Trigger Conditions
B226C	Left Blend Door Motor Stall	The module senses no change in actuator feedback voltage when the actuator motor has been energized.
B226D	Right Blend Door Motor Stall	The module senses no change in actuator feedback voltage when the actuator motor has been energized.
B226E	Mode Blend Door Motor Stall	The module senses no change in actuator feedback voltage when the

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		actuator motor has been energized.
B295A	Driver Blend Door Potentiometer Feedback Circuit Open	The module senses no voltage drop on the actuator feedback circuit, indicating an open or short to voltage.
B295B	Driver Blend Door Potentiometer Feedback Circuit short to ground	The module senses excessive voltage drop on the actuator feedback circuit, indicating a short directly to ground.
B295C	Passenger Blend Door Potentiometer Feedback Circuit Open	The module senses no voltage drop on the actuator feedback circuit, indicating an open or short to voltage.
B295D	Passenger Blend Door Potentiometer Feedback Circuit short to ground	The module senses excessive voltage drop on the actuator feedback circuit, indicating a short directly to ground.
B295E	Mode Door Potentiometer Feedback Circuit Open	The module senses no voltage drop on the actuator feedback circuit, indicating an open or short to voltage.
B295F	Mode Door Potentiometer Feedback Circuit short to ground	The module senses excessive voltage drop on the actuator feedback circuit, indicating a short directly to ground.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Defrost/panel/floor mode door actuator
- LH blend door actuator
- RH blend door actuator
- HVAC module - DATC
- HVAC module - EMTC
- Stuck or bound linkage or door

PINPOINT TEST F: INCORRECT/ERRATIC DIRECTION OF AIRFLOW FROM OUTLET(S)

F1 CHECK THE AIRFLOW

- Key in ON position.
- Check the air discharge temperature in full COOL and full WARM and airflow in the all modes.
- **Is the air discharge temperature and airflow correct in each setting?**

YES : The system is operating normally.

NO : For DATC systems:

DTC B226C, go to F2.

DTC B226D, go to F3.

DTC B226E, go to F4.

DTC B295A, B295C or B295E, go to F9.

DTC B295B, B295D or B295F, go to F11.

If the RECIRC button does not change the air inlet door position, go to **Pinpoint Test E** .

For EMTC systems:

If the airflow is incorrect, follow diagnostics for defrost/panel/floor door actuator. Go to F2.

If the temperature control is inoperative/does not operate correctly, follow diagnostics for the LH blend door actuator. Go to F2.

If the MAX A/C selection does not change the air inlet door position, go to **Pinpoint Test E** .

F2 CHECK THE LH BLEND DOOR ACTUATOR OPERATION

- Key in OFF position.
- Disconnect: HVAC Module - DATC C228a or HVAC Module - EMTC C2357a

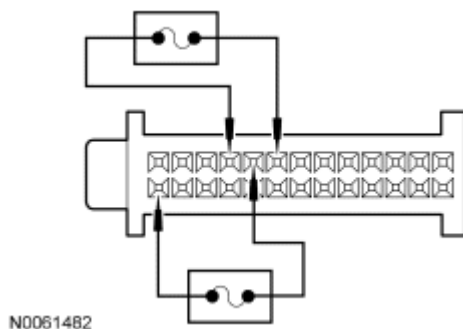


Fig. 29: Connect A Fused Jumper Wire For 5 Seconds Between EATC Module C228A-13, Circuit SBP15 (WH/RD)

Courtesy of FORD MOTOR CO.

NOTE: This step positions the actuator for the following steps.

- Connect a fused jumper wire for 5 seconds between:
 - HVAC module - DATC C228a-13 or HVAC module - EMTC C2357a-13, circuit SBP15

- (WH/RD), harness side and HVAC module - DATC C228a-22, circuit CH238 (YE/OG), harness side or HVAC module - EMTC C2357a-22, circuit CH233 (VT/BN), harness side.
- HVAC module - DATC C228a-23 or HVAC module - EMTC C2357a-23, circuit GD115 (BK/GY), harness side and HVAC module - DATC C228a-21, circuit CH239 (BU/WH), harness side or HVAC module - EMTC C2357a-21, circuit CH234 (YE/GN), harness side.

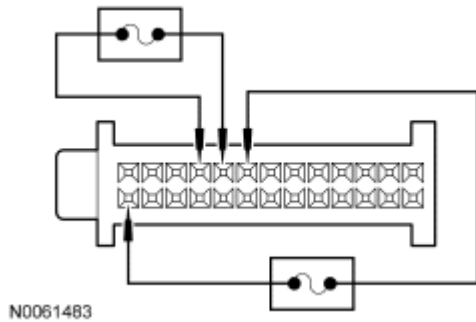


Fig. 30: Connect A Fused Jumper Wire For 5 Seconds Between EATC Module C228A-13, Circuit SBP15 (WH/RD)
 Courtesy of FORD MOTOR CO.

- Connect a fused jumper wire for 5 seconds between:
 - HVAC module - DATC C228a-13 or HVAC module - EMTC C2357a-13, circuit SBP15 (WH/RD), harness side and HVAC module - DATC C228a-21, circuit CH239 (BU/WH), harness side or HVAC module - EMTC C2357a-21, circuit CH234 (YE/GN), harness side.
 - HVAC module - DATC C228a-23 or HVAC module - EMTC C2357a-23, circuit GD115 (BK/GY), harness side and HVAC module - DATC C228a-22, circuit CH238 (YE/OG), harness side or HVAC module - EMTC C2357a-22, circuit CH233 (VT/BN), harness side.

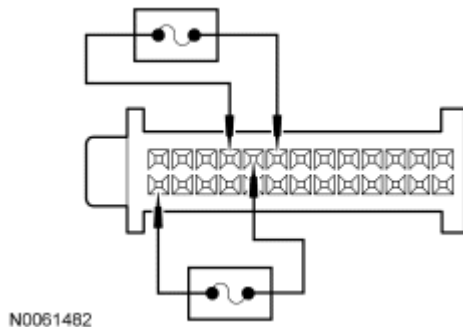


Fig. 31: Connect A Fused Jumper Wire For 5 Seconds Between EATC Module C228A-13, Circuit SBP15 (WH/RD)
 Courtesy of FORD MOTOR CO.

- Connect a fused jumper wire for 5 seconds between:
 - HVAC module - DATC C228a-13 or HVAC module - EMTC C2357a-13, circuit SBP15 (WH/RD), harness side and HVAC module - DATC C228a-22, circuit CH238 (YE/OG), harness side or HVAC module - EMTC C2357a-22, circuit CH233 (VT/BN), harness side.

- HVAC module - DATC C228a-23 or HVAC module - EMTC C2357a-23, circuit GD115 (BK/GY), harness side and HVAC module - DATC C228a-21, circuit CH239 (BU/WH), harness side or HVAC module - EMTC C2357a-21, circuit CH234 (YE/GN), harness side.
- **Does the actuator motor move at least 90 degrees in the clockwise and counterclockwise direction in the previous steps?**

YES : DATC systems, INSPECT for binding or broken linkage. If no condition is found, CARRY OUT the actuator component test in this service information. If the actuator tests OK, go to F14.

EMTC systems, INSPECT for binding or broken linkage. If no condition is found, CARRY OUT the actuator component test in this service information. If the actuator tests OK, go to F9.

NO : Go to F5.

F3 CHECK THE RH BLEND DOOR ACTUATOR OPERATION

- Key in OFF position.
- Disconnect: HVAC Module - DATC C228a
- Disconnect: HVAC Module - DATC C228b

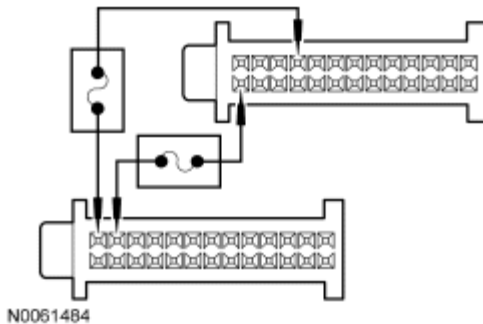


Fig. 32: Connect A Fused Jumper Wire For 5 Seconds Between EATC Module C228A-13, Circuit SBP15 (WH/RD)

Courtesy of FORD MOTOR CO.

NOTE: This step positions the actuator for the following steps.

- Connect a fused jumper wire for 5 seconds between:
 - HVAC module - DATC C228a-13, circuit SBP15 (WH/RD), harness side and HVAC module - DATC C228b-25, circuit CH213 (BN/GN), harness side.
 - HVAC module - DATC C228a-23, circuit GD115 (BK/GY), harness side and HVAC module - DATC C228b-26, circuit CH212 (BU/OG), harness side.

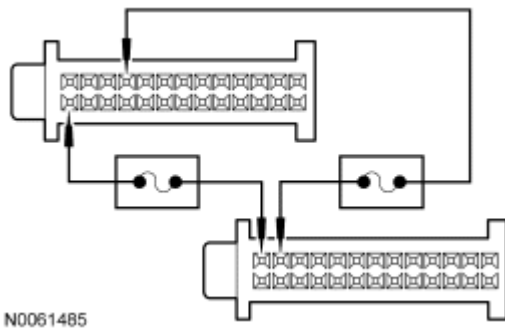


Fig. 33: Connect A Fused Jumper Wire For 5 Seconds Between EATC Module C228A-13, Circuit SBP15 (WH/RD)
Courtesy of FORD MOTOR CO.

- Connect a fused jumper wire for 5 seconds between:
 - HVAC module - DATC C228a-13, circuit SBP15 (WH/RD), harness side and HVAC module - DATC C228b-26, circuit CH212 (BU/OG), harness side.
 - HVAC module - DATC C228a-23, circuit GD115 (BK/GY), harness side and HVAC module - DATC C228b-25, circuit CH213 (BN/GN), harness side.

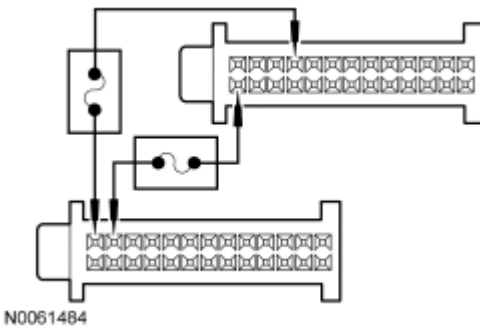


Fig. 34: Connect A Fused Jumper Wire For 5 Seconds Between EATC Module C228A-13, Circuit SBP15 (WH/RD)
Courtesy of FORD MOTOR CO.

- Connect a fused jumper wire for 5 seconds between:
 - HVAC module - DATC C228a-13, circuit SBP15 (WH/RD), harness side and HVAC module - DATC C228b-25, circuit CH213 (BN/GN), harness side.
 - HVAC module - DATC C228a-23, circuit GD115 (BK/GY), harness side and HVAC module - DATC C228b-26, circuit CH212 (BU/OG), harness side.

- **Does the actuator motor move at least 90 degrees in the clockwise and counterclockwise direction in the previous steps?**

YES : INSPECT for binding or broken linkage. If no condition is found, CARRY OUT the actuator component test in this service information. If the actuator tests OK, go to F14.

NO : Go to F5.

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- Key in OFF position.
- Disconnect: HVAC Module - DATC C228a or HVAC Module - EMTC C2357a

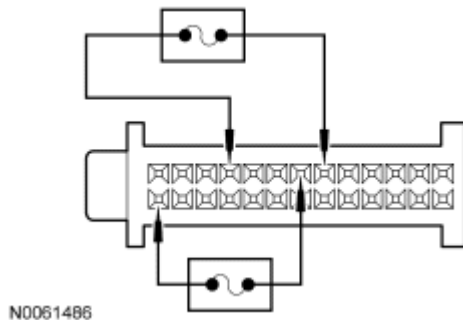


Fig. 35: Connect A Fused Jumper Wire For 5 Seconds Between EATC Module C228A-13, Circuit SBP15 (WH/RD)
Courtesy of FORD MOTOR CO.

NOTE: This step positions the actuator for the following steps.

- Connect a fused jumper wire for 5 seconds between:
 - HVAC module - DATC C228a-13 or HVAC module - EMTC C2357a-13, circuit SBP15 (WH/RD), harness side and HVAC module - DATC C228a-20 or HVAC module - EMTC C2357a-20, circuit CH228 (YE/GY), harness side.
 - HVAC module - DATC C228a-23 or HVAC module - EMTC C2357a-23, circuit GD115 (BK/GY), harness side and HVAC module - DATC C228a-19 or HVAC module - EMTC C2357a-19, circuit CH229 (WH/BU), harness side.

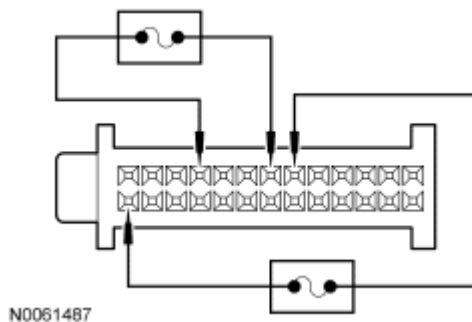


Fig. 36: Connect A Fused Jumper Wire For 5 Seconds Between EATC Module C228A-13, Circuit SBP15 (WH/RD)
Courtesy of FORD MOTOR CO.

- Connect a fused jumper wire for 5 seconds between:
 - HVAC module - DATC C228a-13 or HVAC module - EMTC C2357a-13, circuit SBP15 (WH/RD), harness side and HVAC module - DATC C228a-19 or HVAC module - EMTC C2357a-19, circuit CH229 (WH/BU), harness side.

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- HVAC module - DATC C228a-23 or HVAC module - EMTC C2357a-23, circuit GD115 (BK/GY), harness side and HVAC module - DATC C228a-20 or HVAC module - EMTC C2357a-20, circuit CH228 (YE/GY), harness side.

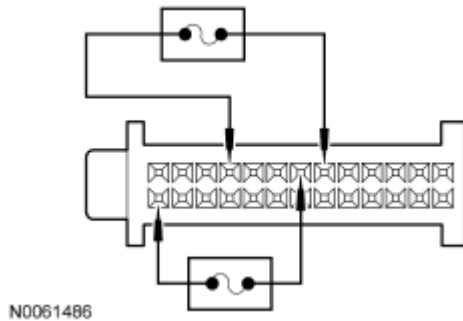


Fig. 37: Connect A Fused Jumper Wire For 5 Seconds Between EATC Module C228A-13, Circuit SBP15 (WH/RD)
Courtesy of FORD MOTOR CO.

- Connect a fused jumper wire for 5 seconds between:
 - HVAC module - DATC C228a-13 or HVAC module - EMTC C2357a-13, circuit SBP15 (WH/RD), harness side and HVAC module - DATC C228a-20 or HVAC module - EMTC C2357a-20, circuit CH228 (YE/GY), harness side.
 - HVAC module - DATC C228a-23 or HVAC module - EMTC C2357a-23, circuit GD115 (BK/GY), harness side and HVAC module - DATC C228a-19 or HVAC module - EMTC C2357a-19, circuit CH229 (WH/BU), harness side.

- **Does the actuator motor move at least 90 degrees in the clockwise and counterclockwise direction in previous steps?**

YES : DATC systems, INSPECT for binding or broken linkage. If no condition is found, CARRY OUT the actuator component test in this service information. If the actuator tests OK, go to F14.

EMTC systems, INSPECT for binding or broken linkage. If no condition is found, CARRY OUT the actuator component test in this service information. If the actuator tests OK, go to F9.

NO : Go to F5.

F5 CHECK THE ACTUATOR MOTOR DRIVE CIRCUITS FOR AN OPEN

- Disconnect: For DTC B226C or LH Blend Door Actuator, LH Blend Door Actuator C2091
- Disconnect: For DTC B226D, RH Blend Door Actuator C2092
- Disconnect: For DTC B226E or Defrost/Panel/Floor Door Actuator, LH Defrost/Panel/Floor Door Actuator C2278

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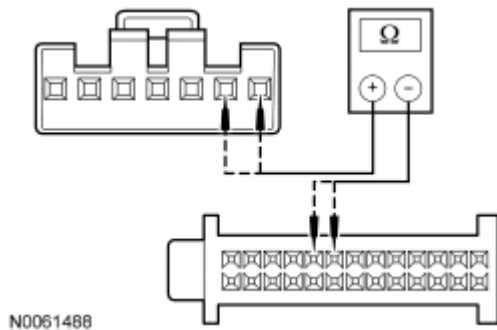


Fig. 38: Measuring Resistance Between EMTC Module C2357A-22 & C2357A-21, Circuit CH233 (VT/BN) & CH234 (YE/GN)
Courtesy of FORD MOTOR CO.

NOTE: For DTC B226C or blend door actuator.

- Measure the resistance between:
 - HVAC module - EMTC C2357a-22, circuit CH233 (VT/BN), harness side and blend door actuator C2091-7, circuit (OG/BK), harness side.
 - HVAC module - EMTC C2357a-21, circuit CH234 (YE/GN), harness side and blend door actuator C2091-6, circuit (VT), harness side.
 - HVAC module - DATC C228a-22, circuit CH238 (YE/OG), harness side and LH blend door actuator C2091-7, circuit (RD/YE), harness side.
 - HVAC module - DATC C228a-21, circuit CH239 (BU/WH), harness side and LH blend door actuator C2091-6, circuit (WH), harness side.

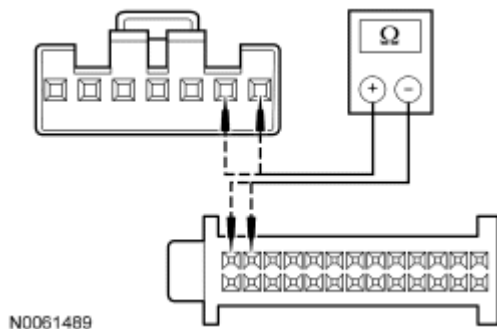


Fig. 39: Measuring Resistance Between EMTC Module C228B-25 & C228B-26, Circuit CH213 (BN/GN) & CH212 (BU/OG)
Courtesy of FORD MOTOR CO.

NOTE: For DTC B226D.

- Measure the resistance between:
 - HVAC module - DATC C228b-26, circuit CH212 (BU/OG), harness side and RH blend door actuator C2092-6, circuit (VT), harness side.

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- HVAC module - DATC C228b-25, circuit CH213 (BN/GN), harness side and RH blend door actuator C2092-7, circuit (OG/BK), harness side.

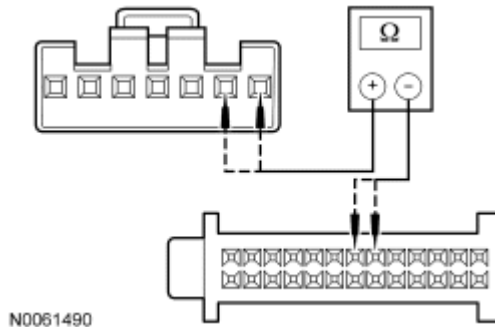


Fig. 40: Measuring Resistance Between EMTC Module C228a-20 & C228a-19, Circuit CH228 (YE/GY) & CH229 (WH/BU)
Courtesy of FORD MOTOR CO.

NOTE: For DTC B226E or defrost/panel/floor door actuator.

- Measure the resistance between:
 - HVAC module - DATC C228a-20 or HVAC module - EMTC C2357a-20, circuit CH228 (YE/GY), harness side and defrost/panel/floor door actuator C2278-7, circuit (GY/BK), harness side.
 - HVAC module - DATC C228a-19 or HVAC module - EMTC C2357a-19, circuit CH229 (WH/BU), harness side and defrost/panel/floor door actuator C2278-6, circuit (BU), harness side.
- **Are the resistances less than 5 ohms?**

YES : Go to F6.

NO : REPAIR the circuit(s) for an open. TEST the system for normal operation.

F6 CHECK THE ACTUATOR MOTOR DRIVE CIRCUITS FOR A SHORT TO GROUND

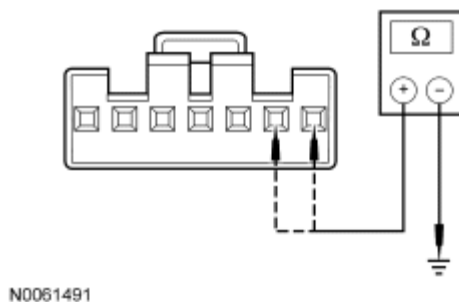


Fig. 41: Measuring Resistance Between Ground (EMTC) Blend Door Actuator C2091-7 & C2091-6, Circuit (OG/BK) & Circuit (VT)
Courtesy of FORD MOTOR CO.

- Measure the resistance between ground and:
 - (EMTC) Blend door actuator C2091-7, circuit (OG/BK), harness side.
 - (EMTC) Blend door actuator C2091-6, circuit (VT), harness side.
 - **DTC B226C** , LH blend door actuator C2091-7, circuit (RD/YE), harness side.
 - **DTC B226C** , LH blend door actuator C2091-6, circuit (WH), harness side.
 - **DTC B226D** , RH blend door actuator C2092-6, circuit (VT), harness side.
 - **DTC B226D** , RH blend door actuator C2092-7, circuit (OG/BK), harness side.
 - **DTC B226E** or defrost/panel/floor door actuator, defrost/panel/floor door actuator C2278-7, circuit (GY/BK), harness side.
 - **DTC B226E** or defrost/panel/floor door actuator, defrost/panel/floor door actuator C2278-6, circuit (BU), harness side.

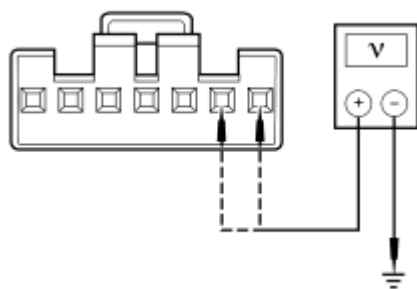
- **Are the resistances greater than 10,000 ohms?**

YES : Go to F7.

NO : REPAIR the circuit(s) for a short to ground. TEST the system for normal operation.

F7 CHECK THE ACTUATOR MOTOR DRIVE CIRCUITS FOR A SHORT TO VOLTAGE

- Key in ON position.



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Fig. 42: Measuring Voltage Between Ground (EMTC) Blend Door Actuator C2091-7 & C2091-6, Circuit (OG/BK) & Circuit (VT)
 Courtesy of FORD MOTOR CO.

- Measure the voltage between ground and:
 - (EMTC) Blend door actuator C2091-7, circuit (OG/BK), harness side.
 - (EMTC) Blend door actuator C2091-6, circuit (VT), harness side.
 - **DTC B226C** , LH blend door actuator C2091-7, circuit (RD/YE), harness side.
 - **DTC B226C** , LH blend door actuator C2091-6, circuit (WH), harness side.
 - **DTC B226D** , RH blend door actuator C2092-6, circuit (VT), harness side.
 - **DTC B226D** , RH blend door actuator C2092-7, circuit (OG/BK), harness side.
 - **DTC B226E** or defrost/panel/floor door actuator, defrost/panel/floor door actuator C2278-7, circuit (GY/BK), harness side.
 - **DTC B226E** or defrost/panel/floor door actuator, defrost/panel/floor door actuator C2278-6,

circuit (BU), harness side.

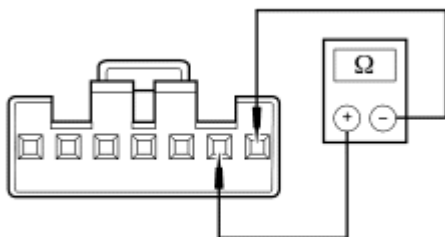
- **Is any voltage present?**

YES : REPAIR the circuit(s) for a short to power. TEST the system for normal operation.

NO : Go to F8.

F8 CHECK THE ACTUATOR MOTOR DRIVE CIRCUITS FOR A SHORT TOGETHER

- Key in OFF position.



N0061493

Fig. 43: Measuring Resistance Between (EMTC) LH Blend Door Actuator C2091-7, Circuit (OG/BK) & Circuit (RD/YE)

Courtesy of FORD MOTOR CO.

- Measure the resistance between:
 - (EMTC) For LH blend door actuator, LH blend door actuator C2091-7, circuit/ (OG/BK), harness side and LH blend door actuator C2091-6, circuit (VT), harness side.
 - **DTC B226C** or LH blend door actuator, LH blend door actuator C2091-7, circuit (RD/YE), harness side and LH blend door actuator C2091-6, circuit (WH), harness side.
 - **DTC B226D** , RH blend door actuator C2092-6, circuit (VT), harness side and RH blend door actuator C2092-7, circuit (OG/BK), harness side.
 - **DTC B226E** or defrost/panel/floor door actuator, defrost/panel/floor door actuator C2278-7, circuit (GY/BK), harness side and defrost/panel/floor door actuator C2278-6, circuit (BU), harness side.

- **Is the resistance greater than 10,000 ohms?**

YES : CARRY OUT the actuator component test in this service information. If the actuator tests OK, go to F13.

NO : REPAIR circuit(s) for a short together. TEST the system for normal operation.

F9 CHECK THE ACTUATOR FEEDBACK CIRCUITS FOR AN OPEN

- Key in OFF position.
- Disconnect: HVAC Module - DATC C228a or HVAC Module - EMTC C2357a
- Disconnect: DATC Systems Only, HVAC Module - DATC C228b
- Disconnect: LH Blend Door Actuator C2091
- Disconnect: DATC Systems Only, RH Blend Door Actuator C2092
- Disconnect: Defrost/Panel/Floor Door Actuator C2278

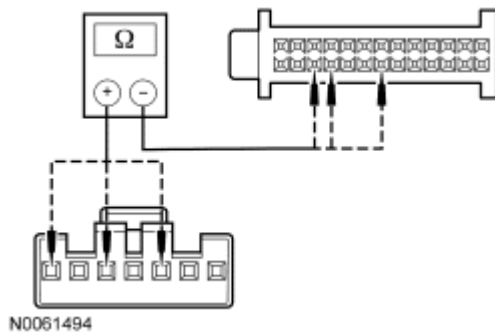


Fig. 44: Measuring Resistance Between EMTC Module C2357A-11 & C2357A-10, Circuit LH111 (BN/WH) & RH111 (GY/BU)
Courtesy of FORD MOTOR CO.

NOTE: For DTC B295A (DATC) or blend door actuator (EMTC).

- Measure the resistance between:
 - HVAC module - EMTC C2357a-11, circuit LH111 (BN/WH), harness side and blend door actuator C2091-3, circuit (OG), harness side.
 - HVAC module - EMTC C2357a-10, circuit RH111 (GY/BU), harness side and blend door actuator C2091-1, circuit (BK), harness side.
 - HVAC module - EMTC C2357a-7, circuit VH439 (GY/VT), harness side and blend door actuator C2091-5, circuit (RD), harness side.
 - HVAC module - DATC C228a-11, circuit LH111 (BN/WH), harness side and LH blend door actuator C2091-3, circuit (OG), harness side.
 - HVAC module - DATC C228a-10, circuit RH111 (GY/BU), harness side and LH blend door actuator C2091-1, circuit (BK), harness side.
 - HVAC module - DATC C228a-7, circuit VH440 (BU/BN), harness side and LH blend door actuator C2091-5, circuit (YE), harness side.

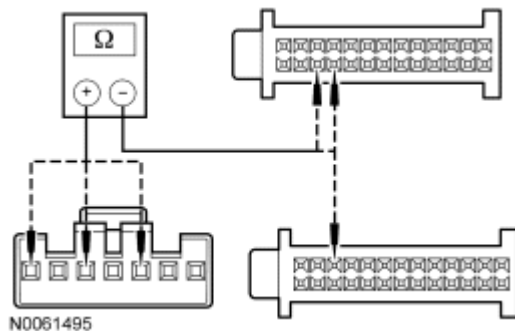


Fig. 45: Measuring Resistance Between RH Blend Door Actuator C2092-3, C2092-1 & C2092-5
Courtesy of FORD MOTOR CO.

NOTE: For DTC B295C.

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- Measure the resistance between:
 - HVAC module - DATC C228a-11 circuit LH111 (BN/WH), harness side and RH blend door actuator C2092-3, circuit (OG), harness side.
 - HVAC module - DATC C228a-10 circuit RH111 (GY/BU), harness side and RH blend door actuator C2092-1, circuit (BK), harness side.
 - HVAC module - DATC C228b-24 circuit VH441 (WH/BN), harness side and RH blend door actuator C2092-5, circuit (RD), harness side.

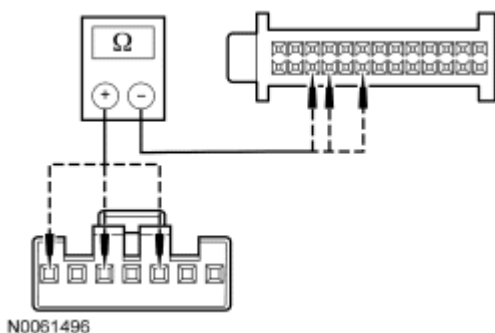


Fig. 46: Measuring Resistance Between Defrost/Panel/Floor Door Actuator C2278-3, C2278-1 & C2278-5

Courtesy of FORD MOTOR CO.

NOTE: For DTC B295E or defrost/panel/floor door actuator.

- Measure the resistance between:
 - HVAC module - DATC C228a-11 or HVAC module - EMTC C2357a-11, circuit LH111 (BN/WH), harness side and defrost/panel/floor door actuator C2278-3, circuit (OG), harness side.
 - HVAC module - DATC C228a-10 or HVAC module - EMTC C2357a-10, circuit RH111 (GY/BU), harness side and defrost/panel/floor door actuator C2278-1, circuit (BK), harness side.
 - HVAC module - DATC C228a-8 or HVAC module - EMTC C2357a-8, circuit VH436 (YE/VT), harness side and defrost/panel/floor door actuator C2278-5, circuit (BU/RD), harness side.

- **Are the resistances less than 5 ohms?**

YES : Go to F10.

NO : REPAIR circuit(s) for an open. TEST the system for normal operation.

F10 CHECK THE ACTUATOR FEEDBACK CIRCUITS FOR A SHORT TO VOLTAGE

- Key in ON position.

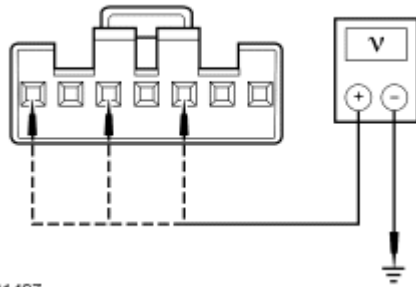


Fig. 47: Measuring Voltage Between (EMTC) Blend Door Actuator C2091-3, C2091-1 & C2091-5

Courtesy of FORD MOTOR CO.

- Measure the voltage between ground and:
 - (EMTC) Blend door actuator C2091-3, circuit (OG), harness side.
 - (EMTC) Blend door actuator C2091-1, circuit (BK), harness side.
 - (EMTC) Blend door actuator C2091-5, circuit (RD), harness side.
 - (EMTC) Defrost/panel/floor door actuator C2092-5, circuit (BU/RD), harness side.
 - **DTC B295A** , LH blend door actuator C2091-5, circuit (YE), harness side.
 - **DTC B295C** , RH blend door actuator C2092-5, circuit (RD), harness side.
 - **DTC B295E** defrost/panel/floor door actuator C2278-5, circuit (BU/RD), harness side.
- **Is any voltage present?**

YES : REPAIR circuit(s) for a short to voltage. TEST the system for normal operation.

NO : DATC systems, go to F12.

EMTC systems, go to F11.

F11 CHECK THE ACTUATOR FEEDBACK CIRCUITS FOR A SHORT TO GROUND

- Key in OFF position.

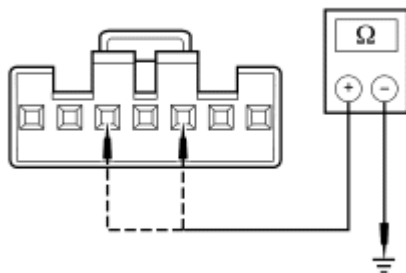


Fig. 48: Measuring Resistance Between (EMTC) Blend Door Actuator C2091-3, C2091-1 & C2091-5

Courtesy of FORD MOTOR CO.

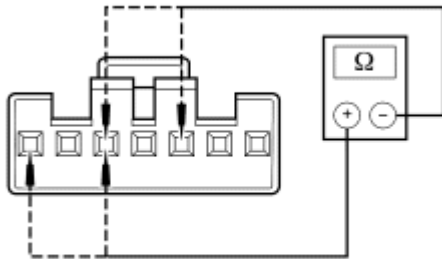
- Measure the resistance between ground and:
 - (EMTC) Blend door actuator C2091-3, circuit (OG), harness side.
 - (EMTC) Blend door actuator C2091-5, circuit (RD), harness side.
 - (EMTC) Defrost/panel/floor door actuator C2092-5, circuit (BU/RD), harness side.
 - **DTC B295A** , LH blend door actuator C2091-5, circuit (YE), harness side.
 - **DTC B295C** , RH blend door actuator C2092-5, circuit (RD), harness side.
 - **DTC B295E** defrost/panel/floor door actuator C2278-5, circuit (BU/RD), harness side.
- **Are the resistances greater than 10,000 ohms?**

YES : Go to F12.

NO : REPAIR circuit(s) for a short to ground. TEST the system for normal operation.

F12 CHECK THE ACTUATOR FEEDBACK CIRCUITS FOR A SHORT TOGETHER

- Key in OFF position.



N0061499

Fig. 49: Measuring Resistance Between (EMTC) Blend Door Actuator C2091-3 & C2091-5, Blend Door Actuator C2091-1
Courtesy of FORD MOTOR CO.

- Measure the resistance between:
 - (EMTC) Blend door actuator C2091-3, circuit (OG), harness side and blend door actuator C2091-1, circuit (BK), harness side.
 - (EMTC) Blend door actuator C2091-5, circuit (RD), harness side and blend door actuator C2091-1, circuit (BK), harness side.
 - (EMTC) Blend door actuator C2091-5, circuit (RD), harness side and blend door actuator C2091-3, circuit (OG), harness side.
 - (EMTC) Defrost/panel/floor door actuator C2278-5, circuit (BU/RD), harness side and defrost/panel/floor door actuator C2278-1, circuit (BK), harness side.
 - (EMTC) Defrost/panel/floor door actuator C2278-5, circuit (BU/RD), harness side and defrost/panel/floor door actuator C2278-3, circuit (OG), harness side.
 - **DTC B295A** , LH blend door actuator C2091-5, circuit (YE), harness side and LH blend door actuator C2091-1, circuit (BK), harness side.
 - **DTC B295A** , LH blend door actuator C2091-5, circuit/ (YE), harness side and LH blend door actuator C2091-3, circuit (OG), harness side.

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- **DTC B295C** , RH blend door actuator C2092-5, circuit (RD), harness side and RH blend door actuator C2092-1, circuit (OG), harness side.
 - **DTC B295C** , RH blend door actuator C2092-5, circuit (RD), harness side and RH blend door actuator C2092-3, circuit (BK), harness side.
 - **DTC B295E** , defrost/panel/floor door actuator C2278-5, circuit (BU/RD), harness side and defrost/panel/floor door actuator C2278-1, circuit (BK), harness side.
 - **DTC B295E** or defrost/panel/floor door actuator, defrost/panel/floor door actuator C2278-5, circuit (BU/RD), harness side and defrost/panel/floor door actuator C2278-3, circuit (OG), harness side.
- **Are the resistances greater than 10,000 ohms?**
YES : Go to F14.
NO : REPAIR circuit(s) for a short together. TEST the system for normal operation.

F13 CHECK THE ACTUATOR CONNECTION

NOTE: **DATC systems only.**

- Clear the DTCs.
- Disconnect the actuator connector.
- Check for:
 - corrosion.
 - pushed-out connectors.
 - incorrectly seated connector.
- Connect and correctly seat the actuator connector.
- Carry out the HVAC Module Cold Boot Process.

NOTE: **DATC systems only.**

- Carry out the HVAC Module - Dual Automatic Temperature Control (DATC) On-Demand Self Test.
- Operate the system.
- **Does the concern return?**
YES : INSTALL a new actuator. REFER to **CLIMATE CONTROL** article. TEST the system for normal operation.
NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs.

F14 CHECK THE MODULE CONNECTION

- Carry out the HVAC Module Cold Boot Process.

NOTE: **DATC systems only.**

- Carry out the HVAC Module - Dual Automatic Temperature Control (DATC) On-Demand Self Test.
- Operate the system.
- **Does the concern return?**

YES : INSTALL a new HVAC module - DATC or HVAC module - EMTC. REFER to **CLIMATE CONTROL** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test G: DTC: B1318 - Battery Voltage Low

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Automatic Climate Control System for schematic and connector information.

Normal Operation

Under normal operation, the HVAC module - dual automatic temperature control (DATC) receives a ground through circuit GD115 (BK/GY). The HVAC module - DATC is supplied constant battery voltage through circuit SBP15 (WH/RD) and ignition switched voltage through circuit CBP37 (WH).

This pinpoint test is intended to diagnose the following:

- Fuse(s)
- Wiring, terminals or connectors
- HVAC module - DATC

PINPOINT TEST G: DTC: B1318 - BATTERY VOLTAGE LOW

G1 CHECK CIRCUIT GD115 (BK/GY) FOR AN OPEN

- Key in OFF position.
- Disconnect: HVAC Module - DATC C228a

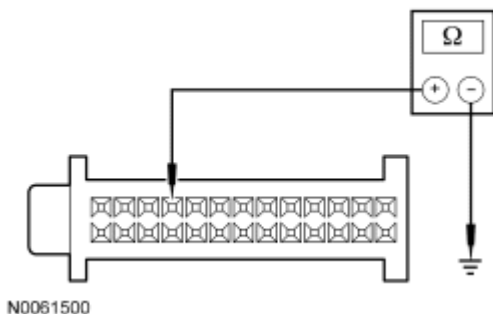


Fig. 50: Measuring Resistance Between EATC Module C228A-23, Circuit GD115 (BK/GY) & Ground

Courtesy of FORD MOTOR CO.

- Measure the resistance between HVAC module - DATC C228a-23, circuit GD115 (BK/GY), harness side and ground.

- **Is the resistance less than 5 ohms?**

YES : Go to G2.

NO : REPAIR circuit GD115 (BK/GY). CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

G2 CHECK CIRCUITS SBP15 (WH/RD) AND CBP37 (WH) FOR AN OPEN

- Key in ON position.

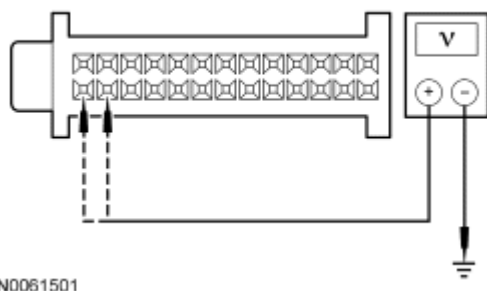


Fig. 51: Measuring Voltage Between EATC Module C228A-13 & C228A-12, Circuit SBP15 (WH/RD) & CBP37 (WH)

Courtesy of FORD MOTOR CO.

- Measure the voltage between HVAC module - DATC C228a-13, circuit SBP15 (WH/RD), harness side, HVAC module - DATC C228a-12, circuit CBP37 (WH), harness side and ground.

- **Are the voltages greater than 10 volts?**

YES : Go to **MODULE COMMUNICATIONS NETWORK** article to diagnose the network concern.

NO : VERIFY smart junction box (SJB) fuses 15 (10A) and 37 (10A) are OK. If OK, REPAIR circuit SBP15 (WH/RD) or CBP37 (WH) for an open. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test H: The DATC System is Inoperative, Intermittent or Incorrect Operation

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Automatic Climate Control System for schematic and connector information.

Normal Operation

Under normal operation, the HVAC module - dual automatic temperature control (DATC) receives a ground through circuit GD115 (BK/GY). The HVAC module - DATC is supplied constant battery voltage through circuit SBP15 (WH/RD) and ignition switched voltage through circuit CBP37 (WH).

NOTE: For a complete explanation of the DATC system functions, refer to Description and Operation in this service information.

This pinpoint test is intended to diagnose the following:

- Fuse(s)
- Wiring, terminals or connectors
- HVAC module - DATC

PINPOINT TEST H: THE DATC SYSTEM IS INOPERATIVE, INTERMITTENT OR INCORRECT OPERATION

H1 VERIFY AUTOMATIC OPERATION

- Key in ON position.
- With the engine running, press the AUTO button.
- **Does the indicator above the AUTO button illuminate and the selected temperature for both the driver and passenger appear in the display window?**

YES : Go to H2.

NO : Go to H11.

H2 CARRY OUT THE HVAC MODULE - DATC SELF-TEST

- Key in ON position.
- Carry out the HVAC Module - Dual Automatic Temperature Control Module (DATC) Self-Test in this service information. Record the DTCs displayed, if any.
- **Were any DTCs displayed as a result of the self-test?**

YES : REFER to the **HVAC MODULE - DUAL AUTOMATIC TEMPERATURE CONTROL DTC CHART**. CARRY OUT the necessary diagnosis and REPAIR as required.

NO : Go to H3.

H3 CHECK THE VACUUM FLUORESCENT DISPLAY

- Exit the self-test. Refer to the HVAC Module - Dual Automatic Temperature Control Module (DATC) Self-Test in this service information. Observe the button indicators and function symbols displayed on the vacuum fluorescent display while changing the temperature, blower speed and functions.
- **Are the display and indicators correct and complete without any missing elements?**

YES : Go to H4.

NO : Go to H14.

H4 CHECK THE BLOWER MANUAL OVERRIDE OPERATION

- Press the blower motor speed override control fully up and fully down.
- **Does the blower motor speed increase from low speed to high and high speed to low?**

YES : Go to H5.

NO : If the blower motor is inoperative, go to **Pinpoint Test N**.

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If the blower motor does not operate correctly, go to **Pinpoint Test O** .

H5 VERIFY THE DEFROST OVERRIDE OPERATION

- Press the override button for defrost operation.
- **Is outside air being discharged from the windshield defroster nozzle and the side window demisters?**
YES : Go to H6.
NO : Go to **Pinpoint Test F** .

H6 VERIFY THE FLOOR OVERRIDE OPERATION

- Press the mode override button until floor mode is indicated.
- **Is outside air being discharged from the floor duct?**
YES : Go to H7.
NO : Go to **Pinpoint Test E** .

H7 VERIFY THE PANEL OVERRIDE OPERATION

- Press the mode override button until panel mode is indicated.
- **Is outside air being discharged from the instrument panel registers?**
YES : Go to H8.
NO : Go to **Pinpoint Test F** .

H8 VERIFY THAT THE A/C CLUTCH DOES NOT ENGAGE IN THE PANEL MODE

- In panel operation and with the A/C OFF, listen for the compressor clutch to engage.
- **Does the A/C clutch engage with the PANEL override button pressed?**
YES : Go to **Pinpoint Test K** .
NO : Go to H9.

H9 VERIFY A/C CLUTCH ENGAGEMENT IN THE A/C MODE

- Make sure the ambient air temperature is above 6°C (43°F).
- Press the override button for A/C operation.
- **Does the A/C clutch engage when the A/C override button is pressed?**
YES : Go to H10.
NO : Go to **Pinpoint Test J** .

H10 VERIFY THE RECIRC OVERRIDE OPERATION

- While in PANEL mode, press the RECIRC override button.
- **Is recirculated air being discharged from the instrument panel registers?**
YES : The test is complete. The system is functioning normally.
NO : Go to **Pinpoint Test E** .

H11 CHECK THE HVAC MODULE - DATC MODULE FUNCTIONS

- Press each function button and observe the display.
- **Does the HVAC module - DATC carry out and display any functions?**
YES : Go to H14.

NO : Go to H12.

H12 CHECK CIRCUIT GD115 (BK/GY) FOR AN OPEN

- Key in OFF position.
- Disconnect: HVAC Module - DATC C228a

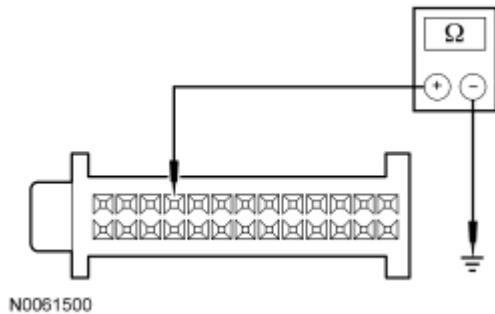


Fig. 52: Measuring Resistance Between EATC Module C228A-23, Circuit GD115 (BK/GY) & Ground

Courtesy of FORD MOTOR CO.

- Measure the resistance between HVAC module - DATC C228a-23, circuit GD115 (BK/GY), harness side and ground.
- **Is the resistance less than 5 ohms?**
YES : Go to H13.

NO : REPAIR circuit GD115 (BK/GY). TEST the system for normal operation.

H13 CHECK CIRCUITS SBP15 (WH/RD) AND CBP37 (WH) FOR AN OPEN

- Key in ON position.

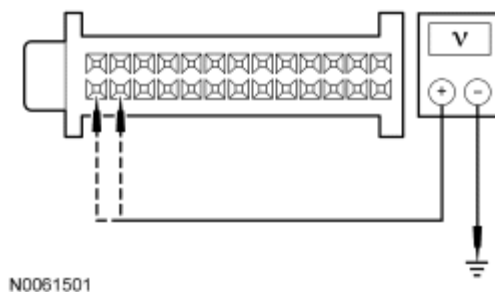


Fig. 53: Measuring Voltage Between EATC Module C228A-13 & C228A-12, Circuit SBP15 (WH/RD) & CBP37 (WH)

Courtesy of FORD MOTOR CO.

- Measure the voltage between HVAC module - DATC C228a-13, circuit SBP15 (WH/RD), harness side, HVAC module - DATC C228a-12, circuit CBP37 (WH), harness side and ground.
- **Are the voltages greater than 10 volts?**

YES : Go to H14.

NO : VERIFY smart junction box (SJB) fuses 15 (10A) and 37 (10A) are OK. If OK, REPAIR circuit SBP15 (WH/RD) or CBP37 (WH) for an open. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

H14 CHECK THE MODULE CONNECTION

- Carry out the HVAC Module Cold Boot Process.
- Carry out the HVAC Module - Dual Automatic Temperature Control (DATC) On-Demand Self Test.
- Operate the system.
- **Does the concern return?**

YES : INSTALL a new HVAC module - DATC. REFER to **CLIMATE CONTROL** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test I: Insufficient, Erratic or No Heat

Normal Operation

Under normal operation, warm coolant flows from the engine through the heater core and back to the engine.

This pinpoint test is intended to diagnose the following:

- Plugged heater core
- Coolant level
- Temperature blend door

PINPOINT TEST I: INSUFFICIENT, ERRATIC OR NO HEAT

I1 CHECK FOR CORRECT ENGINE COOLANT LEVEL

- Check the engine coolant level when hot and cold.
- **Is the engine coolant at the correct level (hot/cold) as indicated on the engine coolant recovery reservoir?**

YES : Go to I3.

NO : Go to I2.

I2 CHECK THE ENGINE COOLING SYSTEM FOR LEAKS

- Pressure test the cooling system for leaks. Refer to **ENGINE COOLING** article.
- **Does the engine cooling system leak?**

YES : REPAIR the engine coolant leak. TEST the system for normal operation.

NO : Go to I3.

I3 CHECK FOR COOLANT FLOW TO THE HEATER CORE

- Run the engine until it reaches normal operation temperature. Select the FLOOR position on the control assembly. Set the temperature control to full warm and the blower to the lowest setting.

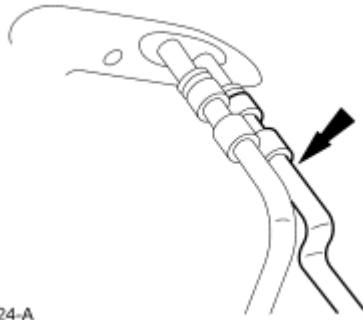


Fig. 54: Checking Coolant Flow To Heater Core
Courtesy of FORD MOTOR CO.

- Using a suitable thermo-couple temperature measuring device, check the heater core inlet hose to see if it is hot.
- **Is the heater core inlet hose hot?**
YES : Go to I4.
NO : REFER to **ENGINE COOLING** article to check cooling system function.

I4 CHECK FOR A PLUGGED OR RESTRICTED HEATER CORE

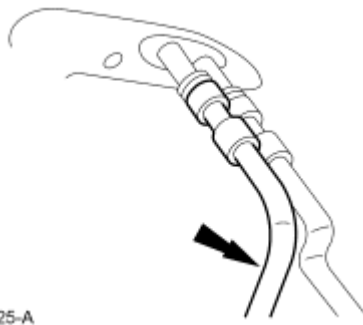


Fig. 55: Check Plugged Or Restricted Heater Core
Courtesy of FORD MOTOR CO.

- Using a suitable thermo-couple temperature measuring device, measure the heater core outlet hose temperature.
- **Is the heater core outlet hose temperature similar to the inlet hose temperature (within approximately 6-17°C [10-30°F])?**
YES : Go to **Pinpoint Test F** and diagnose for a blend door actuator.
NO : INSTALL a new heater core. TEST the system for normal operation.

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Refer to appropriate SYSTEM WIRING DIAGRAMS article, Manual Climate Control System for schematic and connector information.

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Automatic Climate Control System for schematic and connector information.

Normal Operation

Under normal operation, when A/C is requested, in the dual automatic temperature control (DATC) system, a message is sent over the medium-speed controller area network (MS-CAN) bus to the instrument cluster, then from the instrument cluster through the high-speed controller area network (HS-CAN) bus to the PCM; or in electronic manual temperature control (EMTC) systems, voltage is sent to the instrument cluster through circuit CH434 (GY/YE), then a message is sent from the instrument cluster through the HS-CAN bus to the PCM.

Ignition voltage is provided to the A/C clutch relay switch. When the PCM energizes the relay, ignition voltage is supplied to the A/C clutch through circuit CH401 (VT/WH). Ground is supplied for the A/C clutch through circuit GD120 (BK/GN).

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- PCM
- HVAC module - DATC
- HVAC module - EMTC
- A/C evaporator discharge temperature sensor
- A/C pressure transducer
- A/C compressor
- A/C control relay
- Battery junction box (BJB)

PINPOINT TEST J: THE AIR CONDITIONING (A/C) IS INOPERATIVE/DOES NOT OPERATE CORRECTLY

CAUTION: It is important to install relays in their correct position in the battery junction box (BJB). Installing a relay incorrectly may cause wiring shorts or damage to modules. While carrying out diagnostics on BJB relays, have only one BJB relay removed at a time. Failure to follow these instructions may result in damage to the vehicle circuitry or to control modules.

NOTE: Some PCM DTCs may inhibit A/C operation. If any PCM DTCs are retrieved, diagnose those first. Refer to [Introduction - Gasoline Engines](#) article.

NOTE: When disconnecting and reconnecting the pressure transducer electrical

connector, make sure that the connector-locking device is in place and that the locking device and connector are correctly and fully seated.

NOTE: Before carrying out the following test, check that the A/C system pressure is above 290 kPa (42 psi). If the pressure is below 290 kPa (42 psi), refer to **Fluorescent Dye Leak Detection**.

J1 CHECK THE A/C PRESSURE SENSOR (ACP_PSI) PCM PID

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: ACP_PSI PCM PID
- With the manifold gauge set connected, compare the pressure readings of the manifold gauge set and the ACP_PSI PID.
- **Are the pressure values of the manifold gauge set and the ACP_PSI PID similar?**

YES : Go to J2.

NO : INSTALL a new A/C pressure transducer. TEST the system for normal operation.

J2 CHECK THE POST EVAPORATOR TEMPERATURE SENSOR INPUT (PEVAP_TMP) HVAC PID

- Allow the vehicle exterior and interior to stabilize to an ambient temperature above 16°C (60°F).
- Enter the following diagnostic mode on the diagnostic tool: PEVAP_TMP HVAC PID
- **Does the PEVAP_TMP HVAC PID read similar to the ambient temperature?**

YES : DATC systems, go to J3.

EMTC systems, go to J4.

NO : CARRY OUT the A/C evaporator discharge temperature sensor component test in this service information. If the sensor tests OK, go to J13. If the sensor does not test OK, INSTALL a new heater core and evaporator core housing. REFER to **CLIMATE CONTROL** article. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

J3 CHECK THE AIR CONDITIONING SWITCH STATUS (CC_AC) HVAC PID WITH THE A/C ON

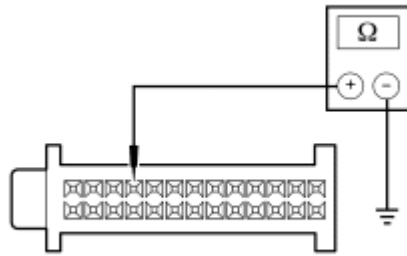
- Enter the following diagnostic mode on the diagnostic tool: CC_AC HVAC PID
- Press the PANEL and A/C manual override buttons on the HVAC module - DATC.
- **Does the CC_AC HVAC PID read ON?**

YES : Go to J5.

NO : Go to J13.

J4 CHECK THE HVAC MODULE - EMTC A/C SIGNAL WITH THE A/C ON

- Key in OFF position.
- Disconnect: Instrument Cluster C220
- Key in ON position.
- Select PANEL mode and press the A/C button on the HVAC module - EMTC.



N0041895

Fig. 56: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between instrument cluster C220-23, circuit CH434 (GY/YE), harness side and ground.
- **Is the resistance less than 30 ohms?**
YES : Go to J5.
NO : Go to J7.

J5 CHECK THE AIR CONDITIONING COMPRESSOR CYCLING SWITCH (ACCS) PCM PID WITH THE A/C ON

- Key in OFF position.

NOTE: EMTC systems only.

- Connect: Instrument Cluster C220

NOTE: EMTC systems only.

- Key in ON position.

NOTE: EMTC systems only.

- Enter the following diagnostic mode on the diagnostic tool: ACCS PCM PID
- With the engine running, press the PANEL and A/C manual override buttons on the HVAC module - DATC or select MAX A/C on the HVAC module - EMTC.
- **Does the ACCS PCM PID read ON?**
YES : Go to J6.
NO : Go to J14.

J6 CHECK THE AIR CONDITIONING CLUTCH (WAC/ACCR) PCM PID WITH THE A/C ON

- Enter the following diagnostic mode on the diagnostic tool: WAC/ACCR PCM PID
- **Does the WAC/ACCR PCM PID read ON?**
YES : Go to J9.

NO : Go to J14.

J7 CHECK CIRCUIT CH434 (GY/YE) FOR AN OPEN

- Key in OFF position.
- Disconnect: HVAC Module - EMTC C2357a

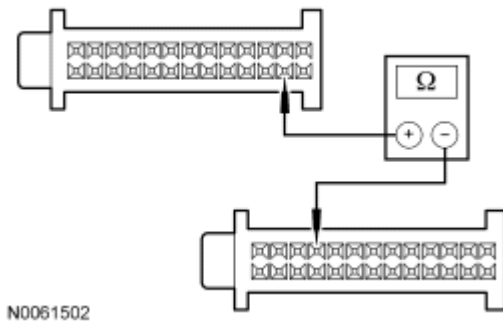


Fig. 57: Measuring Resistance Between EMTC Module C2357A-2, Circuit CH434 (GY/YE) & IC C220-23

Courtesy of FORD MOTOR CO.

- Measure the resistance between HVAC module - EMTC C2357a-2, circuit CH434 (GY/YE), harness side and IC C220-23, circuit CH434 (GY/YE), harness side.

- **Is the resistance less than 5 ohms?**

YES : Go to J8.

NO : REPAIR circuit CH434 (GY/YE) for an open. TEST the system for normal operation.

J8 CHECK CIRCUIT CH434 (GY/YE) FOR A SHORT TO VOLTAGE

- Key in ON position.

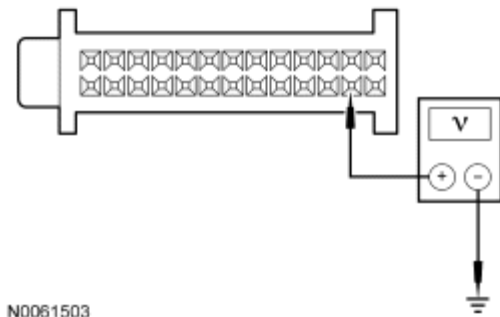


Fig. 58: Measuring Voltage Between EMTC Module C2357A-2, Circuit CH434 (GY/YE) & Ground

Courtesy of FORD MOTOR CO.

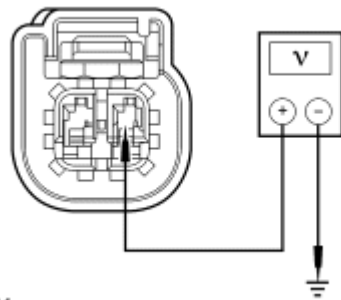
- Measure the voltage between HVAC module - EMTC C2357a-2, circuit CH434 (GY/YE), harness side and ground.
- **Is the voltage greater than 10 volts?**

YES : Go to J14.

NO : REPAIR circuit CH434 (GY/YE) for a short to voltage. TEST the system for normal operation.

J9 CHECK THE VOLTAGE AT THE A/C COMPRESSOR CLUTCH FIELD COIL

- Key in OFF position.
- Disconnect: A/C Compressor Clutch Field Coil C100
- Key in ON position.
- With the engine running, press the PANEL and A/C manual override buttons on the HVAC module - DATC or select MAX A/C on the HVAC module - EMTC.



N0061504

Fig. 59: Measuring Voltage Between A/C Compressor Clutch Field Coil C100-1, Circuit CH401 (VT/WH) & Ground
Courtesy of FORD MOTOR CO.

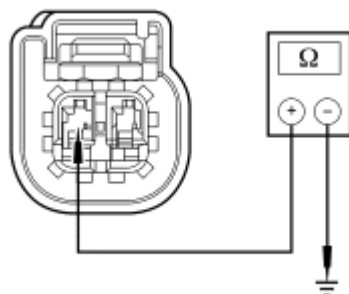
- Measure the voltage between A/C compressor clutch field coil C100-1, circuit CH401 (VT/WH), harness side and ground.
- **Is the voltage greater than 10 volts?**

YES : Go to J10.

NO : CARRY OUT the A/C Control Relay Component Test. Refer to appropriate COMPONENT TESTING article. If the relay tests OK, go to J11.

J10 CHECK THE GROUND AT THE A/C COMPRESSOR CLUTCH FIELD COIL

- Key in OFF position.



N0061505

Fig. 60: Measuring Resistance Between A/C Compressor Clutch Field Coil C100-2, Circuit

GD120 (BK/GN) & Ground

Courtesy of FORD MOTOR CO.

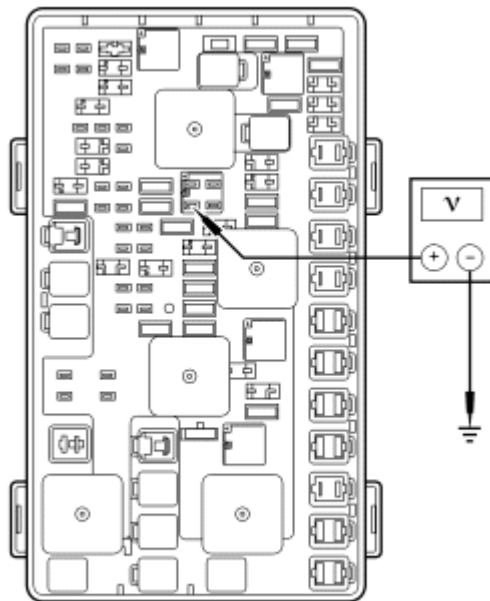
- Measure the resistance between A/C compressor clutch field coil C100-2, circuit GD120 (BK/GN), harness side and ground.
- **Is the resistance less than 5 ohms?**

YES : INSTALL a new A/C compressor. REFER to **CLIMATE CONTROL** article. TEST the system for normal operation.

NO : REPAIR circuit GD120 (BK/GN) for an open. TEST the system for normal operation.

J11 CHECK THE RELAY SWITCH FOR VOLTAGE

- Key in OFF position.
- Disconnect: A/C Clutch Relay
- Key in ON position.



N0061506

Fig. 61: Measuring Voltage Between A/C Clutch Relay Socket & Ground

Courtesy of FORD MOTOR CO.

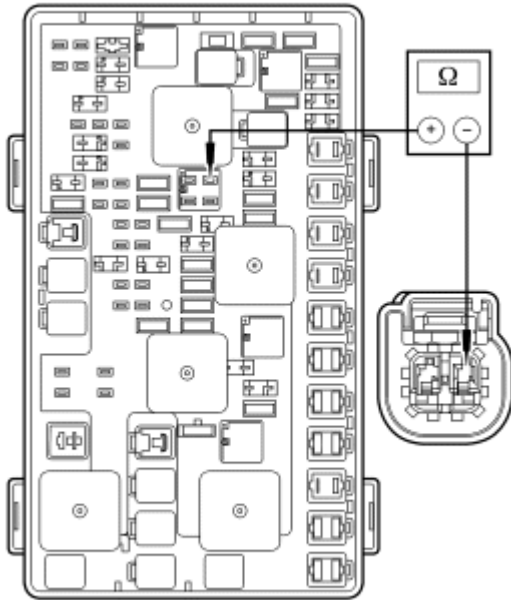
- Measure the voltage between A/C clutch relay socket and ground.
- **Is the voltage greater than 10 volts?**

YES : Go to J12.

NO : VERIFY BJB fuse 35 (10A) is OK. If OK, INSTALL a new BJB. TEST the system for normal operation.

J12 CHECK CIRCUIT CH401 (VT/WH) FOR AN OPEN

- Key in OFF position.



N0061507

Fig. 62: Measuring Resistance Between A/C Clutch Relay Socket, Circuit CH401 (VT/WH) & A/C Compressor Clutch Field Coil C100-1

Courtesy of FORD MOTOR CO.

- Measure the resistance between A/C clutch relay socket, circuit CH401 (VT/WH) and A/C compressor clutch field coil C100-1, circuit CH401 (VT/WH), harness side.

- **Is the resistance less than 5 ohms?**

YES : Go to J14.

NO : REPAIR circuit CH401 (VT/WH) for an open. TEST the system for normal operation.

J13 CHECK THE MODULE CONNECTION

- Carry out the HVAC Module Cold Boot Process.

NOTE: DATC systems only.

- Carry out the HVAC Module - Dual Automatic Temperature Control (DATC) On-Demand Self Test.
- Operate the system.
- **Does the concern return?**

YES : INSTALL a new HVAC module - DATC or HVAC module - EMTC. REFER to **CLIMATE CONTROL** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose

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or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

J14 CHECK THE PCM MODULE CONNECTION

- Check the DTCs.
- Disconnect all the PCM connectors.
- Check for:
 - corrosion.
 - pushed-out pins.
 - incorrectly seated connector.
- Connect and correctly seat all the PCM connectors.
- Operate the system.
- **Does the concern return?**

YES : INSTALL a new PCM. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test K: The Air Conditioning (A/C) is Always On

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Manual Climate Control System for schematic and connector information.

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Automatic Climate Control System for schematic and connector information.

Normal Operation

Under normal operation, when A/C is requested, in the dual automatic temperature control (DATC) system, a message is sent over the medium-speed controller area network (MS-CAN) bus to the instrument cluster, then from the instrument cluster through the high-speed controller area network (HS-CAN) bus to the PCM; or in electronic manual temperature control (EMTC) systems, voltage is sent to the instrument cluster through circuit CH434 (GY/YE), then a message is sent from the instrument cluster through the HS-CAN bus to the PCM.

Ignition voltage is provided to the A/C clutch relay switch. When the PCM energizes the relay, ignition voltage is supplied to the A/C clutch through circuit CH401 (VT/WH). Ground is supplied for the A/C clutch through circuit GD120 (BK/GN).

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- PCM
- HVAC module - DATC
- HVAC module - EMTC
- A/C evaporator discharge temperature sensor

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- A/C pressure transducer
- A/C control relay
- A/C compressor

PINPOINT TEST K: THE AIR CONDITIONING (A/C) IS ALWAYS ON

NOTE: When disconnecting and reconnecting the pressure transducer electrical connector, make sure that the connector-locking device is in place and that the locking device and connector are correctly and fully seated.

K1 CHECK THE A/C PRESSURE SENSOR (ACP_PSI) PCM PID

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: A/C Pressure PCM PID
- With the manifold gauge set connected, compare the pressure readings of the manifold gauge set and the ACP_PSI PID.
- **Are the pressure values of the manifold gauge set and the ACP_PSI PCM PID similar?**
YES : Go to K2.

NO : INSTALL a new A/C pressure transducer. TEST the system for normal operation.

K2 CHECK THE POST EVAPORATOR TEMPERATURE SENSOR INPUT (PEVAP_TMP) HVAC PID

- Allow the vehicle exterior and interior to stabilize to an ambient temperature above 16°C (60°F).
- Enter the following diagnostic mode on the diagnostic tool: PEVAP_TMP HVAC PID
- **Does the PEVAP_TMP HVAC PID read similar to the ambient temperature?**
YES : Go to K3.

NO : CARRY OUT the A/C evaporator discharge temperature sensor component test in this service information. If the sensor tests OK, go to K11. If the sensor does not test OK, INSTALL a new heater core and evaporator core housing. REFER to **CLIMATE CONTROL** article. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

K3 CHECK THE AIR CONDITIONING COMPRESSOR CYCLING SWITCH (ACCS) PCM PID WITH THE A/C OFF

- Enter the following diagnostic mode on the diagnostic tool: ACCS PCM PID
- **Does the ACCS PCM PID read ON?**
YES : DATC systems, go to K4.

EMTC systems, go to K5.

NO : Go to K7.

K4 CHECK THE AIR CONDITIONING SWITCH STATUS (CC_AC) HVAC PID WITH THE A/C OFF

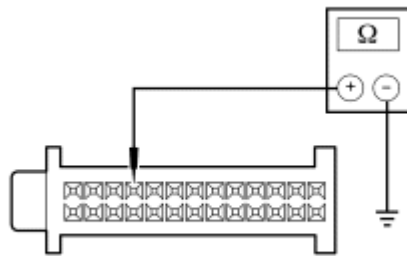
- Enter the following diagnostic mode on the diagnostic tool: CC_AC HVAC PID
- **Does the CC_AC HVAC PID read ON?**

YES : Go to K10.

NO : Go to K11.

K5 CHECK HVAC - EMTC A/C REQUEST OUTPUT

- Key in OFF position.
- Disconnect: Instrument Cluster (IC) C220
- Key in ON position.
- Select the OFF position on the HVAC module - EMTC.



N0041895

Fig. 63: Measuring Resistance
Courtesy of FORD MOTOR CO.

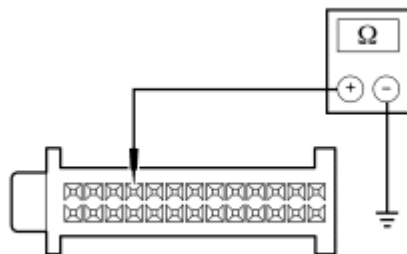
- Measure the resistance between IC C220-23, circuit CH434 (GY/YE), harness side and ground.
- **Is the resistance greater than 10,000 ohms?**

YES : Go to K11.

NO : Go to K6.

K6 CHECK CIRCUIT CH434 (GY/YE) FOR A SHORT TO GROUND

- Key in OFF position.
- Disconnect: HVAC Module - EMTC C2357a



N0041895

Fig. 64: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between IC C220-23, circuit CH434 (GY/YE), harness side and ground.

- **Is the resistance greater than 10,000 ohms?**

YES : Go to K10.

NO : REPAIR circuit CH434 (GY/YE) for a short to ground. TEST the system for normal operation.

K7 CHECK THE AIR CONDITIONING CLUTCH (WAC_ACCR) PCM PID WITH THE A/C OFF

- Enter the following diagnostic mode on the diagnostic tool: WAC_ACCR PCM PID

- **Does the WAC_ACCR PCM PID read ON?**

YES : REFER to **Introduction - Gasoline Engines** article.

NO : Go to K8.

K8 CHECK FOR VOLTAGE TO THE A/C COMPRESSOR CLUTCH FIELD COIL

- Key in OFF position.
- Disconnect: A/C Compressor Clutch Field Coil C100
- Key in ON position.
- Select the OFF position on the HVAC module - EMTC.

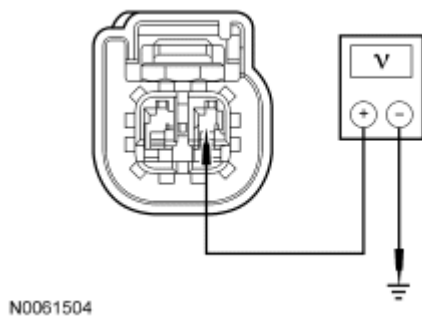


Fig. 65: Measuring Voltage Between A/C Compressor Clutch Field Coil C100-1, Circuit CH401 (VT/WH) & Ground
Courtesy of FORD MOTOR CO.

- Measure the voltage between A/C compressor clutch field coil C100-1, circuit CH401 (VT/WH), harness side and ground.

- **Is the voltage greater than 10 volts?**

YES : Go to K9.

NO : INSTALL a new A/C compressor. REFER to **CLIMATE CONTROL** article. TEST the system for normal operation.

K9 CHECK CIRCUIT CH401 (VT/WH) FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: A/C Clutch Relay
- Key in ON position.

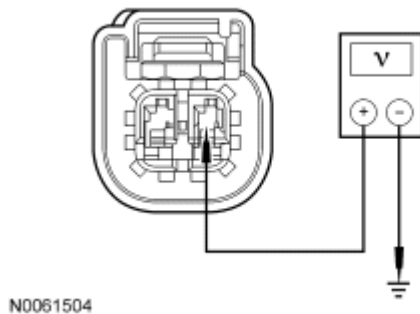


Fig. 66: Measuring Voltage Between A/C Compressor Clutch Field Coil C100-1, Circuit CH401 (VT/WH) & Ground
Courtesy of FORD MOTOR CO.

- Measure the voltage between A/C compressor clutch field coil C100-1, circuit CH401 (VT/WH), harness side and ground.
- **Is the voltage greater than 10 volts?**
YES : REPAIR circuit CH401 (VT/WH) for a short to voltage. TEST the system for normal operation.
NO : INSTALL a new A/C clutch relay. Check the A/C clutch diode. INSTALL a new diode if necessary. TEST the system for normal operation.

K10 CHECK THE MODULE CONNECTION

- Carry out the HVAC Module Cold Boot Process.

NOTE: DATC systems only.

- Carry out the HVAC Module - Dual Automatic Temperature Control (DATC) On-Demand Self Test.
- Operate the system.
- **Does the concern return?**
YES : INSTALL a new HVAC module - DATC or HVAC module - EMTC. REFER to **CLIMATE CONTROL** article. TEST the system for normal operation.
NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

K11 CHECK THE PCM CONNECTION

- Clear the DTCs.
- Disconnect all the PCM connectors.
- Check for:
 - corrosion.
 - pushed-out pins.
 - incorrectly seated connector.

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- Connect and correctly seat all the PCM connectors.
- Operate the system.
- **Does the concern return?**

YES : INSTALL a new PCM. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test L: The Blower Motor is Inoperative - EMTC

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Manual Climate Control System for schematic and connector information.

Normal Operation

Under normal operation, the blower motor relay coil receives ignition voltage through circuit CBP45 (YE). The coil receives ground from the HVAC module - electronic manual temperature control (EMTC) through circuit CH123 (VT/GN) if any function selector position but OFF is selected. Voltage is supplied to the relay switch contact through circuit CH402 (YE/GN). When the relay coil is energized, voltage is delivered to the blower motor through circuit CH402 (YE/GN)/(BK). Ground for the blower motor is provided through circuit CH430 (VT/OG)/(BK/RD) from the blower resistor or the blower switch (HI). The blower resistor and blower switch are grounded through circuit GD116 (BK/VT).

This pinpoint test is intended to diagnose the following:

- Fuse(s)
- Wiring, terminals or connectors
- HVAC module - EMTC
- Blower motor
- Blower motor relay
- Blower motor switch

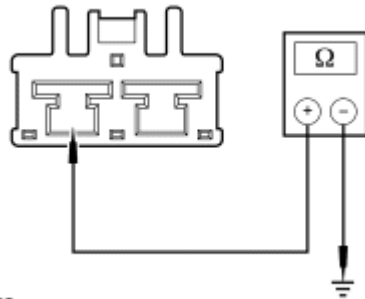
PINPOINT TEST L: THE BLOWER MOTOR IS INOPERATIVE - EMTC

CAUTION: It is important to install relays in their correct position in the battery junction box (BJB). Installing a relay incorrectly may cause wiring shorts or damage to modules. While carrying out diagnostics on BJB relays, have only one BJB relay removed at a time. Failure to follow these instructions may result in damage to the vehicle circuitry or to control modules.

L1 CHECK CIRCUIT (BK/RD) FOR GROUND

- Key in OFF position.
- Disconnect: Blower Motor C288
- Turn the function selector switch to the PANEL position.

- Turn the blower motor switch to the HI position.



N0061508

Fig. 67: Measuring Resistance Between A/C Blower Motor C288-2, Circuit (BK/RD) & Ground

Courtesy of FORD MOTOR CO.

- Measure the resistance between A/C blower motor C288-1, circuit (BK/RD), harness side and ground.

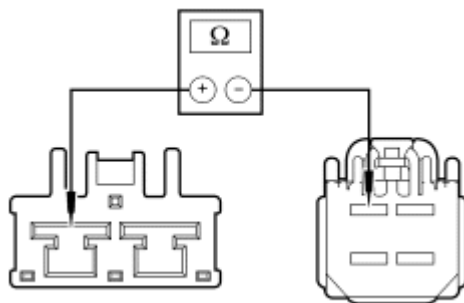
- **Is the resistance less than 5 ohms?**

YES : Go to L4.

NO : Go to L2.

L2 CHECK CIRCUIT CH430 (VT/OG)/(BK/RD) FOR AN OPEN

- Disconnect: Blower Motor Switch C2357b



N0061509

Fig. 68: Measuring Resistance

Courtesy of FORD MOTOR CO.

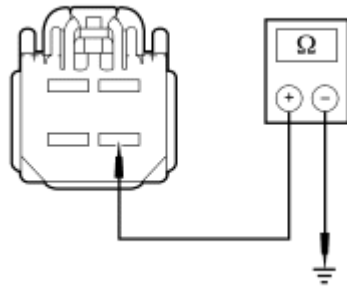
- Measure the resistance between blower motor switch C2357b-2, circuit CH430 (VT/OG), harness side and blower motor C288-1, circuit (BK/RD), harness side.

- **Is the resistance less than 5 ohms?**

YES : Go to L3.

NO : REPAIR circuit CH430 (VT/OG)/(BK/RD) for an open. TEST the system for normal operation.

L3 CHECK CIRCUIT GD116 (BK/VT) FOR AN OPEN



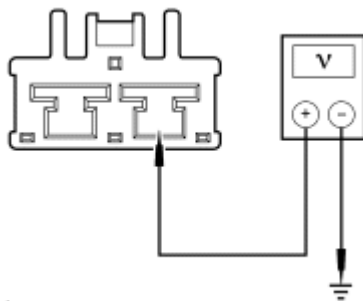
N0061510

Fig. 69: Measuring Resistance Between Blower Motor Switch C2357B-3, Circuit GD116 (BK/VT) & Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between blower motor switch C2357b-3, circuit GD116 (BK/VT), harness side and ground.
- **Is the resistance less than 5 ohms?**
YES : INSTALL a new blower motor switch. TEST the system for normal operation.
NO : REPAIR circuit GD116 (BK/VT) for an open. TEST the system for normal operation.

L4 CHECK A/C BLOWER MOTOR CIRCUIT (BK) FOR VOLTAGE

- Key in ON position.



N0061511

Fig. 70: Measuring Voltage
Courtesy of FORD MOTOR CO.

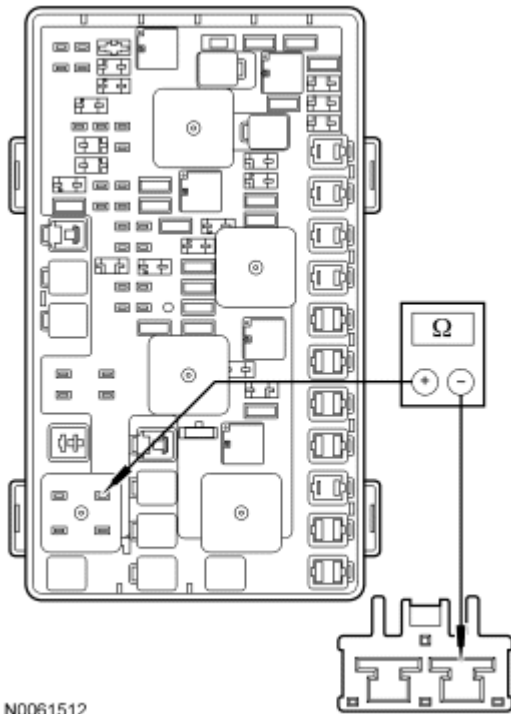
- Measure the voltage between A/C blower motor connector C288-1, circuit (BK), harness side and ground.
- **Is the voltage greater than 10 volts?**
YES : INSTALL a new blower motor. TEST the system for normal operation.
NO : Go to L5.

L5 CHECK CIRCUIT CH402 (YE/GN)/(BK) FOR AN OPEN

- Key in OFF position.
- Disconnect: Blower Motor Relay

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N0061512

Fig. 71: Measuring Resistance Between Blower Motor Relay Socket, Circuit CH402 (YE/GN) & Blower Motor C288-1

Courtesy of FORD MOTOR CO.

- Measure the resistance between blower motor relay socket, circuit CH402 (YE/GN) and blower motor C288-1, circuit (BK), harness side.

- **Is the resistance less than 5 ohms?**

YES : Go to L6.

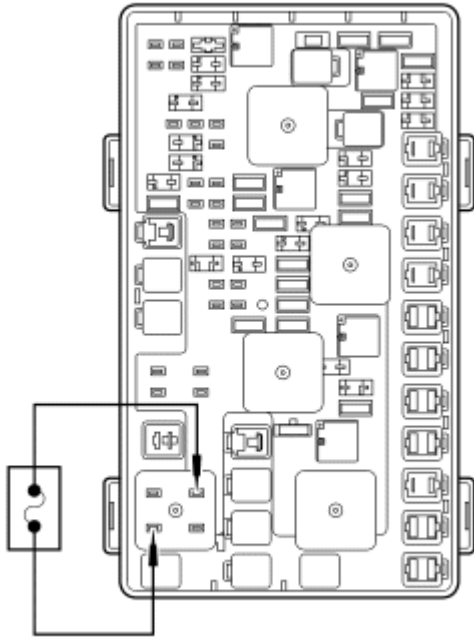
NO : REPAIR circuit CH402 (YE/GN)/(BK) for an open. TEST the system for normal operation.

L6 CHECK THE BLOWER MOTOR RELAY

- Connect: Blower Motor C288
- Key in ON position.

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N0061513

Fig. 72: Connecting Fused Jumper Lead Between Blower Motor Relay Socket, Circuit CH402 (YE/GN)

Courtesy of FORD MOTOR CO.

- Connect a fused jumper lead between blower motor relay socket, circuit CH402 (YE/GN) and circuit CH402 (YE/GN).

- **Does the blower motor operate?**

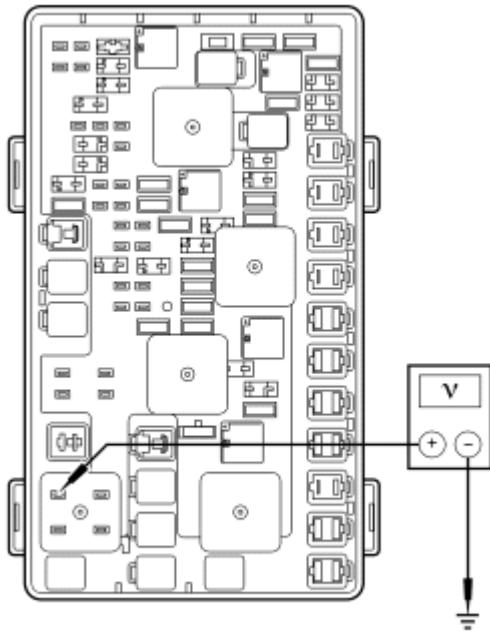
YES : Go to L7.

NO : VERIFY battery junction box (BJB) fuse 66 (40A) is OK. If OK, REPAIR circuit CH402 (YE/GN) for an open. TEST the system for normal operation.

L7 CHECK THE RELAY COIL SUPPLY VOLTAGE

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N0061514

Fig. 73: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between blower motor relay socket, circuit CBP45 (YE) and ground.

- **Is the voltage greater than 10 volts?**

YES : Go to L8.

NO : VERIFY smart junction box (SJB) fuse 45 (5A) is OK. If OK, REPAIR circuit CBP45 (YE) for an open. TEST the system for normal operation.

L8 CHECK THE BLOWER MOTOR RELAY

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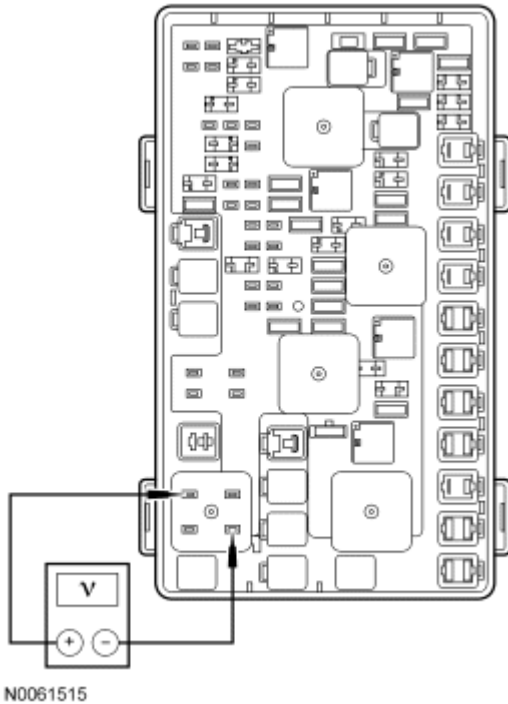


Fig. 74: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between blower motor relay socket, circuit CH123 (VT/GN) and circuit CBP45 (YE).
- **Is the voltage greater than 10 volts?**
YES : INSTALL a new blower motor relay. TEST the system for normal operation.
NO : Go to L9.

L9 CHECK CIRCUIT CH123 (VT/GN) FOR AN OPEN

- Key in OFF position.
- Disconnect: HVAC - Module EMTC C2357a

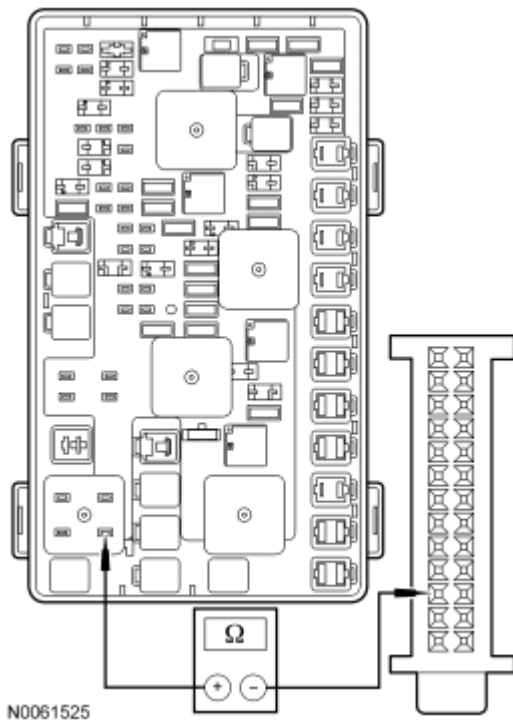


Fig. 75: Measuring Resistance Between & Blower Motor Relay Socket, Circuit CH123 (VT/GN) EATC Module C228A-24
 Courtesy of FORD MOTOR CO.

- Measure the resistance between and blower motor relay socket, circuit CH123 (VT/GN) HVAC module - EMTC C2357a-24, circuit CH123 (VT/GN), harness side.

- **Is the resistance less than 5 ohms?**

YES : Go to L10.

NO : REPAIR circuit CH123 (VT/GN) for an open. TEST the system for normal operation.

L10 CHECK THE MODULE CONNECTION

- Carry out the HVAC Module Cold Boot Process.
- Operate the system.
- **Does the concern return?**

YES : INSTALL a new HVAC module - EMTC. REFER to **CLIMATE CONTROL** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. TEST the system for normal operation.

Pinpoint Test M: The Blower Motor Does Not Operate Correctly - EMTC

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Manual Climate Control System for schematic and connector information.

Normal Operation

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Under normal operation, the blower motor is provided a ground from the blower resistor through circuit CH430 (VT/OG)/(BK/RD). The resistor gets a ground from circuit GD116 (BK/VT) in the lowest blower setting. In MED-LO and MED-HI the resistor gets a ground from the blower motor switch through circuit CH428 (GN/WH)/(RD) or CH429 (GY/BN)/(BN), depending on selected speed. In HI, the blower motor is grounded directly through the blower motor switch from circuit CH430 (VT/OG)/(BK/RD) to circuit GD116 (BK/VT). The blower switch receives its ground from circuit GD116 (BK/VT)/(GN).

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Blower motor resistor
- Blower motor switch

PINPOINT TEST M: THE BLOWER MOTOR DOES NOT OPERATE CORRECTLY - EMTC

M1 CHECK THE BLOWER MOTOR OPERATION

- Key in ON position.
- Turn the function selector switch to the PANEL position.
- Select all blower speed positions.
- **Does the blower motor operate in any position?**
YES : If the blower motor does not operate in HI, go to M2.

If the blower motor does not operate in MED-HI, go to M3.

If the blower motor does not operate in MED-LO, go to M4.

If the blower motor does not operate in LO, go to M5.

For all other symptoms, go to M8.

NO : Go to **Pinpoint Test L** .

M2 CHECK CIRCUIT CH430 (VT/OG)/(BK/RD) FOR AN OPEN

- Key in OFF position.
- Disconnect: Blower Motor Switch C2357b
- Disconnect: Blower Motor C288
- Disconnect: Blower Motor Resistor C2185

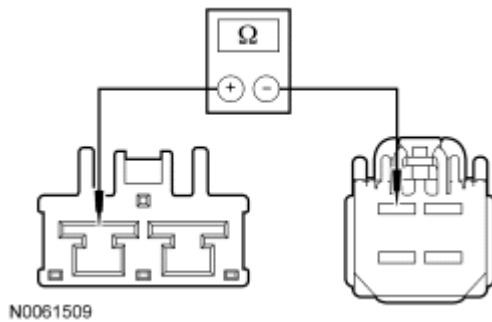


Fig. 76: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between blower motor switch C2357b-2, circuit CH430 (VT/OG), harness side and blower motor C288-1, circuit (BK/RD), harness side.
- **Is the resistance less than 5 ohms?**
YES : Go to M6.
NO : REPAIR circuit CH430 (VT/OG)/(BK/RD) for an open. TEST the system for normal operation.

M3 CHECK CIRCUIT CH429 (GY/BN)/(BN) FOR AN OPEN

- Key in OFF position.
- Disconnect: Blower Motor Switch C2357b
- Disconnect: Blower Motor Resistor C2185

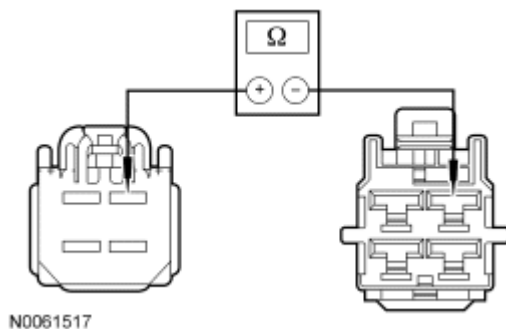


Fig. 77: Measuring Resistance Between Blower Motor Switch C2357B-1 & C2185-2, Circuit CH429 (GY/BN) & Circuit (BN)
Courtesy of FORD MOTOR CO.

- Measure the resistance between blower motor switch C2357b-1, circuit CH429 (GY/BN), harness side and blower motor resistor C2185-2, circuit (BN), harness side.
- **Is the resistance less than 5 ohms?**
YES : Go to M6.
NO : REPAIR circuit CH429 (GY/BN)/(BN) for an open. TEST the system for normal operation.

M4 CHECK CIRCUIT CH428 (GN/WH)/(RD) FOR AN OPEN

- Key in OFF position.
- Disconnect: Blower Motor Switch C2357b
- Disconnect: Blower Motor Resistor C2185

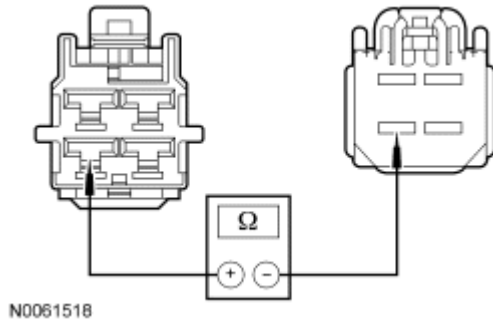


Fig. 78: Measuring Resistance Between Blower Motor Switch C2357B-4 & C2185-3, Circuit CH428 (GN/WH) & Circuit (RD)
 Courtesy of FORD MOTOR CO.

- Measure the resistance between blower motor switch C2357b-4, circuit CH428 (GN/WH), harness side and blower motor resistor C2185-3, circuit (RD), harness side.

- **Is the resistance less than 5 ohms?**

YES : Go to M6.

NO : REPAIR circuit CH428 (GN/WH)/(RD) for an open. TEST the system for normal operation.

M5 CHECK BLOWER MOTOR RESISTOR GROUND CIRCUIT GD116 (BK/VT)/(GN) FOR AN OPEN

- Key in OFF position.
- Disconnect: Blower Motor Resistor C2185

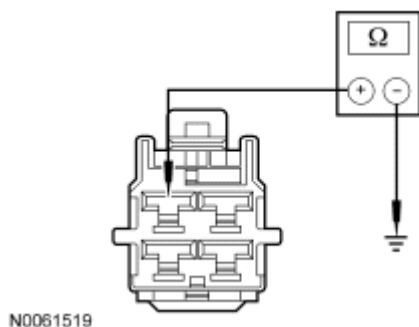


Fig. 79: Measuring Resistance Between Blower Motor Resistor C2185-1, Circuit GD116 (BK/VT)/(GN) & Ground
 Courtesy of FORD MOTOR CO.

- Measure the resistance between blower motor resistor C2185-1, circuit GD116 (BK/VT)/(GN), harness side and ground.

- **Is the resistance less than 5 ohms?**

YES : Go to M6.

NO : REPAIR circuit GD116 (BK/VT)/(GN) for an open. TEST the system for normal operation.

M6 CHECK BLOWER MOTOR RESISTOR CIRCUIT (BK/RD) FOR AN OPEN

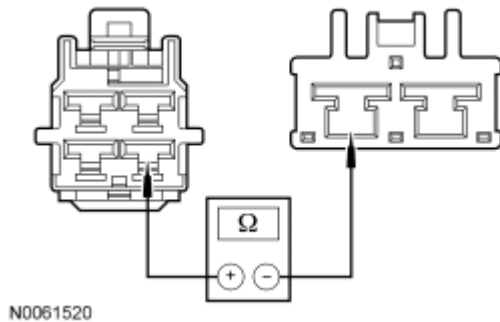


Fig. 80: Measuring Resistance Between Blower Motor Resistor C2185-4, Circuit (BK/RD) & Blower Motor C288-2

Courtesy of FORD MOTOR CO.

- Measure the resistance between blower motor resistor C2185-4, circuit (BK/RD), harness side and blower motor C288-1, circuit (BK/RD), harness side.
- **Is the resistance less than 5 ohms?**

YES : CARRY OUT the blower motor resistor component test in this service information. If the resistor tests OK, go to M7.

NO : REPAIR circuit (BK/RD) for an open. TEST the system for normal operation.

M7 CHECK BLOWER MOTOR SWITCH CIRCUIT GD116 (BK/VT) FOR AN OPEN

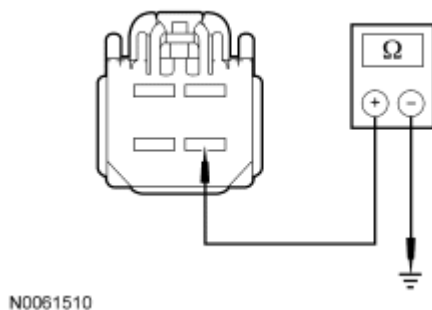


Fig. 81: Measuring Resistance Between Blower Motor Switch C2357B-3, Circuit GD116 (BK/VT) & Ground

Courtesy of FORD MOTOR CO.

- Measure the resistance between blower motor switch C2357b-3, circuit GD116 (BK/VT), harness side and ground.
- **Is the resistance less than 5 ohms?**

YES : INSTALL a new blower motor switch. TEST the system for normal operation.

NO : REPAIR circuit GD116 (BK/VT) for an open. TEST the system for normal operation.

M8 CHECK CIRCUITS CH430 (VT/OG)/(BK/RD), CH428 (GN/WH)/(RD) AND CH429 (GY/BN)/(BN) FOR A SHORT TO GROUND

- Key in OFF position.
- Disconnect: Blower Motor C288
- Disconnect: Blower Motor Switch C2357b
- Disconnect: Blower Motor Resistor C2185
- Place the blower switch in the lowest setting.

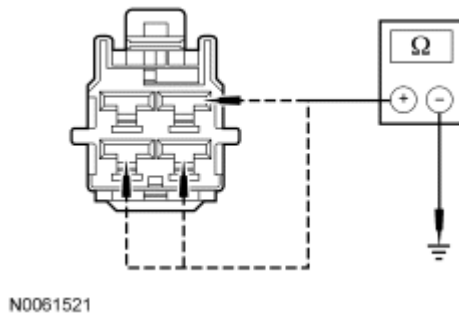


Fig. 82: Measuring Resistance Between Ground & Blower Motor Resistor, C2185-3, C2185-2 & C2185-4

Courtesy of FORD MOTOR CO.

- Measure the resistance between ground and blower motor resistor:
 - C2185-3, circuit (RD), harness side.
 - C2185-2, circuit (BN), harness side.
 - C2185-4, circuit (BK/RD), harness side.

- **Are the resistances greater than 10,000 ohms?**

YES : Go to M9.

NO : REPAIR the affected circuit. TEST the system for normal operation.

M9 CHECK THE BLOWER MOTOR CIRCUITS FOR SHORTS TOGETHER

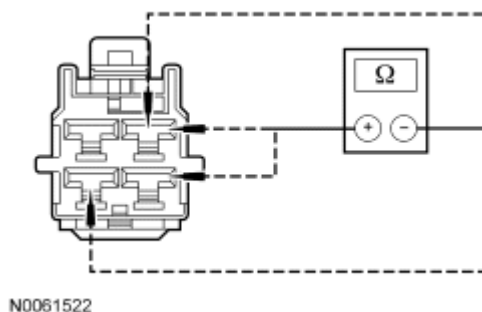


Fig. 83: Measuring Resistance Between Blower Motor Resistor C2185-2, C2185-4 & C2185-4
Courtesy of FORD MOTOR CO.

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- Measure the resistance between blower motor resistor:
 - C2185-3, circuit (RD), harness side and C2185-2, circuit (BN), harness side.
 - C2185-3, circuit (RD), harness side and C2185-4, circuit (BK/RD), harness side.
 - C2185-2, circuit (BN), harness side and C2185-4, circuit (BK/RD), harness side.
- **Are the resistances greater than 10,000 ohms?**
YES : INSTALL a new blower motor switch. TEST the system for normal operation.
NO : REPAIR the affected circuits. TEST the system for normal operation.

Pinpoint Test N: The Blower Motor is Inoperative - DATC

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Automatic Climate Control System for schematic and connector information.

Normal Operation

Under normal operation, the blower motor relay coil receives a ground from the HVAC module - dual automatic temperature control (DATC) through circuit CH123 (VT/GN). The coil receives ignition voltage through circuit CBP45 (YE). Voltage is supplied to the relay switch contact through circuit CH402 (YE/GN). When the relay coil is energized, voltage is delivered to the blower motor control module through circuit CH402 (YE/GN)/(BK). Ground for the blower motor speed control is provided by circuit GD116 (BK/VT)/(BK/RD). The HVAC module - DATC provides a blower target speed signal to the blower motor speed control through circuit VH101 (WH/VT)/(RD) to control the blower speed. The actual blower speed signal is provided to the HVAC module - DATC through circuit CH102 (YE/BU)/(BN) from the blower motor speed control. Ground for the motor is provided through circuit (BU) from the blower motor speed control.

This pinpoint test is intended to diagnose the following:

- Fuse(s)
- Wiring, terminals or connectors
- Blower motor relay
- Blower motor speed control
- HVAC module - DATC
- Blower motor

PINPOINT TEST N: THE BLOWER MOTOR IS INOPERATIVE - DATC

CAUTION: It is important to install relays in their correct position in the battery junction box (BJB). Installing a relay incorrectly may cause wiring shorts or damage to modules. While carrying out diagnostics on BJB relays, have only one BJB relay removed at a time. Failure to follow these instructions may result in damage to the vehicle circuitry or to control modules.

N1 VERIFY THE BLOWER MOTOR OPERATION

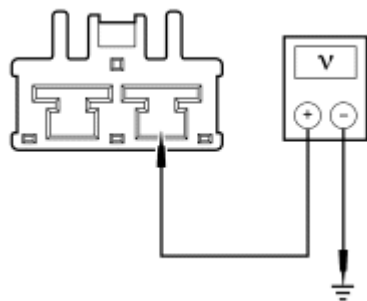
- Key in ON position.
- Press the PANEL button on the HVAC module - DATC. Adjust the blower motor setting to LO and then to HI.
- **Is the blower motor inoperative in all settings?**
YES : Go to N2.
NO : Go to **Pinpoint Test O** .

N2 CHECK THE BLOWER MOTOR SPEED TARGET POSITION (BLWR_T#) HVAC PID

- Enter the following diagnostic mode on the diagnostic tool: BLWR_T# HVAC PID
- Observe the BLWR_T# HVAC PID while increasing and decreasing the blower speed.
- **Does the BLWR_T# HVAC PID reading change in all speeds between 0% and 100%?**
YES : Go to N3.
NO : Go to N16.

N3 CHECK FOR VOLTAGE TO THE BLOWER MOTOR

- Key in OFF position.
- Disconnect: Blower Motor C288
- Key in ON position.



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Fig. 84: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between ground and blower motor C288-2, circuit (BK), harness side.
- **Is the voltage greater than 10 volts?**
YES : Go to N9.

NO : CARRY OUT the blower motor relay component test. Refer to appropriate COMPONENT TESTING article. If the relay tests OK, go to N4.

N4 CHECK THE RELAY SWITCH POWER CIRCUIT FOR AN OPEN

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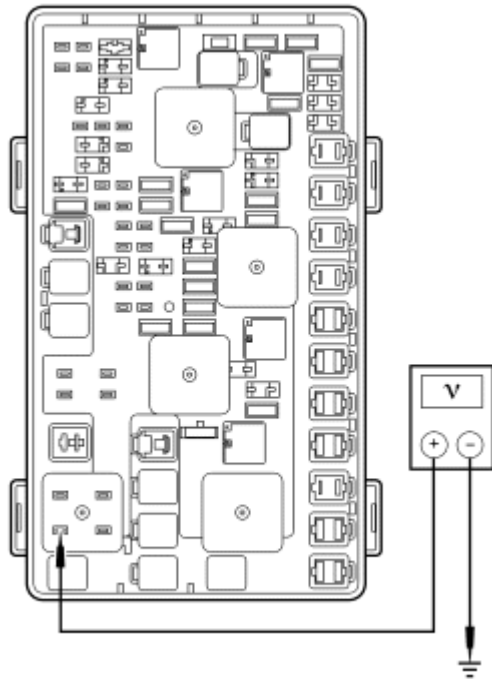


Fig. 85: Measuring Voltage Between Ground & Blower Motor Relay Socket, Circuit CH402 (YE/GN)

Courtesy of FORD MOTOR CO.

- Measure the voltage between ground and blower motor relay socket, circuit CH402 (YE/GN).
- **Is the voltage greater than 10 volts?**

YES : Go to N5.

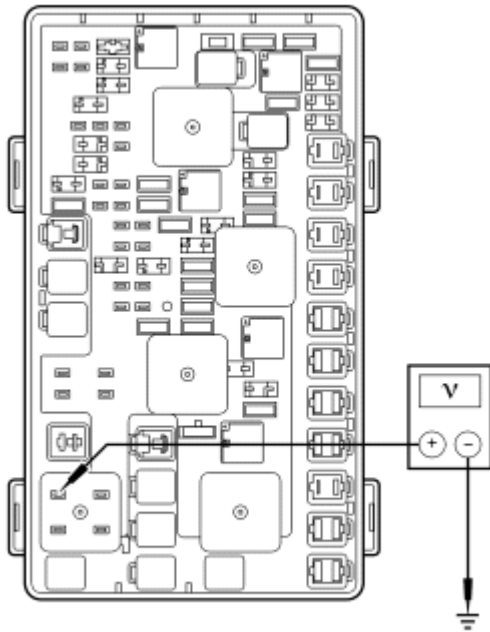
NO : VERIFY battery junction box (BJB) fuse 66 (40A) is OK. If OK, REPAIR circuit CH402 (YE/GN) for an open. TEST the system for normal operation.

N5 CHECK THE RELAY COIL SUPPLY VOLTAGE

- Key in OFF position.
- Disconnect: Blower Motor Relay
- Key in ON position.

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N0061514

Fig. 86: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between ground and blower motor relay socket, circuit CBP45 (YE).

- **Is the voltage greater than 10 volts?**

YES : Go to N6.

NO : VERIFY smart junction box (SJB) fuse 45 (5A) is OK. If OK, REPAIR circuit CBP45 (YE) for an open. TEST the system for normal operation.

N6 CHECK THE HVAC - DATC MODULE OUTPUT

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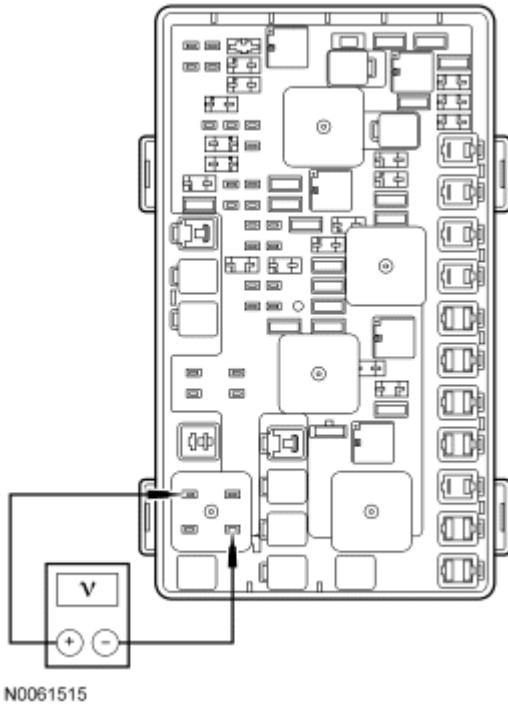


Fig. 87: Measuring Voltage
Courtesy of FORD MOTOR CO.

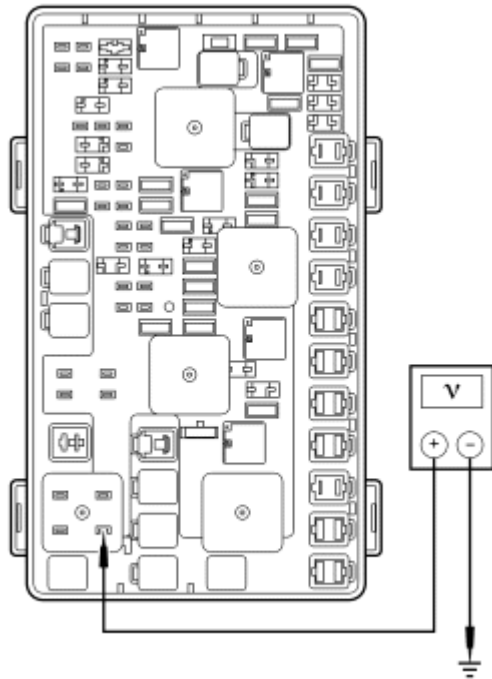
- Measure the voltage between blower motor relay socket, circuit CBP45 (YE) and blower motor relay socket, circuit CH123 (VT/GN).
- **Is the voltage greater than 10 volts?**
YES : REPAIR circuit CH402 (YE/GN)/(BK) for an open. TEST the system for normal operation.
NO : Go to N7.

N7 CHECK CIRCUIT CH123 (VT/GN) FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: HVAC Module - DATC C228a
- Key in ON position.

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N0061524

Fig. 88: Measuring Voltage Between Ground & Blower Motor Relay Socket, Circuit CH123 (VT/GN) & Ground

Courtesy of FORD MOTOR CO.

- Measure the voltage between ground and blower motor relay socket, circuit CH123 (VT/GN) and ground.
- **Is any voltage present?**
YES : REPAIR circuit CH123 (VT/GN) for a short to voltage. TEST the system for normal operation.

NO : Go to N8.

N8 CHECK CIRCUIT CH123 (VT/GN) FOR AN OPEN

- Key in OFF position.

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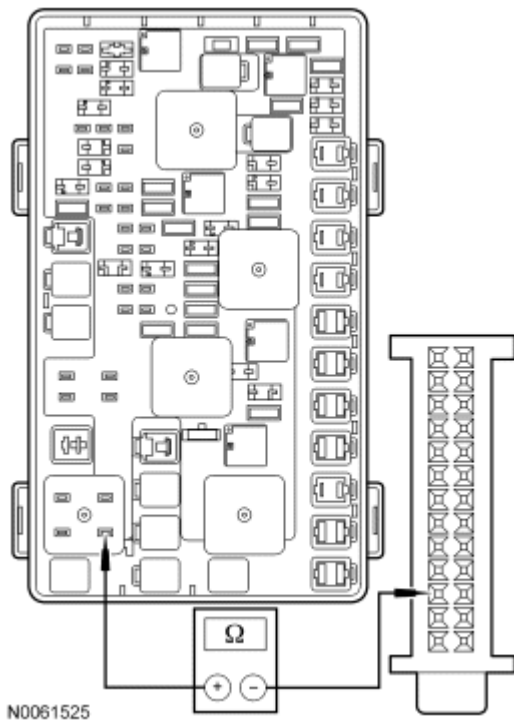
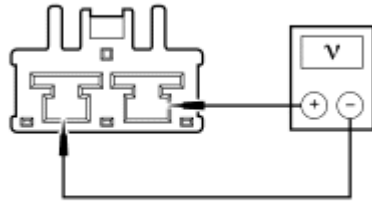


Fig. 89: Measuring Resistance Between & Blower Motor Relay Socket, Circuit CH123 (VT/GN) EATC Module C228A-24
Courtesy of FORD MOTOR CO.

- Measure the resistance between and blower motor relay socket, circuit CH123 (VT/GN) HVAC module - DATC C228a-24, circuit CH123 (VT/GN), harness side.
- **Is the resistance less than 5 ohms?**
YES : Go to N16.
NO : REPAIR circuit CH123 (VT/GN) for an open. TEST the system for normal operation.

N9 CHECK FOR GROUND TO THE BLOWER MOTOR

- Key in OFF position.
- Disconnect: Blower Motor C288
- Key in ON position.
- Select the highest blower speed setting.



N0061602

Fig. 90: Measuring Voltage Between Blower Motor C288-1 & C288-2, Circuit (BK) & Circuit (BU)

Courtesy of FORD MOTOR CO.

- Measure the voltage between blower motor C288-2, circuit (BK), harness side and blower motor C288-1, circuit (BU), harness side.

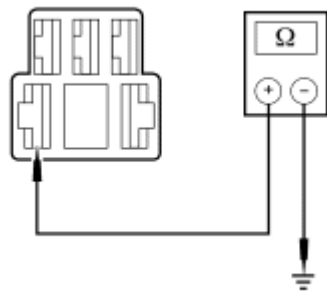
- **Is the voltage greater than 10 volts?**

YES : INSTALL a new blower motor. REFER to CLIMATE CONTROL article. TEST the system for normal operation.

NO : Go to N10.

N10 CHECK CIRCUIT GD116 (BK/VT)/(BK/RD) FOR AN OPEN

- Key in OFF position.
- Disconnect: Blower Motor Speed Control C271



N0061526

Fig. 91: Measuring Resistance Between Blower Motor Speed Control Module C271-4, Circuit (BK/RD) & Ground

Courtesy of FORD MOTOR CO.

- Measure the resistance between blower motor speed control C271-4, circuit (BK/RD), harness side and ground.

- **Is the resistance less than 5 ohms?**

YES : Go to N11.

NO : REPAIR circuit GD116 (BK/VT)/(BK/RD) for an open. TEST the system for normal operation.

N11 CHECK CIRCUITS VH101 (WH/VT)/(RD) AND CH102 (YE/BU)/(BN) FOR A SHORT TO

GROUND

- Disconnect: HVAC Module - DATC C228b

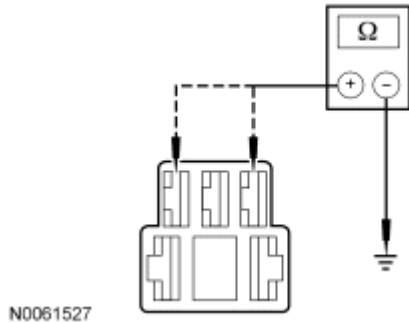


Fig. 92: Measuring Resistance Between Blower Motor Speed Control Module C271-1 & C271-3

Courtesy of FORD MOTOR CO.

- Measure the resistance between ground and:
 - blower motor speed control C271-1, circuit (RD), harness side.
 - blower motor speed control C271-3, circuit (BN), harness side.

- **Are the resistances greater than 10,000 ohms?**

YES : Go to N12.

NO : REPAIR circuit VH101 (WH/VT)/(RD) or CH102 (YE/BU)/(BN) for a short to ground.

TEST the system for normal operation.

N12 CHECK CIRCUITS (BU), VH101 (WH/VT)/(RD) AND CH102 (YE/BU)/(BN) FOR A SHORT TO VOLTAGE

- Key in ON position.

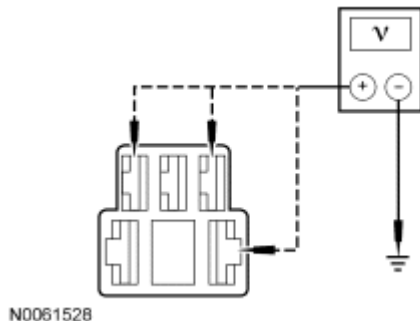


Fig. 93: Measuring Voltage

Courtesy of FORD MOTOR CO.

- Measure the voltage between ground and:
 - blower motor speed control C271-1, circuit (RD), harness side.
 - blower motor speed control C271-3, circuit (BN), harness side.

- blower motor speed control C271-5, circuit (BU), harness side.
- **Is any voltage present?**
YES : REPAIR circuit VH101 (WH/VT)/(RD), CH102 (YE/BU)/(BN) or (BU) for a short to voltage. TEST the system for normal operation.
NO : Go to N13.

N13 CHECK CIRCUITS VH101 (WH/VT)/(RD) AND CH102 (YE/BU)/(BN) FOR A SHORT TOGETHER

- Key in OFF position.

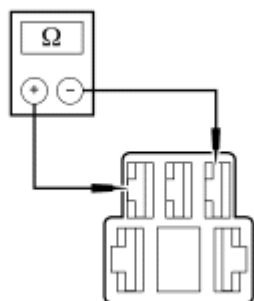


Fig. 94: Measuring Resistance Between Blower Motor Speed Control Module C271-1 & C271-3, Circuit (RD) & Circuit (BN)
 Courtesy of FORD MOTOR CO.

- Measure the resistance between blower motor speed control C271-1, circuit (RD), harness side and blower motor speed control module C271-3, circuit (BN), harness side.
- **Is the resistance greater than 10,000 ohms?**
YES : Go to N14.
NO : REPAIR circuit VH101 (WH/VT)/(RD) or CH102 (YE/BU)/(BN) for a short together. TEST the system for normal operation.

N14 CHECK CIRCUIT (BU) FOR AN OPEN

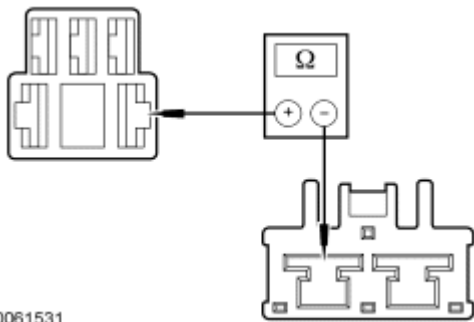


Fig. 95: Measuring Resistance Between Blower Motor C288-2 & C271-5, Circuit (BU)
 Courtesy of FORD MOTOR CO.

- Measure the resistance between blower motor C288-1, circuit (BU), harness side and blower motor

speed control C271-5, circuit (BU), harness side.

- **Is the resistance less than 5 ohms?**

YES : Go to N15.

NO : REPAIR circuit (BU) for an open. TEST the system for normal operation.

N15 CHECK CIRCUITS VH101 (WH/VT)/(RD) AND CH102 (YE/BU)/(BN) FOR AN OPEN

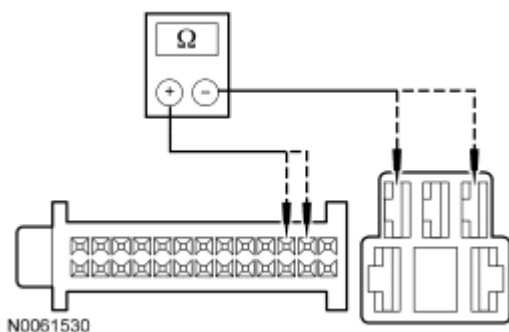


Fig. 96: Measuring Resistance Between Blower Motor Speed Control Module C271-1 & C271-3, EATC Module C228B-16 & C228B-15

Courtesy of FORD MOTOR CO.

- Measure the resistance between:
 - blower motor speed control C271-1, circuit (RD), harness side and HVAC module - DATC C228b-16, circuit VH101 (WH/VT), harness side.
 - blower motor speed control C271-3, circuit (BN), harness side and HVAC module - DATC C228b-15, circuit CH102 (YE/BU), harness side.

- **Are the resistances less than 5 ohms?**

YES : INSTALL a new blower motor speed control. REFER to **CLIMATE CONTROL** article. TEST the system for normal operation. If the condition returns, INSTALL a new HVAC module - DATC. REFER to **CLIMATE CONTROL** article. TEST the system for normal operation.

NO : REPAIR circuit VH101 (WH/VT)/(RD) or CH102 (YE/BU)/(BN) for an open. TEST the system for normal operation.

N16 CHECK THE MODULE CONNECTION

- Carry out the HVAC Module Cold Boot Process.
- Carry out the HVAC Module - Dual Automatic Temperature Control (DATC) On-Demand Self Test.
- Operate the system.
- **Does the concern return?**

YES : INSTALL a new HVAC module - DATC. REFER to **CLIMATE CONTROL** article. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test O: The Blower Motor Does Not Operate Correctly - DATC

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Automatic Climate Control System for schematic and connector information.

Normal Operation

Under normal operation, the blower motor relay coil receives a ground from the HVAC module - dual automatic temperature control (DATC) through circuit CH123 (VT/GN). The coil receives ignition voltage through circuit CBP45 (YE). Voltage is supplied to the relay switch contact through circuit CH402 (YE/GN)/(BK). When the relay coil is energized, voltage is delivered to the blower motor speed control through circuit CH402 (YE/GN)/(BK). Ground for the blower motor speed control is provided by circuit GD116 (BK/VT)/(BK/RD). The HVAC module - DATC provides a blower target speed signal to the blower motor speed control through circuit VH101 (WH/VT)/(RD) to control the blower speed. The actual blower speed signal is provided to the HVAC module - DATC through circuit CH102 (YE/BU)/(BN) from the blower motor speed control. Ground for the motor is provided through circuit (BU) from the blower motor speed control.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Blower motor speed control
- Blower motor relay
- HVAC module - DATC

PINPOINT TEST O: THE BLOWER MOTOR DOES NOT OPERATE CORRECTLY - HVAC - DATC

CAUTION: It is important to install relays in their correct position in the battery junction box (BJB). Installing a relay incorrectly may cause wiring shorts or damage to modules. While carrying out diagnostics on BJB relays, have only one BJB relay removed at a time. Failure to follow these instructions may result in damage to the vehicle circuitry or to control modules.

O1 VERIFY THE BLOWER MOTOR OPERATION

- Key in ON position.
- Press the panel button on the HVAC module - DATC. Adjust the blower motor setting to LO and then to HI.
- **Does the blower motor operate at any setting?**
YES : Go to O2.
NO : Go to **Pinpoint Test N**.

O2 CHECK THE BLOWER MOTOR SPEED TARGET POSITION (BLWR_T#) HVAC PID

- Enter the following diagnostic mode on the diagnostic tool: BLWR_T# HVAC PID
- Observe the BLWR_T# HVAC PID while increasing and decreasing the blower speed.
- **Does the BLWR_T# HVAC PID reading change in all speeds between 0% and 100%?**

YES : If the blower motor is always on, CARRY OUT the blower motor relay component test. Refer to appropriate COMPONENT TESTING article. If the relay tests OK, go to O7.

All other symptoms, go to O3.

NO : Go to O9.

O3 CHECK CIRCUITS (BU), VH101 (WH/VT)/(RD) AND CH102 (YE/BU)/(BN) FOR A SHORT TO VOLTAGE

- Disconnect: HVAC Module - DATC C228a
- Disconnect: Blower Motor Speed Control C271
- Disconnect: Blower Motor C288
- Key in ON position.

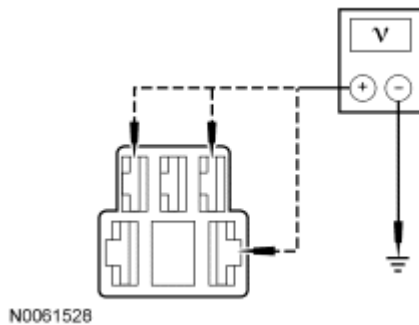


Fig. 97: Measuring Voltage
Courtesy of FORD MOTOR CO.

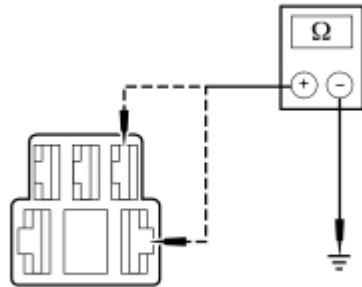
- Measure the voltage between ground and:
 - blower motor speed control C271-1, circuit (RD), harness side.
 - blower motor speed control C271-3, circuit (BN), harness side.
 - blower motor speed control C271-5, circuit (BU), harness side.
- **Is any voltage present?**

YES : REPAIR circuit VH101 (WH/VT)/(RD), CH102 (YE/BU)/(BN) or (BU) for a short to voltage. TEST the system for normal operation.

NO : Go to O4.

O4 CHECK CIRCUITS (BU) AND CH102 (YE/BU)/(BN) FOR A SHORT TO GROUND

- Key in OFF position.



N0061532

Fig. 98: Measuring Resistance Between Blower Motor Speed Control Module C271-5 & C271-3

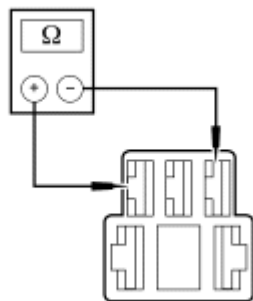
Courtesy of FORD MOTOR CO.

- Measure the resistance between ground and:
 - blower motor speed control C271-5, circuit (BU), harness side.
 - blower motor speed control C271-3, circuit (BN), harness side.
- **Are the resistances greater than 10,000 ohms?**

YES : Go to O5.

NO : REPAIR circuit (BU) or CH102 (YE/BU)/(BN) for a short to ground. TEST the system for normal operation.

O5 CHECK CIRCUITS VH101 (WH/VT)/(RD) AND CH102 (YE/BU)/(BN) FOR A SHORT TOGETHER



N0061529

Fig. 99: Measuring Resistance Between Blower Motor Speed Control Module C271-1 & C271-3, Circuit (RD) & Circuit (BN)

Courtesy of FORD MOTOR CO.

- Measure the resistance between blower motor speed control C271-1, circuit (RD), harness side and blower motor speed control C271-3, circuit (BN), harness side.
- **Is the resistance greater than 10,000 ohms?**

YES : Go to O6.

NO : REPAIR circuit VH101 (WH/VT)/(RD) or CH102 (YE/BU)/(BN) for a short together. TEST the system for normal operation.

O6 CHECK CIRCUIT CH102 (YE/BU)/(BN) FOR AN OPEN

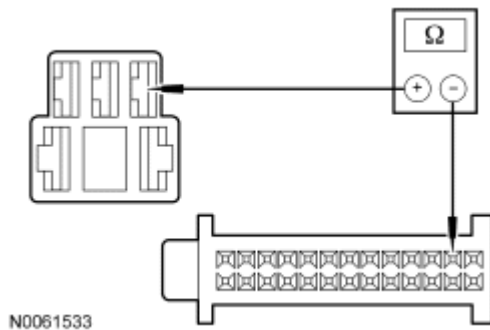


Fig. 100: Measuring Resistance Between Blower Motor Speed Control Module C271-3 & C228B-15, Circuit (BN) & CH102 (YE/BU)
Courtesy of FORD MOTOR CO.

- Measure the resistance between blower motor speed control C271-3, circuit (BN), harness side and HVAC module - DATC C228b-15, circuit CH102 (YE/BU), harness side.
- **Are the resistances less than 5 ohms?**

YES : INSTALL a new blower motor speed control. REFER to **CLIMATE CONTROL** article. TEST the system for normal operation. If the condition returns, INSTALL a new HVAC module - DATC. REFER to **CLIMATE CONTROL** article. TEST the system for normal operation.

NO : REPAIR circuit CH102 (YE/BU)/(BN) for an open. TEST the system for normal operation.

07 CHECK CIRCUIT CH402 (YE/GN)/(BK) FOR A SHORT TO VOLTAGE

- Key in OFF position.
- Disconnect: Blower Motor Relay
- Disconnect: Blower Motor C288
- Key in ON position.

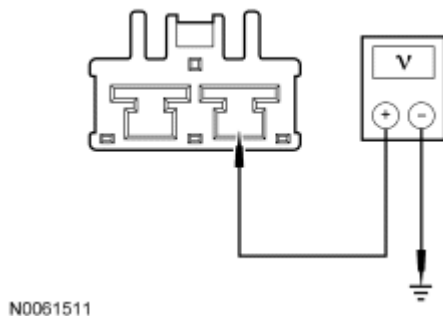


Fig. 101: Measuring Voltage
Courtesy of FORD MOTOR CO.

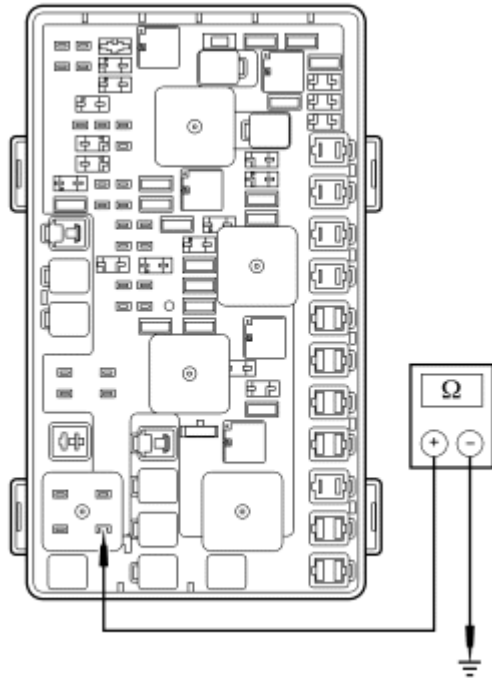
- Measure the voltage between ground and blower motor C288-2, circuit (BK), harness side.
- **Is any voltage present?**

YES : REPAIR circuit CH402 (YE/GN)/(BK) for a short to voltage. TEST the system for normal operation.

NO : Go to O8.

O8 CHECK CIRCUIT CH123 (VT/GN) FOR A SHORT TO GROUND

- Key in OFF position.
- Disconnect: HVAC Module - DATC C228a



N0061958

Fig. 102: Measuring Resistance Between Ground & Blower Motor Relay Socket, Circuit CH123 (VT/GN) & Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between ground and blower motor relay socket, circuit CH123 (VT/GN) and ground.
- **Is the resistance greater than 10,000 ohms?**

YES : Go to O9.

NO : REPAIR circuit CH123 (VT/GN) for a short to ground. TEST the system for normal operation.

O9 CHECK MODULE CONNECTION

- Carry out the HVAC Module Cold Boot Process.
- Carry out the HVAC Module - Dual Automatic Temperature Control (DATC) On-Demand Self Test.
- Operate the system.
- **Does the concern return?**

YES : INSTALL a new HVAC module - DATC. REFER to **CLIMATE CONTROL** article. TEST the system for normal operation.

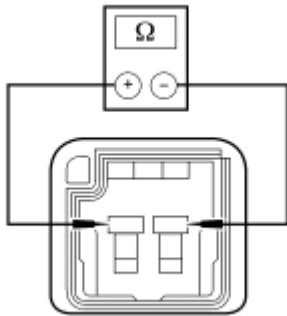
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NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Component Tests

Temperature Sensor - Evaporator Discharge



N0061538

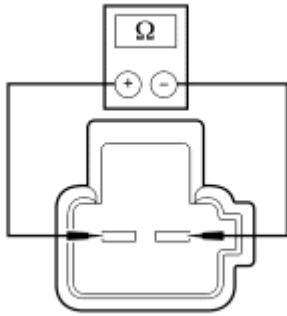
Fig. 103: Temperature Sensor - Evaporator Discharge
Courtesy of FORD MOTOR CO.

Ambient Temperature	Approximate Resistance
-10°C (14°F)	8,255 ohms
-5°C (23°F)	6,298 ohms
0°C (32°F)	4,852 ohms
5°C (41°F)	3,773 ohms
10°C (50°F)	2,961 ohms
15°C (59°F)	2,343 ohms
20°C (68°F)	1,869 ohms
25°C (77°F)	1,502 ohms
30°C (86°F)	1,216 ohms
35°C (95°F)	991 ohms
40°C (104°F)	813 ohms
45°C (113°F)	671 ohms
50°C (122°F)	557 ohms
55°C (131°F)	466 ohms
60°C (140°F)	391 ohms

In-Vehicle Temperature Sensor

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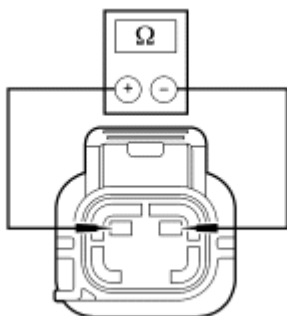


N0061540

Fig. 104: In-Vehicle Temperature Sensor
Courtesy of FORD MOTOR CO.

Ambient Temperature	Approximate Resistance
-10°C (14°F)	9,683 ohms
-5°C (23°F)	7,345 ohms
0°C (32°F)	5,628 ohms
5°C (41°F)	4,354 ohms
10°C (50°F)	3,399 ohms
15°C (59°F)	2,677 ohms
20°C (68°F)	2,125 ohms
25°C (77°F)	1,700 ohms
30°C (86°F)	1,370 ohms
35°C (95°F)	1,112 ohms
40°C (104°F)	909 ohms
45°C (113°F)	747 ohms
50°C (122°F)	618 ohms
55°C (131°F)	514 ohms
60°C (140°F)	430 ohms

Ambient Temperature Sensor



N0061541

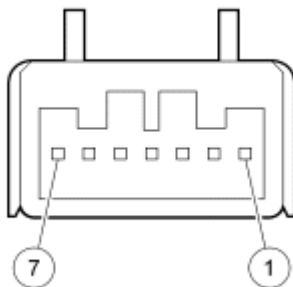
Fig. 105: Ambient Temperature Sensor
Courtesy of FORD MOTOR CO.

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Ambient Temperature	Approximate Resistance
-50°C (-58°F)	19,336 ohms
-40°C (-40°F)	10,732 ohms
-30°C (-22°F)	6,177 ohms
-20°C (-4°F)	3,672 ohms
-10°C (14°F)	2,248 ohms
0°C (32°F)	1,413 ohms
10°C (50°F)	915 ohms
20°C (68°F)	608 ohms
25°C (77°F)	500 ohms
30°C (86°F)	413 ohms
40°C (104°F)	287 ohms
50°C (122°F)	203 ohms
60°C (140°F)	147 ohms
70°C (158°F)	108 ohms
80°C (176°F)	80 ohms
90°C (194°F)	61 ohms
100°C (212°F)	47 ohms
110°C (231°F)	36 ohms
120°C (248°F)	28 ohms

Defrost/Panel/Floor Door and Blend Door Mode Actuator



N0061542

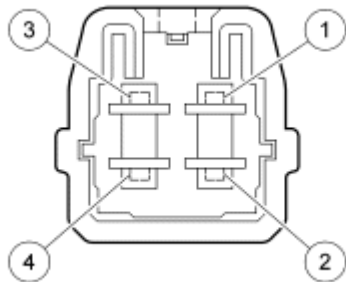
Fig. 106: Defrost/Panel/Floor Door and Blend Door Mode Actuator
Courtesy of FORD MOTOR CO.

Actuator Pins	Approx. Resistance
3 and 5	1.2K-4.8K ohms
1 and 5	1.2K-4.8K ohms
1 and 3	6K ohms

Blower Motor Resistor

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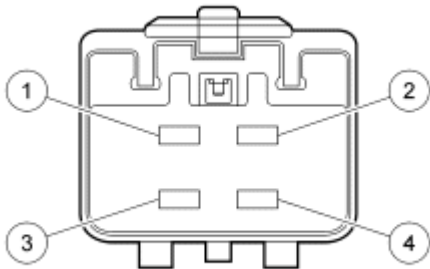


N0061539

Fig. 107: Blower Motor Resistor
Courtesy of FORD MOTOR CO.

Blower Motor Resistor Pins	Approx. Resistance
4 and 3	0.25-0.29 ohm
4 and 2	0.88-1.02 ohm
4 and 1	2.0-2.30 ohms

Blower Motor Switch - HVAC - Electronic Manual Temperature Control (EMTC)



N0061543

Fig. 108: Blower Motor Switch - EMTC
Courtesy of FORD MOTOR CO.

Blower Motor Switch Position	Continuity Between Pins
LOW	None
MED-LO	3 and 4
MED-HI	3 and 1
HI	3 and 2

Heater Core

NOTE: Testing of returned heater cores reveals that a large percentage of heater cores are good and did not require the installation of a new heater core. If a heater core leak is suspected, the heater core must be tested by following the plugged heater core component test before the heater core

pressure test. Carry out a system inspection by checking the heater system thoroughly as follows:

1. Inspect for evidence of coolant leakage at the heater hose to heater core attachments. A coolant leak in the heater hose could follow the heater core tube to the heater core and appear as a leak in the heater core.

NOTE: **Spring-type clamps are installed as original equipment. Installation and overtightening of non-specified clamps can cause leakage at the heater hose connection and damage the heater core.**

2. Check the integrity of the heater hose clamps.

Heater Core - Plugged

1. Check to see that the engine coolant is at the correct level.
2. Start the engine and turn on the heater.
3. When the engine coolant reaches operating temperature, feel the heater core inlet and outlet hoses to see if they are hot.
4. If the outlet only is not hot:
 - the heater core may have an air pocket.
 - the heater core may be plugged.
5. If the inlet only is not hot:
 - the thermostat may not be working correctly.

Heater Core - Pressure Test

Use the Pressure Test Kit to carry out the pressure test.

NOTE: **Due to space limitations, a bench test may be necessary for pressure testing.**

1. Drain the coolant from the cooling system.
2. Disconnect the heater hoses from the heater core.
3. Install a short piece of heater hose, approximately 101 mm (4 in) long on each heater core tube.
4. Fill the heater core and heater hoses with water and install the 2 plugs BT-7422-B and the adapter BT-7422-A from the Pressure Test Kit. Secure the heater hoses, plug and adapter with hose clamps.

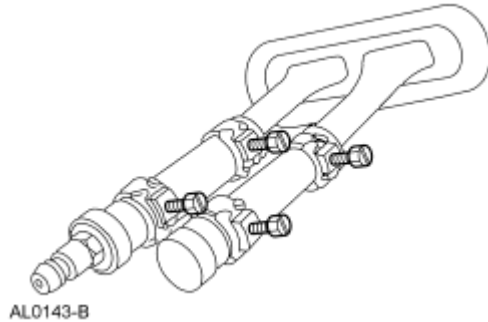


Fig. 109: Heater Core With Adapter & Plug
Courtesy of FORD MOTOR CO.

5. Attach the pump and gauge assembly from the Pressure Test Kit to the adapter.
6. Close the bleed valve at the base of the gauge. Pump 138 kPa (20 psi) of air pressure into the heater core.
7. Observe the pressure gauge for a minimum of 3 minutes.
8. If the pressure drops, check the heater hose connections to the core tubes for leaks. If the heater hoses do not leak, remove the heater core from the vehicle and carry out the bench test.

Heater Core - Bench Test

1. Remove the heater core from the vehicle.
2. Drain all of the coolant from the heater core.
3. Connect the 101 mm (4 in) test heater hoses with plug and adapter to the core tubes. Then connect the Pressure Test Kit to the adapter.
4. Apply 138 kPa (20 psi) of air pressure to the heater core. Submerge the heater core in water.
5. If a leak is observed, install a new heater core.

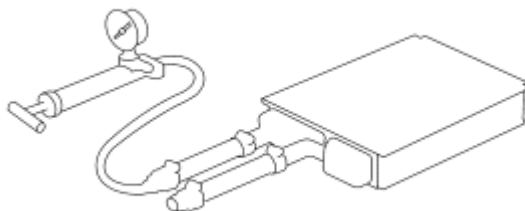


Fig. 110: Bench Testing Heater Core
Courtesy of FORD MOTOR CO.

Evaporator/Condenser Core - On-Vehicle Leak Test

1. Recover the refrigerant. Refer to **Air Conditioning (A/C) System Recovery, Evacuation and Charging**.

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NOTE: For evaporator core testing only.

2. Remove the thermostatic expansion valve (TXV) from the evaporator core.

NOTE: For condenser core testing only.

3. Disconnect the condenser-to-evaporator line spring lock coupling and the compressor-to-condenser discharge line middle fitting.

NOTE: For evaporator core testing only.

4. Connect the appropriate test fitting to the evaporator core TXV connection.

NOTE: For condenser core testing only.

5. Connect the appropriate test fittings to the condenser-to-evaporator line spring lock coupling, condenser side and the compressor-to-condenser discharge line middle fitting, condenser side.

NOTE: The automatic shut-off valves on some gauge set hoses do not open when connected to the test fittings. If available, use hoses without shut-off valves. If hoses with shut-off valves are used, make sure the valve opens when attached to the test fittings or install an adapter that will activate the valve. The test is not valid if the shut-off valve does not open.

6. Connect the red and blue hoses from the R-134a Manifold Gauge Set to the test fittings on the evaporator core or condenser core. Connect the yellow hose to a known good vacuum pump.
7. Open both gauge set valves and start the vacuum pump. Allow the vacuum pump to operate for a minimum of 45 minutes after the gauge set low pressure gauge indicates 101 kPa (30 in-Hg). The 45-minute evacuation is necessary to remove any refrigerant from oil left in the evaporator core or condenser core. If the refrigerant is not completely removed from the oil, outgassing will degrade the vacuum and appear as a refrigerant leak.
8. If the low pressure gauge reading will not drop to 101 kPa (30 in-Hg) when the valves on the gauge and manifold set are open and the vacuum pump is operating, close the gauge set valves and observe the low pressure gauge. If the pressure rises rapidly to zero, a large leak is indicated. Recheck the test fitting connections and gauge set connections before installing a new evaporator core or condenser core.
9. After evacuating for 45 minutes, close the gauge set valves and stop the vacuum pump. Observe the low pressure gauge; it should remain at the 101 kPa (30 in-Hg) mark.
 - If the low pressure gauge reading rises 34 or more kPa (10 or more in-Hg) of vacuum from the 101 kPa (30 in-Hg) position in 10 minutes, a leak is indicated.
 - If a very small leak is suspected, wait 30 minutes and observe the vacuum gauge.
 - If a small amount of vacuum is lost, operate the vacuum pump with gauge valves open for an additional 30 minutes to remove any remaining refrigerant from the oil in the evaporator core or condenser core. Then recheck for loss of vacuum.

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- If a very small leak is suspected, allow the system to sit overnight with vacuum applied and check for vacuum loss.

NOTE: For evaporator core testing only.

10. If the evaporator core does leak, as verified by the above procedure, install a new evaporator core.

NOTE: For condenser core testing only.

11. If the test indicates a leak, as verified by the above procedure, inspect and install new O-rings at the condenser line fittings as necessary. Retest the system. If the test again indicates a leak, install a new condenser core.

A/C Compressor - External Leak Test

1. Disconnect the evaporator outlet line spring lock coupling and the compressor-to-condenser discharge line middle fitting.
2. Connect the appropriate test fittings to the evaporator outlet line spring lock coupling, compressor side and the compressor-to-condenser discharge line middle fitting, compressor side.
3. Connect the high and low pressure lines of a manifold gauge set or a refrigerant recovery/recycling station such as the R-134a A/C service center to the corresponding fittings on the A/C Pressure Test Adapter.
4. Attach the center hose of a manifold gauge set to a refrigerant container standing in an upright position.
5. Open the low pressure gauge valve, the high pressure gauge valve and the valve on the refrigerant container to allow the refrigerant vapor to flow into the A/C compressor.
6. Using the Refrigerant Leak Detector, check for external compressor leaks.
7. If an external leak is found, inspect and install new as necessary the O-rings at the compressor line fittings. Retest the system. If the test again indicates a leak, install a new compressor.
8. When the leak test is complete, recover the refrigerant from the compressor.

GENERAL PROCEDURES

REFRIGERANT SYSTEM TESTS

Special Tools

Illustration	Tool Name	Tool Number
 ST2738-A	R-134a Refrigerant Center	176-00002 or equivalent

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 ST2739A	R-134a Refrigerant Center	023-00174 or equivalent
 ST2570-A	R-134a Manifold Gauge Set	023-00047 or equivalent

Procedure 1 - Ambient Temperature at or Below 38°C (100°F)

NOTE: The system performance can be evaluated and diagnosed by analysis of the compressor suction and discharge pressures. The following procedure is used to determine if the system is operating at normal pressures.

NOTE: The procedure varies depending on the ambient (shop) temperature. If the ambient temperature is 38°C (100°F) or lower, follow Procedure 1. If the ambient temperature is over 38°C (100°F), follow Procedure 2.

NOTE: If the A/C compressor cycles at any time during this test, refer to the diagnostic table.

1. Drive the vehicle or run the engine until it reaches normal operating temperature.
2. Connect a manifold gauge set or refrigerant service center with high-pressure and low-pressure gauges to the refrigerant system.
3. Set the climate controls.
 - If equipped with manual climate control, set the A/C controls for normal A/C-PANEL mode, full COOL temperature, FRESH air, HI blower. If the vehicle has a fresh air/recirc button, set it to FRESH. If the vehicle has an A/C switch or compressor on switch, set it to A/C ON.
 - If equipped with electronic automatic temperature control (EATC), set temperature to 15°C (60°F) (lowest possible temp setting) with the dual function disabled (if equipped). Manually set blower on HI. If the vehicle has a fresh air/recirc button, set it to FRESH. If the vehicle has an A/C switch or compressor on switch, set it to A/C ON.
4. Open all vehicle windows and leave the hood open for the test. Open the rear doors.
5. Confirm the compressor clutch is engaged and the engine cooling fan(s) are operating or engaged. Allow the vehicle to idle until the suction (low-side) and discharge (high-side) pressures are stable or fluctuate in a range that repeats.
6. Record the ambient (shop) temperature.
7. Record the discharge pressure. If the pressure is fluctuating, record the average value.
8. Determine if the discharge pressure falls within the normal operating limits using the Normal Refrigerant

Discharge Pressures chart.

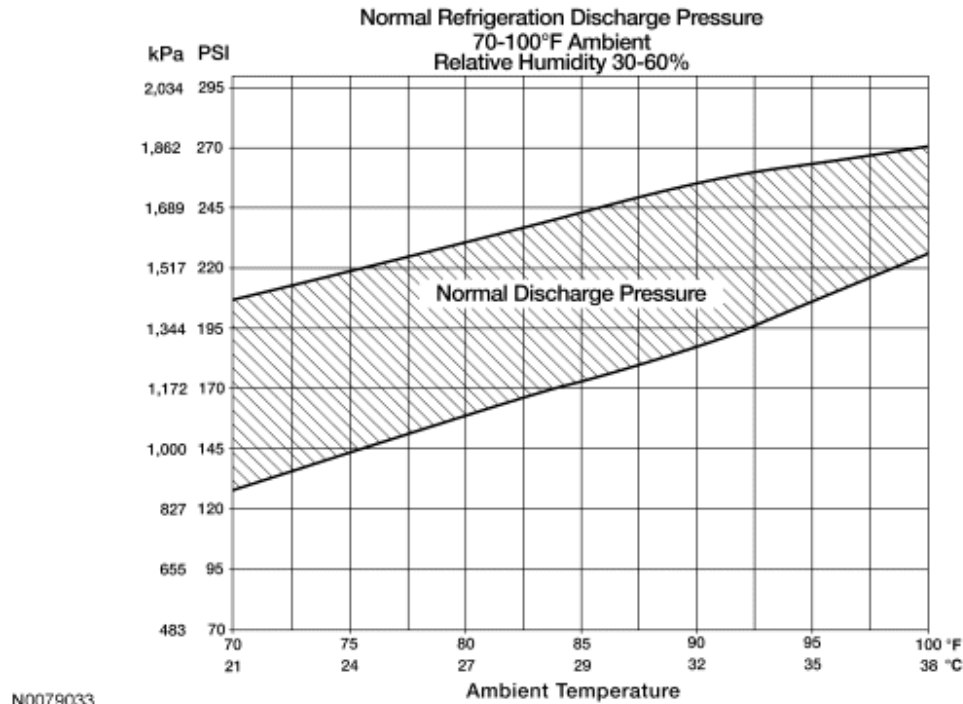


Fig. 111: Normal Refrigerant Discharge Pressures Chart - Relative Humidity 30-60%
Courtesy of FORD MOTOR CO.

9. Record the suction pressure. If the pressure is fluctuating, record the average value.
10. Determine if the suction pressure falls between normal operating limits using the Normal Refrigerant Suction Pressures chart.

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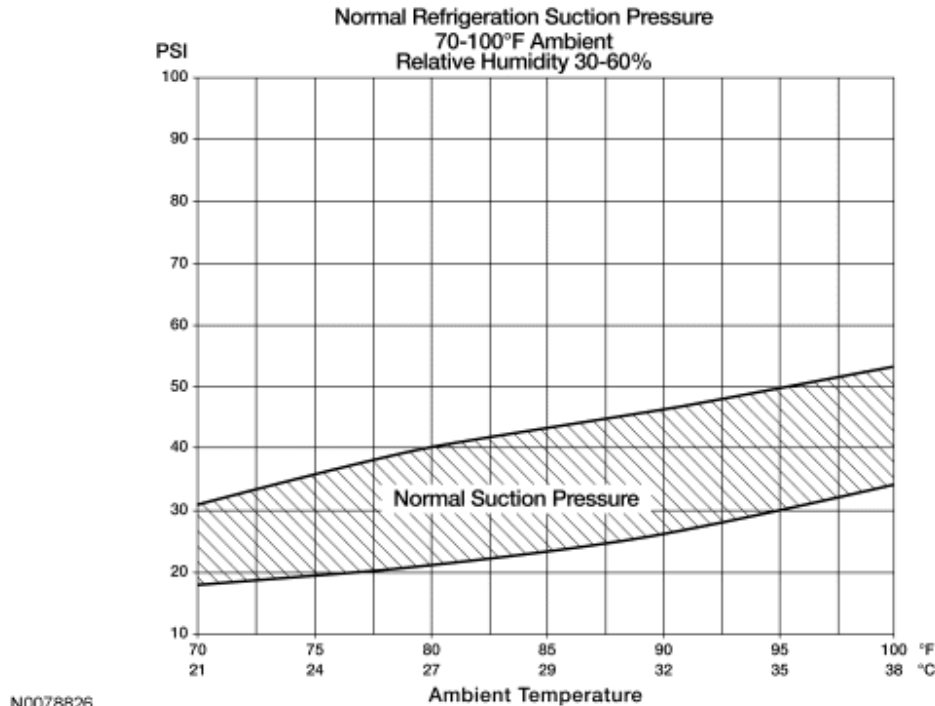


Fig. 112: Normal Refrigerant Suction Pressures Chart - Relative Humidity 30-60%
Courtesy of FORD MOTOR CO.

11. Proceed to the diagnostic table.

Procedure 2 - Ambient Temperature Above 38°C (100°F)

NOTE: The system performance can be evaluated and diagnosed by analysis of the compressor suction and discharge pressures. The following procedure is used to determine if the system is operating at normal pressures.

NOTE: The procedure varies depending on the ambient (shop) temperature. If the ambient temperature is 38°C (100°F) or lower, follow Procedure 1. If the ambient temperature is over 38°C (100°F), follow Procedure 2.

1. Drive the vehicle or run the engine until it reaches normal operating temperature.
2. Connect a manifold gauge set or refrigerant service center with high-pressure and low-pressure gauges to the refrigerant system.
3. Set the climate controls.
 - If equipped with manual climate control, set the A/C controls for normal A/C-PANEL mode, full COOL temperature, FRESH air, MED LO blower. If the vehicle has a fresh air/recirc button, set it to FRESH. If the vehicle has an A/C switch or compressor on switch, set it to A/C ON.
 - If equipped with EATC, set temperature to 15°C (60°F) (lowest possible temp setting). Manually set blower to MED LO (3 to 4 bars). If the vehicle has a fresh air/recirc button, set it to FRESH. If the vehicle has an A/C switch or compressor on switch, set it to A/C ON.

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4. Open all vehicle windows and leave the hood open for the test. Open the rear hatch and/or rear doors (if equipped).
5. Confirm the compressor clutch is engaged and the engine cooling fan(s) are operating or engaged. Allow the vehicle to idle until the suction (low-side) and discharge (high-side) pressures are stable or fluctuate in a range that repeats.
6. Record the ambient (shop) temperature.
7. Record the discharge pressure. If the pressure is fluctuating, record the average value.
8. Determine if the discharge pressure falls within the normal operating limits using the Normal Refrigerant Discharge Pressures chart.

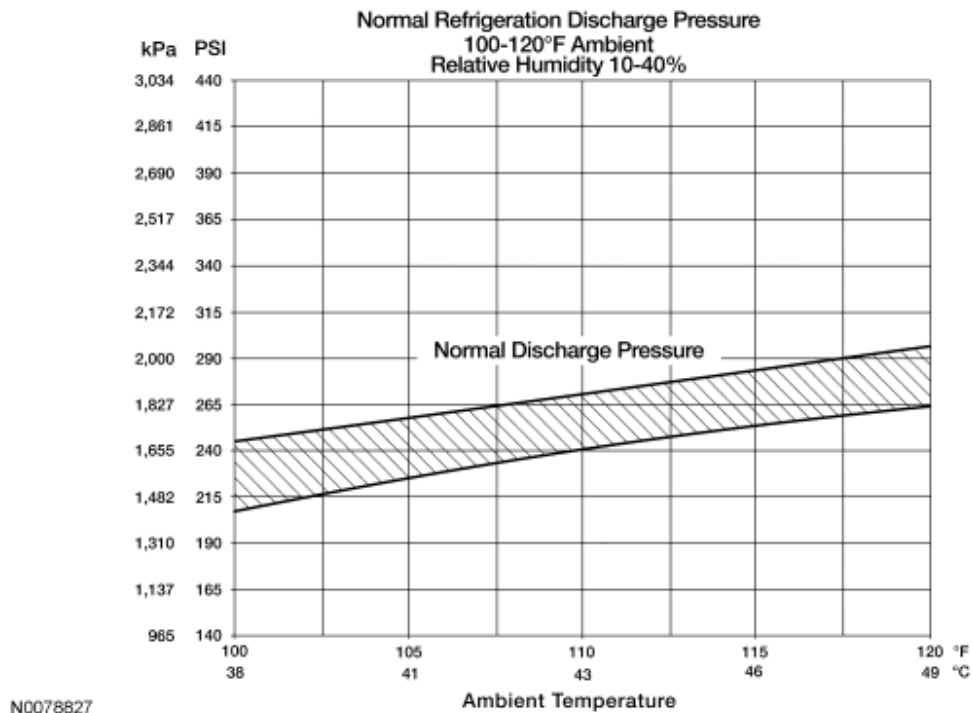


Fig. 113: Normal Refrigerant Discharge Pressures Chart - Relative Humidity 10-40%
Courtesy of FORD MOTOR CO.

9. Record the suction pressure. If the pressure is fluctuating, record the average value.
10. Determine if the suction pressure falls between normal operating limits using the Normal Refrigerant Suction Pressures chart.

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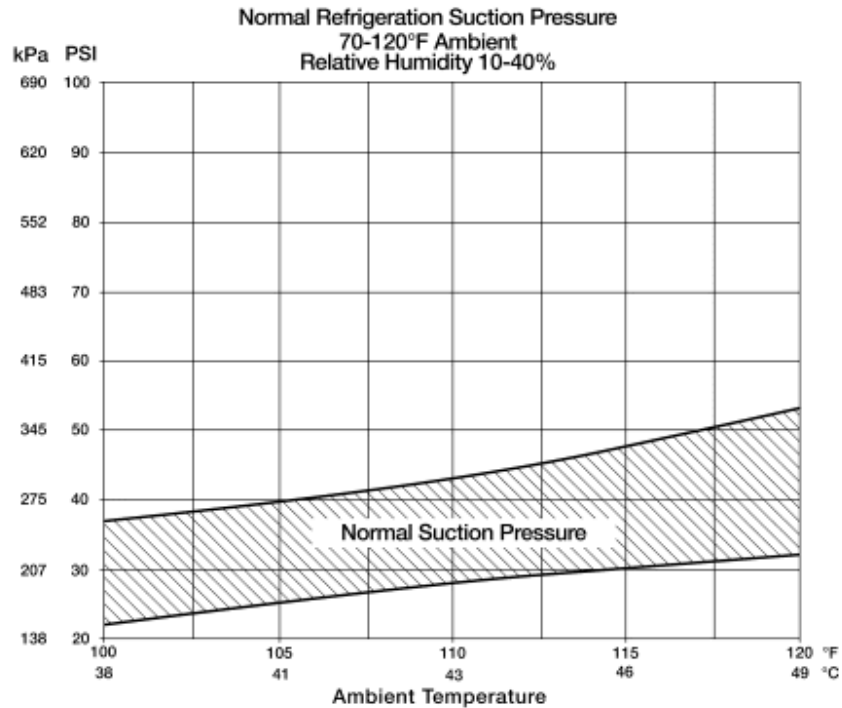


Fig. 114: Normal Refrigerant Suction Pressures Chart - Relative Humidity 10-40%
Courtesy of FORD MOTOR CO.

11. Proceed to the diagnostic table.

Diagnostic Table

NOTE: The following table is used to guide diagnosis of the refrigerant system if operating pressures are outside normal limits.

1. Refer to the chart below.

High (Discharge) Pressure	Low (Suction) Pressure	Component - Causes
High or Clutch Cycling	High	Condenser - inadequate airflow.
High	Normal to High	Engine - overheating.
Normal to High	Normal	Refrigerant overcharge - air in refrigerant.
		A/C suction

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Normal to Low	Normal to High	line - partially restricted or plugged ^a .
Normal to Low	Low or Clutch Cycling	Low refrigerant charge, A/C suction line - partially restricted or plugged ^b A/C cycling switch - sticking closed (if equipped).
Erratic Operation or Compressor Not Running		Evaporator discharge air temperature sensor or A/C pressure transducer - poor connection at A/C clutch connector. A/C electrical circuit erratic - see A/C Electrical Circuit Wiring Diagram. Refer to appropriate SYSTEM WIRING DIAGRAMS article.
Normal to Low	High	Compressor - low performance.

Additional Possible Cause Components Associated With

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Inadequate Compressor Operation
• Compressor Drive Belt - loose • Compressor Clutch - slipping • Clutch Coil Open - shorted, or loose mounting • Control Assembly Switch - dirty contacts or sticking open • Clutch Wiring Circuit - high resistance, open or blown fuse • Compressor Operation Interrupted by Engine Computer
Additional Possible Cause Components Associated With a Damaged Compressor
• Incorrect Clutch Air-gap • Refrigerant Leaks

^a Low pressure reading will be normal to high if pressure is taken at accumulator and if restriction is downstream of service access valve.

^b Low pressure reading will be low if pressure is taken near the compressor and restriction is upstream of service access valve.

FLUORESCENT DYE LEAK DETECTION

Special Tools

Illustration	Tool Name	Tool Number
 ST3073-A	Cordless/Rechargeable True UV LED Light ES	023-00182 or equivalent
 ST2650-A	R-134a Leak Detection Dye	164-R6060

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 ST2649-A	R-134a Loop/Add On Injector Kit-Set	219-00069 or equivalent
 ST2670-A	R-134a Manifold Gauge Set	023-00047 or equivalent
 ST2739A	R-134a Refrigerant Center	023-00174 or equivalent
 ST2738-A	R-134a Refrigerant Center	176-00002 or equivalent

Fluorescent Dye Detection

NOTE: Ford Motor Company vehicles are produced with R-134a fluorescent dye installed in the refrigerant system from the factory. The location of leaks can be pinpointed by the bright yellow-green glow of the fluorescent dye under a UV lamp. Since more than one leak can exist, make sure to inspect each component, line and fitting in the refrigerant system for a leak.

1. Check for leaks using a Rotunda-approved UV lamp.
 - Inspect all components, lines and fittings of the refrigerant system.
2. If a leak is found, recover the refrigerant. For additional information, refer to **Air Conditioning (A/C) System Recovery, Evacuation and Charging**.
3. Repair the refrigerant system leak(s).
4. Evacuate and charge the refrigerant system. For additional information, refer to **Air Conditioning (A/C) System Recovery, Evacuation and Charging**.
5. After the leak(s) is/are repaired, remove any traces of fluorescent dye with a general purpose oil solvent.
6. Verify the repair by running the vehicle for a short period of time and rechecking the area of the leak with a Rotunda-approved UV lamp.

Fluorescent Dye Injection - Using an R-134a Refrigerant Management Center and Dye/Lubricant Injector

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NOTE: Fluorescent refrigerant system dye is added to the refrigerant system at the factory to assist in refrigerant system leak diagnosis using a Rotunda-approved ultraviolet blacklight. It is not necessary to add additional dye to the refrigerant system before diagnosing leaks, even if a significant amount of refrigerant has been removed from the system. Additional refrigerant system dye should only be added if more than 50% of the refrigerant system lubricant capacity has been lost due to a fitting separation, hose rupture or other damage.

NOTE: Before using the dye/lubricant injector from the R-134a Loop/Add On Injector Kit-Set for the first time, refer to the manufacturer's instructions on evacuation of any non-condensable gases from the hoses.

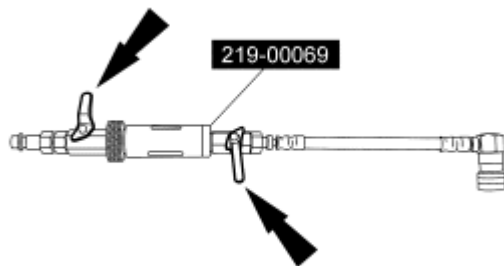
NOTE: Only connect the dye/lubricant injector when fluorescent dye is to be injected. The injector has a one-way check valve that will prevent refrigerant system recovery and evacuation.

NOTE: Refrigerant system pressure should be between 413-551 kPa (60-80 psi) at 24°C (75°F) with the engine off.

1. Connect an R-134a Refrigerant Center or a R-134a Manifold Gauge Set to the refrigerant system service port valves.

NOTE: The dye/lubricant injector is included as part of the 219-00069 R-134a Loop/Add On Injector Kit-Set.

2. Verify that the valves on the dye/lubricant injector are closed.



N0085274

Fig. 115: Verifying Valves On Dye/Lubricant Injector Are Closed
Courtesy of FORD MOTOR CO.

3. Fill the fluorescent dye injector reservoir with 7 ml (0.25 oz) of fluorescent dye.



Fig. 116: Filling Fluorescent Dye Injector Reservoir With 7 ml (0.25 oz.) Of Fluorescent Dye
Courtesy of FORD MOTOR CO.

4. Install the dye/lubricant injector between the low-pressure service gauge port valve and the R-134a Refrigerant Center or R-134a Manifold Gauge Set.
5. Open all valves and inject the fluorescent dye into the refrigerant system.
6. When fluorescent dye injection is complete, close all valves.
7. Recover the refrigerant from the dye/lubricant injector.
8. Remove the dye/lubricant injector from the low-pressure service gauge port valve and the R-134a Refrigerant Center or R-134a Manifold Gauge Set.

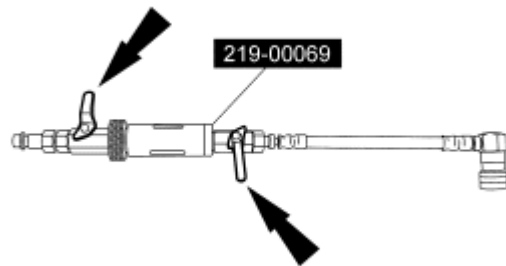
Fluorescent Dye Injection - Using an R-134a Loop/Add On Injector Kit-Set

NOTE: Fluorescent refrigerant system dye is added to the refrigerant system at the factory to assist in refrigerant system leak diagnosis using a Rotunda-approved ultraviolet blacklight. It is not necessary to add additional dye to the refrigerant system before diagnosing leaks, even if a significant amount of refrigerant has been removed from the system. Additional refrigerant system dye should only be added if more than 50% of the refrigerant system lubricant capacity has been lost due to a fitting separation, hose rupture or other damage.

NOTE: Before using the R-134a Loop/Add On Injector Kit-Set for the first time, refer to the equipment manufacturer's instructions on evacuation of non-condensable gases from the hoses.

NOTE: Refrigerant system pressure should be between 413-551 kPa (60-80 psi) at 24°C (75°F).

1. Verify that the valves on the deluxe injector loop kit are closed.



N0085274

Fig. 117: Verifying Valves On Dye/Lubricant Injector Are Closed
Courtesy of FORD MOTOR CO.

2. Fill the R-134a Loop/Add On Injector Kit-Set reservoir with 7 ml (0.25 oz) of fluorescent dye.



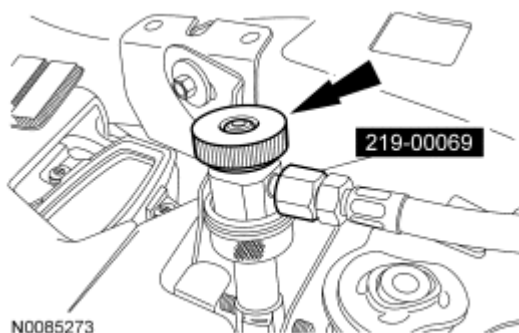
N0085275

Fig. 118: Filling Fluorescent Dye Injector Reservoir With 7 ml (0.25 oz.) Of Fluorescent Dye
Courtesy of FORD MOTOR CO.

3. Install the R-134a Loop/Add On Injector Kit-Set between the high-pressure and low-pressure service gauge port valves.

NOTE: Make sure all tools and hoses are clear of the engine cooling fan and drive belt before starting the engine.

4. Start the engine.
5. Open the high-pressure service valve.



N0085273

Fig. 119: Installing R-134A Loop/Add On Injector Kit-Set Between High-Pressure & Low-Pressure Service Gauge Port Valves
Courtesy of FORD MOTOR CO.

6. Open all valves and inject the fluorescent dye into the refrigerant system.

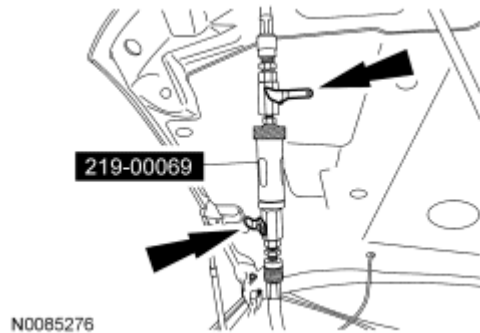


Fig. 120: Opening All Valves & Inject Fluorescent Dye Into Refrigerant System
Courtesy of FORD MOTOR CO.

7. Close the high-pressure service valve to allow the pressure inside the R-134a Loop/Add On Injector Kit-Set to equalize with the suction side of the refrigerant system.

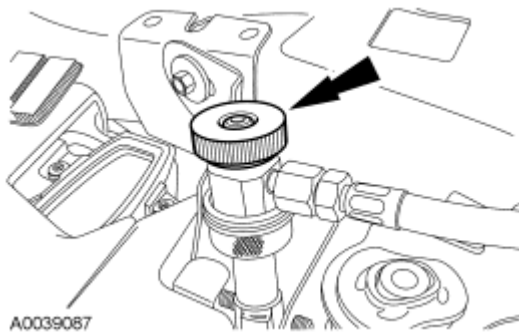


Fig. 121: Identifying High-Pressure Service Valve
Courtesy of FORD MOTOR CO.

NOTE: Close the valves on the R-134a Loop/Add On Injector Kit-Set while the A/C compressor is operating.

8. Close the valves on the R-134a Loop/Add On Injector Kit-Set.

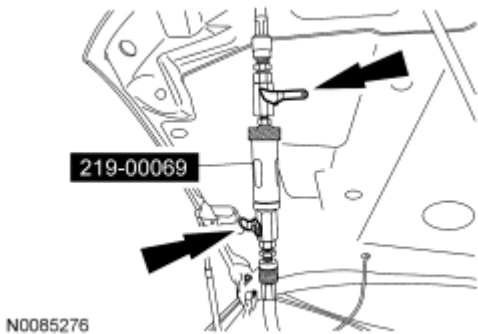


Fig. 122: Closing Valves On R-134A Loop/Add On Injector Kit-Set
Courtesy of FORD MOTOR CO.

NOTE: Leave all valves on the R-134a Loop/Add On Injector Kit-Set closed when not in use.

9. Disconnect the high-pressure and low-pressure service valves and remove the R-134a Loop/Add On Injector Kit-Set from the vehicle.

ELECTRONIC LEAK DETECTION

Special Tools

Illustration	Tool Name	Tool Number
 ST3072-A	Heated Pentode Halogen Leak Detector	023-00178 or equivalent

NOTE: Good ventilation is necessary in the area where electronic A/C leak testing is to be carried out. If the surrounding air is contaminated with refrigerant gas, the Heated Pentode Halogen Leak Detector will indicate this gas all the time. Odors from other chemicals such as antifreeze, diesel fuel, disc brake cleaner or other cleaning solvents can cause the same problem. Using a fan to ventilate the area to be tested before proceeding with the leak detection procedure is helpful in removing small traces of contamination from the air, but the fan should be turned off during actual testing.

NOTE: R-134a is heavier than air, and will tend to move downward from the source of the leak if present. It is possible that a leak may not be detected if the leak detector tip is held above the leaking fitting, line or component. Always be sure to thoroughly leak test below the fitting, line or component for the presence of R-134a as well as leak testing above and around.

NOTE: The system pressure should be between 413-551 kPa (60-80 psi) at 24°C

(75°F) with the engine off.

1. Leak test the refrigerant system using the leak detector. Follow the instructions included with the leak detector for handling and operation techniques.

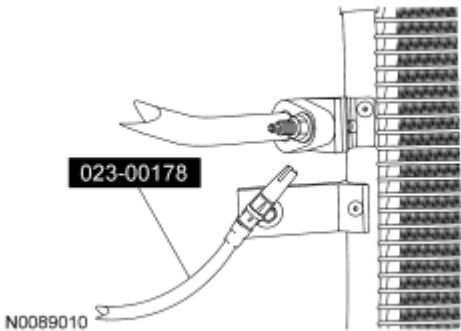


Fig. 123: Leak Test Refrigerant System Using Leak Detector
Courtesy of FORD MOTOR CO.

2. If a leak is found, recover the refrigerant. For additional information, refer to **Air Conditioning (A/C) System Recovery, Evacuation and Charging**.
 - Repair the system.
 - Test the system for normal operation.

AIR CONDITIONING (A/C) SYSTEM RECOVERY, EVACUATION AND CHARGING

Special Tools

Illustration	Tool Name	Tool Number
A line drawing of a 1.2 CFM vacuum pump, model ST1685-A. It is a compact, cylindrical unit with a handle on top and a hose connection on the side. ST1685-A	1.2 CFM Vacuum Pump	023-00162 or equivalent
A line drawing of a 4.0 CFM vacuum pump, model ST1686-A. It is a larger, rectangular unit with a handle on top and a hose connection on the side. ST1686-A	4.0 CFM Vacuum Pump	023-00163 or equivalent
A line drawing of an automatic refrigerant charging meter, model ST2742-A. It is a small, rectangular device with a circular dial and a handle. ST2742-A	Automatic Refrigerant Charging Meter	023-00155 or equivalent

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 ST2670-A	R-134a Manifold Gauge Set	023-00047 or equivalent
 ST3079A	R-134a Refrigerant Management Machine (SAE J-2788 Compliant)	023-00181 or equivalent
 ST3081-A	R-134a Refrigerant Management Machine (SAE J-2788 Compliant)	199-00067 or equivalent
 ST3080-A	R-134a Refrigerant Management Machine (SAE J-2788 Compliant)	265-00012 or equivalent

Material

Item	Specification
PAG Refrigerant Compressor Oil (R-134a Systems) YN-12-D	WSH-M1C231-B

Refrigerant System Recovery

NOTE: An air conditioning (A/C) refrigerant analyzer must be used before the recovery of any vehicle's A/C refrigerant. Failure to do so puts the shop's bulk refrigerant at risk of contamination. If the vehicle's A/C refrigerant is contaminated, refer the customer to the service facility that carried out the last A/C service. If the customer wishes to pay the additional cost, use the A/C recovery equipment that is designated for recovering contaminated A/C refrigerant. All contaminated A/C refrigerant must be disposed of as hazardous waste. For all equipment, follow the equipment manufacturer procedures and instructions.

NOTE: Ford Motor Company recommends use of an R-134a Refrigerant Management Machine to carry out recovery. If an R-134a Refrigerant Management Machine is not available, refrigerant system recovery may be accomplished using a separate recovery station.

NOTE: Leaks in refrigerant system service equipment, hoses or gauges can cause a

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leak in vacuum that may be misinterpreted as a problem with the vehicle's refrigerant system. It is necessary to leak-test all refrigerant system service equipment, hoses and gauges on a weekly basis to verify that no leaks are present.

1. Prior to recovering, the purity of the refrigerant must be verified. For additional information, refer to **Refrigerant Identification Testing**.
2. Connect an R-134a Refrigerant Management Machine to the low- and high-pressure service gauge port valves following the operating instructions provided by the equipment manufacturer.
3. Recover the refrigerant from the system following the operating instructions provided by the equipment manufacturer. Note the amount of oil removed during the refrigerant recovery (if any). Add that same amount back into the system once repairs are complete.
4. Once the R-134a Refrigerant Center has recovered the refrigerant, switch OFF the power supply.
5. Allow the system to set for about 2 minutes, and observe the system vacuum reading. If the vacuum is not lost, disconnect the recovery equipment.
6. If the system does lose vacuum, repeat Steps 3 through 5 until the vacuum level remains stable for 2 minutes.
7. Carry out the required repairs.

Refrigerant System Evacuation Using an R-134a Refrigerant Management Machine

NOTE: Ford Motor Company recommends use of an R-134a Refrigerant Management Machine to carry out recovery, evacuation and charging of the refrigerant system. If an R-134a Refrigerant Management Machine is not available, evacuation may be accomplished using a separate Vacuum Pump and R-134a Manifold Gauge Set.

NOTE: Leaks in refrigerant system service equipment, hoses or gauges can cause a leak in vacuum that may be misinterpreted as a problem with the vehicle's refrigerant system. It is necessary to leak-test all refrigerant system service equipment, hoses and gauges on a weekly basis to verify that no leaks are present.

1. Connect an R-134a Refrigerant Management Machine to the low- and high-pressure service gauge port valves following the operating instructions provided by the equipment manufacturer.
2. Evacuate the system until the low-pressure gauge reads at least 99.4 kPa (29.5 in-Hg) of vacuum and as close to 101.1 kPa (30 in-Hg) as possible. Continue to operate the Vacuum Pump for a minimum of 45 minutes.
3. Turn OFF the Vacuum Pump. Observe the low-pressure gauge for 5 minutes to make sure that the system vacuum is held. If vacuum is not held for 5 minutes, leak test the system, repair the leak and evacuate the system again.

Refrigerant System Evacuation Using an R-134a Manifold Gauge Set

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NOTE: Ford Motor Company recommends use of an R-134a Refrigerant Management Machine to carry out evacuation of the refrigerant system. If an R-134a Refrigerant Management Machine is not available, refrigerant system evacuation may be accomplished using a separate Vacuum Pump and R-134a Manifold Gauge Set.

NOTE: Leaks in refrigerant system service equipment, hoses or gauges can cause a leak in vacuum that may be misinterpreted as a problem with the vehicle's refrigerant system. It is necessary to leak-test all refrigerant system service equipment, hoses and gauges on a weekly basis to verify that no leaks are present.

1. Connect the R-134a Manifold Gauge Set to the low-side and high-side service gauge port valves.
2. Connect the center (yellow) hose from the R-134a Manifold Gauge Set to the suction port on the Vacuum Pump.
3. Open all valves on the R-134a Manifold Gauge Set and both service gauge port valves.
4. Turn on the Vacuum Pump and evacuate the system until the low-pressure gauge reads at least 99.4 kPa (29.5 in-Hg) of vacuum and as close to 101.1 kPa (30 in-Hg) as possible. Continue to operate the Vacuum Pump for a minimum of 45 minutes.
5. Close the high-side and low-side valves on the R-134a Manifold Gauge Set (not the service gauge port valves) and turn OFF the Vacuum Pump.
6. Observe the low-pressure gauge for 5 minutes to make sure that the system vacuum is held. If vacuum is not held for 5 minutes, leak test the system, repair the leak and evacuate the system again.

Refrigerant System Charging Using an R-134a Refrigerant Management Machine

NOTE: Ford Motor Company recommends use of an R-134a Refrigerant Management Machine to carry out charging of the refrigerant system. If an R-134a Refrigerant Management Machine is not available, refrigerant system charging may be accomplished using a separate Automatic Refrigerant Charging Meter and R-134a Manifold Gauge Set.

NOTE: Leaks in refrigerant system service equipment, hoses or gauges can cause a leak that may be misinterpreted as a problem with the vehicle's refrigerant system. It is necessary to leak-test all refrigerant system service equipment, hoses and gauges on a weekly basis to verify that no leaks are present.

1. Lubricate the refrigerant system with the correct amount of clean PAG oil. For additional information, refer to **Refrigerant Oil Adding**.
2. Connect an R-134a Refrigerant Management Machine to the low-side and high-side service gauge port valves following the operating instructions provided by the equipment manufacturer.
3. Set the refrigerant charge amount, and charge the refrigerant system following the instructions provided by the equipment manufacturer.

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Refrigerant System Charging Using an R-134a Manifold Gauge Set and Automatic Refrigerant Charging Motor

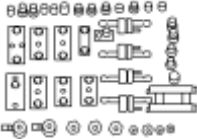
NOTE: Ford Motor Company recommends use of an R-134a Refrigerant Management Machine to carry out charging of the refrigerant system. If an R-134a Refrigerant Management Machine is not available, refrigerant system charging may be accomplished using a separate Automatic Refrigerant Charging Meter and R-134a Manifold Gauge Set.

NOTE: Leaks in refrigerant system service equipment, hoses or gauges can cause a leak that may be misinterpreted as a problem with the vehicle's refrigerant system. It is necessary to leak-test all refrigerant system service equipment, hoses and gauges on a weekly basis to verify that no leaks are present.

1. Lubricate the refrigerant system with the correct amount of clean PAG oil. For additional information, refer to **Refrigerant Oil Adding**.
2. Assemble the R-134a Manifold Gauge Set, Automatic Refrigerant Charging Meter and R-134a supply tank following the Automatic Refrigerant Charging Meter operating instructions.
3. Charge the refrigerant system following the Automatic Refrigerant Charging Meter operating instructions.
4. If the refrigerant flow stops before the refrigerant charge is complete, start the engine, select MAX A/C operation and allow the refrigerant charge to complete.

AIR CONDITIONING (A/C) SYSTEM FLUSHING

Special Tools

Illustration	Tool Name	Tool Number
 ST2826-A	A/C Flush Adapter Kit	219-00074 or equivalent
 ST2469-A	A/C Flush and Purge Fitting Kit	219-00024 or equivalent
 ST2466-A	A/C Flush and Purge Machine	219-00022 (part of 219-00023) or equivalent

Material

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Item	Specification
A/C System Flushing Solvent YN-23	-
PAG Refrigerant Compressor Oil (R-134a Systems) YN-12-D	WSH-M1C231-B

WARNING: Take the following precautions when repairing an air conditioning system containing R-134a:

- Always wear safety goggles.
- Avoid contact with liquid refrigerant R-134a. R-134a vaporizes at approximately -25°C (-13°F) under atmospheric pressure and will freeze skin tissue.
- Never allow refrigerant R-134a gas to escape in quantity in an occupied space. It will displace the oxygen needed to support life.
- Never use a torch in an atmosphere containing R-134a gas. R-134a is non-toxic at all normal conditions, but it decomposes when exposed to high temperatures such as a torch flame. During decomposition it releases irritating and toxic gasses (as described in the MSDS sheet from the manufacturer). Decomposition products are hydrofluoric acid, carbon dioxide and water.

Failure to follow these instructions may result in serious personal injury.

CAUTION: An air conditioning (A/C) refrigerant analyzer must be used before the recovery of any vehicle's A/C refrigerant. Failure to do so puts the shop's bulk refrigerant at risk of contamination. If the vehicle's A/C refrigerant is contaminated, refer the customer to the service facility that carried out the last A/C service. If the customer wishes to pay the additional cost, use the A/C recovery equipment that is designated for recovering contaminated A/C refrigerant. All contaminated A/C refrigerant must be disposed of as hazardous waste. For all equipment, follow the equipment manufacturer procedures and instructions.

CAUTION: Receiver/drier desiccant bag, thermostatic expansion valve (TXV), and hoses with mufflers should be removed when flushing the air conditioning (A/C) system. Internal plumbing of these devices makes it impossible to correctly remove any residual-flushing agent. These components are typically discarded after A/C system contamination. Hoses without mufflers can normally be reused unless they are clogged with foreign material. The 3.785L (1 gal) of A/C System Flushing Solvent YN-23 and FL1-A filter used in A/C Flush and Purge Machine 219-00022 are intended for use on one vehicle only. They may be used to flush both the A/C condenser core and the A/C evaporator core on an individual vehicle, but under no circumstances should they be used on more than one vehicle.

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CAUTION: Only the A/C Flush and Purge Machine 219-00022 and A/C Systems Flushing Solvent, is approved for use on Ford vehicles. No other flushing device or solvent is approved for flushing heat exchangers (air conditioning [A/C] condenser, A/C evaporator). Use of any other flusher or solvent may cause damage to the A/C system and the flushing unit.

NOTE: Prior to using the A/C Flush and Purge Machine 219-00022 for the first time, review the operating instructions.

NOTE: Ford Motor Company has approved a procedure to provide technicians with a non-CFC method of flushing contaminated A/C system heat exchangers. The procedure allows the specific components to be cleaned and flushed. The types of contamination flushed include particle matter that results from A/C compressor or desiccant failure and gummy residue that can form when refrigerant oil is overheated during A/C compressor seizure. The flushing process is a 2-step procedure that involves the use of an A/C Flush and Purge Machine 219-00022 to:

- circulate the flushing solvent through the heat exchanger in the reverse direction of normal refrigerant flow (back-flushing). Particulate matter picked up during flushing is filtered from the returning solvent before the solvent is returned to the reservoir for continued circulation.
- remove the flushing solvent from the heat exchanger. In this step of the procedure, pressurized air 621-862 kPa (90-125 psi) is used to push and evaporate any remaining flush solvent from the heat exchanger.

1. Remove the desiccant bag. For additional information, refer to **CLIMATE CONTROL** article.
2. Disconnect the compressor-to-condenser discharge line middle fitting and remove the condenser-side condenser-to-evaporator line.
3. Disconnect the condenser-to-evaporator line spring lock coupling.
4. Remove the condenser-to-evaporator line bracket bolt and remove the condenser side condenser-to-evaporator line.
5. With the condenser and lines out of the vehicle, reconnect the A/C lines to the condenser inlet and outlet fittings.
6. Connect the A/C Flush and Purge Machine to the A/C line fittings using the correct adapters from the A/C Flush and Purge Fitting Kit.

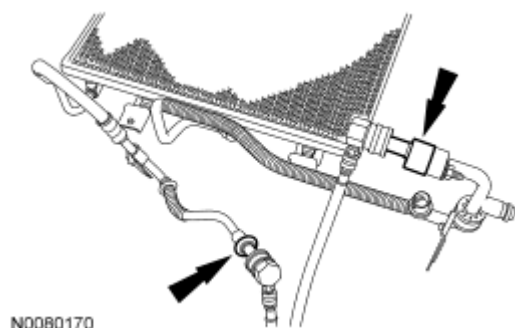


Fig. 124: Identifying A/C line fittings
Courtesy of FORD MOTOR CO.

NOTE: The condenser core must be flushed with the desiccant bag removed. Do not install a new desiccant bag at this time.

7. With the desiccant bag removed, reinstall the receiver/drier plug into the condenser core.
8. Flush the condenser core for a minimum of 30 minutes.
9. Turn the A/C Flush and Purge machine to the PURGE mode and run for a minimum of 20 minutes.
10. Disconnect the A/C Flush and Purge machine and adapters from the condenser core.
11. Disconnect the A/C lines from the condenser core.
12. Install the condenser-to-evaporator line and the condenser-to-evaporator line bracket bolt.
13. Connect the condenser-to-evaporator line spring lock coupling.
14. Install the compressor-to-condenser discharge line and connect the compressor-to-condenser discharge line middle fitting.
15. Remove the receiver/drier plug.

NOTE: Do not charge the refrigerant system after the condenser core is installed.

16. Install a new desiccant bag and reinstall the condenser core. For additional information, refer to **CLIMATE CONTROL** article.
17. Remove the front thermostatic expansion valve (TXV). For additional information, refer to **CLIMATE CONTROL** article.
18. Connect the A/C Flush and Purge Machine to the evaporator core fitting using the correct adapter from the A/C Flush Adapter Kit.
19. Flush the evaporator core for a minimum of 20 minutes.
20. Turn the A/C Flush and Purge Machine to the PURGE mode and run for a minimum of 20 minutes.
21. Disconnect the A/C Flush and Purge Machine and adapter from the evaporator core fitting.
22. Install a new front TXV. For additional information, refer to **CLIMATE CONTROL** article.

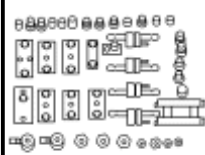
NOTE: It is necessary to evacuate the refrigerant system for a minimum of one hour to evaporate any trace amounts of flush solvent that may remain in

the system.

- 23. Evacuate the refrigerant system for one hour. For additional information, refer to **Air Conditioning (A/C) System Recovery, Evacuation and Charging**.
- 24. If a new A/C compressor is not to be installed, lubricate the refrigerant system with the correct amount of clean PAG oil. For additional information, refer to **Refrigerant Oil Adding**.
- 25. If a new A/C compressor is not to be installed, evacuate, leak test and charge the refrigerant system. For additional information, refer to **Air Conditioning (A/C) System Recovery, Evacuation and Charging**.

REFRIGERANT SYSTEM FILTERING FOLLOWING AIR CONDITIONING (A/C) COMPRESSOR INSTALLATION

Special Tools

Illustration	Tool Name	Tool Number
 ST2469-A	A/C Flush and Purge Fitting Kit	219-00024 or equivalent

CAUTION: On vehicles being serviced for an internal compressor or desiccant failure, a new thermostatic expansion valve (TXV) and any hoses containing mufflers must be installed prior to filtering the air conditioning (A/C) system. Internal plumbing of these devices makes it impossible to correctly remove any foreign material/debris. These components are typically discarded after A/C system contamination. Hoses without mufflers can normally be reused unless they are clogged with foreign material. The F8VZ-19E773-AB filter is intended for use on one vehicle only.

- 1. Remove the desiccant bag. For additional information, refer to **CLIMATE CONTROL** article.

NOTE: The refrigerant system must be filtered with the desiccant bag removed. Do not install a new desiccant bag at this time.

- 2. With the desiccant bag removed, reinstall the receiver/drier plug into the condenser core.

NOTE: Do not install the front bumper cover or evacuate and charge the refrigerant system at this time.

- 3. Install the condenser core back into the vehicle. For additional information, refer to **CLIMATE CONTROL** article.
- 4. Disconnect the condenser-to-evaporator line spring lock coupling.

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- Discard the O-ring seals.

NOTE: Use flexible refrigerant hose of 17,238 kPa (2,500 psi) burst rating.

NOTE: Orient the filter inlet toward the A/C condenser core.

5. Using the correct adapters from the A/C Flush and Purge Fitting Kit and suitable A/C service hoses, install the pancake filter between the two halves of the condenser-to-evaporator line spring lock coupling.
6. Lubricate the refrigerant system with the correct amount of clean PAG oil. For additional information, refer to **Refrigerant Oil Adding**.
7. Evacuate and charge the refrigerant system. For additional information, refer to **Air Conditioning (A/C) System Recovery, Evacuation and Charging**.
8. Provide adequate airflow to the front of the vehicle (with a fan, if necessary). Start the engine and allow it to idle briefly. Select A/C operation and set the blower speed to HI. Verify that the A/C is operating correctly.
9. Gradually bring the engine up to 1,200 RPM by running it at lower RPM for short periods (first at 800 RPM, then at 1,000 RPM). Set the engine at 1,200 RPM and run it for one hour with the A/C system operating.
10. Stop the engine.
11. Recover the refrigerant. For additional information, refer to **Air Conditioning (A/C) System Recovery, Evacuation and Charging**.
12. Remove the pancake filter, hoses and adapters from the vehicle.
13. Install new O-ring seals and connect the condenser-to-evaporator line spring lock coupling.
14. Remove the condenser core. For additional information, refer to **CLIMATE CONTROL** article.
15. Remove the receiver/drier plug and install a new desiccant bag.

NOTE: Lubricate the receiver/drier plug O-ring seal using residual PAG oil from inside the integrated receiver/drier.

16. Install the new receiver/drier plug included in the desiccant bag service kit.
17. Install the condenser core. For additional information, refer to **CLIMATE CONTROL** article.
18. Evacuate, leak test and charge the refrigerant system. For additional information refer to **Air Conditioning (A/C) System Recovery, Evacuation and Charging**.

REFRIGERANT OIL ADDING

Special Tools

Illustration	Tool Name	Tool Number
	Dye/Lubricant Injector	164-R0775

 ST2740A		
 ST2737-A	Add-On Module Kit	219-00069

Material

Item	Specification
PAG Refrigerant Compressor Oil (R-134a Systems) YN-12-D	WSH-M1C231-B

Refrigerant Oil Adding

CAUTION: During normal air conditioning (A/C) operation, oil is circulated through the system with the refrigerant, and a small amount is retained in each component. If certain components of the system are removed, some of the refrigerant oil will go with the component. To maintain the original total oil charge, it is necessary to compensate for the oil lost by adding oil to the system with the new part.

1. Refer to the chart below for refrigerant oil adding amounts and methods of installation.

Component	PAG Oil Amount	Method of Adding
A/C Compressor	See heading below	Add directly to A/C compressor before installation
Receiver/Drier Desiccant Cartridge	60 ml (2 fl oz) added to the amount collected during refrigerant recovery	Inject to low-side service port during system charging

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Evaporator Core	45 ml (1.5 fl oz) added to the amount collected during refrigerant recovery	Add directly to evaporator core inlet tube or inject to low-side service port during system charging
Condenser Core	60 ml (2 fl oz) added to the amount collected during refrigerant recovery	Add directly to condenser core inlet or inject to low-side service port during system charging
Evaporator Core Orifice or Thermostatic Expansion Valve	The amount collected during refrigerant recovery	Inject to low-side service port during system charging
A/C Pressure Relief Valve	60 ml (2 fl oz) added to the amount collected during refrigerant recovery	Inject to low-side service port during system charging
Refrigerant Hose/Line	60 ml (2 fl oz) added to the amount collected during refrigerant recovery ^a	Inject to low-side service port during system charging
	60 ml (2	

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O-ring Leak Repair	fl oz) added to the amount collected during refrigerant recovery ^b	Inject to low-side service port during system charging
Service Port Leak Repair	60 ml (2 fl oz) added to the amount collected during refrigerant recovery	Inject to low-side service port during system charging

^a If an excessive amount of refrigerant oil is lost due to a hose rupture/separation or other damage, the total system refrigerant oil capacity must be added.

^b The amount specified may be used for one or multiple O-ring leak repairs. Do not multiply the refrigerant oil amount by the number of O-ring leaks being repaired.

Refrigerant Oil Adding for New A/C Compressor Installation

NOTE: The service A/C compressor for this vehicle is shipped from the factory with PAG oil installed. The refrigerant oil adding procedure specified for this vehicle must be followed to avoid installing excessive PAG oil in the refrigerant system.

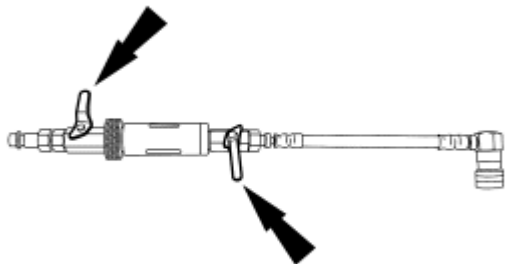
1. Rotate the new A/C compressor shaft 8 to 10 revolutions while collecting the PAG oil in a clean measuring cup.
2. Rotate the oil A/C compressor shaft 8 to 10 revolutions while collecting the PAG oil in a separate clean measuring cup.
3. Remove enough PAG oil from the new compressor measuring cup to make the remaining amount of oil in the cup match the amount remove from the oil A/C compressor.
4. Add the PAG oil from the new A/C compressor measuring cup to the suction side of the A/C compressor.

Oil Injection Using a Dye/Lubricant Injector

NOTE: If fluorescent leak detection dye is also to be added during A/C charging, the dye may be added to the dye/lubricant injector along with the refrigerant oil.

1. Evacuate the refrigerant system. For additional information, refer to **Air Conditioning (A/C) System Recovery, Evacuation and Charging.**

2. Assemble the dye/lubricant injector using the correct adapters to match the amount of refrigerant compressor oil to be injected.
3. Verify that all the valves on the dye/lubricant injector are closed.



A0039089

Fig. 125: Identifying Valves On Fluorescent Dye/Lubricant Injector
Courtesy of FORD MOTOR CO.

4. Fill the dye/lubricant injector with the correct amount of new refrigerant compressor oil.



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Fig. 126: Identifying Deluxe Injector Loop Kit Reservoir
Courtesy of FORD MOTOR CO.

5. Install the dye/lubricant injector between the low-side service gauge port valve and the refrigerant service station or manifold gauge set.
6. Open all valves and charge the refrigerant system. For additional information, refer to **Air Conditioning (A/C) System Recovery, Evacuation and Charging**.

REFRIGERANT IDENTIFICATION TESTING

Special Tools

Illustration	Tool Name	Tool Number
	Refrigerant Identifier with Air-Radiator	198-00003 or equivalent

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ST1457-A

Refrigerant Identification

NOTE: A Refrigerant Identifier with Air-Radiator must be used to identify gas samples taken directly from the refrigeration system or storage containers prior to recovering or charging the refrigerant system.

1. Follow the instructions included with the Refrigerant Identifier with Air-Radiator to obtain the sample for testing.
2. The Refrigerant Identifier with Air-Radiator will display one of the following:
 - If the purity level of R-134a is 98% or greater by weight, the green PASS LED will light. The weight concentrations of R-134a, R-12, R-22, hydrocarbons and air will be displayed on the digital display.
 - If refrigerant R-134a does not meet the 98% purity level, the red FAIL LED will light and an alarm will sound alerting the user of potential hazards. The weight concentrations of R-134a, R-12, R-22 and hydrocarbons will be displayed on the digital display.
 - If hydrocarbon concentrations are 2% or greater by weight, the red FAIL LED will light, "Hydrocarbon High" will be displayed on the digital display, and an alarm will sound alerting the user of potential hazards. The weight concentrations of R-134a, R-12, R-22 and hydrocarbons will also be displayed on the digital display.
3. The percentage of air contained in the sample will be displayed if the R-134a content is 98% or greater. The scan tool eliminates the effect of air when determining the refrigerant sample content because air is not considered a contaminant, although air can affect A/C system performance. When the scan tool has determined that a refrigerant source is pure (R-134a is 98% or greater by weight) and air concentration levels are 2% or greater by weight, the scan tool will prompt the user if an air purge is desired.
4. If contaminated refrigerant is detected, repeat the refrigerant identification test to verify that the refrigerant is indeed contaminated.

Contaminated Refrigerant Handling

NOTE: If contaminated refrigerant is detected, **DO NOT** recover the refrigerant into R-134a recovery/recycling equipment. Recovery of contaminated refrigerant will contaminate the recovered refrigerant supply and may damage the recovery/recycling equipment.

NOTE: A new suction accumulator or receiver/drier must be installed as directed by the A/C system flushing procedure.

1. Recover the contaminated refrigerant using suitable recovery-only equipment designed for capturing and

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storing contaminated refrigerant only.

- If this equipment is not available, contact an A/C service facility in the area with the correct equipment to carry out this service.
2. Determine and correct the cause of the customers initial concern.
 3. Flush the A/C system.
 4. Dispose of the contaminated refrigerant in accordance with all federal, state and local regulations.

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SPECIFICATIONS

MATERIAL

Item	Specification	Fill Capacity
PAG Refrigerant Compressor Oil (R-134a Systems) YN-12-D	WSH-M1C231-B	118 ml (4 fl oz)
R-134a Refrigerant YN-19 (US); CYN-16-P or CYN-16-R (Canada)	WSH-M17B19-A	0.60 kg (21 oz)

TORQUE SPECIFICATIONS

Description	N.m	lb-ft	lb-in
A/C compressor bolt	25	18	-
A/C compressor discharge fitting bolt	9	-	80
A/C compressor nut	25	18	-
A/C compressor stud	9	-	80
A/C compressor suction fitting bolt	9	-	80
Compressor-to-condenser discharge line bracket bolt	7	-	62
Compressor-to-condenser discharge line fitting nut	8	-	71
Condenser bolt	6	-	53
Condenser inlet fitting bolt	6	-	53
Condenser outlet fitting bolt	6	-	53
Condenser-to-evaporator line bracket bolt	9	-	80
Evaporator outlet line bracket stud bolt	11	8	-
Heater core and evaporator core housing nuts	8	-	71
High-pressure Schrader-type valve	3.4	-	30
High-pressure service gauge port valve cap	0.8	-	7
Low-pressure Schrader-type valve	2.26	-	20
Low-pressure service gauge port valve cap	0.8	-	7
Receiver/drier plug	3	-	27
Thermostatic expansion valve (TXV) fitting nut	8	-	71

DESCRIPTION AND OPERATION

AIR CONDITIONING

The refrigerant system components include the following:

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- A/C compressor and clutch assembly
- A/C condenser core
- A/C evaporator core
- Receiver/drier desiccant bag
- Connecting refrigerant lines
- Thermostatic expansion valve (TXV)

The refrigerant system incorporates an A/C compressor controlled by the PCM through an A/C clutch relay. The HVAC module sends an A/C request signal to the instrument cluster, which relays the request to the PCM. An evaporator temperature sensor is used to cycle the A/C compressor off when the evaporator core temperature falls below an acceptable temperature.

The HVAC module will send an A/C request signal to the instrument cluster if the following conditions are met:

- A mode which requires A/C operation has been manually selected by the driver or automatically selected by the HVAC module in AUTO mode.
- The A/C evaporator temperature sensor is reading an acceptable evaporator core temperature.

When an A/C request is received, the A/C compressor clutch will only be engaged by the PCM through the A/C clutch relay if all of the following conditions are met:

- The engine has been running for more than 5 seconds.
- Engine speed is above 425 RPM and below 5,800 RPM.
- The A/C pressure transducer is reading an acceptable pressure in the high side of the refrigerant system.
- The engine coolant temperature is not excessively high.
- The PCM has not detected a wide open throttle (WOT) condition.

An A/C pressure relief valve is installed in the A/C compressor to protect the refrigerant system against excessively high refrigerant pressures.

Refrigerant flow into the evaporator core is metered by a TXV.

A/C Compressor and Clutch Assembly

NOTE: **A/C compressor components are not serviced separately. The A/C compressor is serviced only as an assembly.**

The A/C compressor has the following characteristics:

- A non-serviceable shaft seal.
- A pressure relief valve is installed in the rear of the compressor to protect the refrigerant system against excessively high refrigerant pressures.
- The A/C compressor uses PAG oil or equivalent. This oil contains special additives required for the A/C

compressor.

- The A/C compressor oil may have some slightly dark-colored streaks while maintaining normal oil viscosity. This is normal for this A/C compressor because of break-in wear that can discolor the oil.
- Use the oil matching procedure specified for this vehicle when installing a new A/C compressor.

When battery voltage is applied to the A/C compressor clutch field coil, the clutch disc and hub assembly is drawn toward the A/C clutch pulley. The magnetic force locks the clutch plate and hub assembly and the A/C clutch pulley together as one unit, causing the compressor shaft to rotate. When battery voltage is removed from the A/C compressor clutch field coil, springs in the clutch plate and hub assembly move the clutch plate away from the A/C clutch pulley.

A/C Condenser

NOTE: The A/C condenser for this vehicle is produced from the factory with a receiver/drier desiccant bag installed in the integrated receiver/drier. This receiver/drier desiccant bag may become contaminated with moisture if the A/C condenser is not installed for an extended period of time. Since it cannot be determined how much time has elapsed between production and installation of the unit as a service part, the receiver/drier desiccant bag must be removed and replaced with the separate receiver/drier desiccant bag, which is supplied together with the A/C condenser service part kit.

The A/C condenser is an aluminum fin-and-tube design heat exchanger located in front of the vehicle radiator. It cools compressed refrigerant gas by allowing air to pass over fins and tubes to extract heat and by condensing gas to liquid refrigerant as it is cooled.

The receiver/drier is incorporated onto the LH side of the A/C condenser. The receiver/drier desiccant bag is a separate component and can be separately removed and installed with the A/C condenser removed from the vehicle.

Evaporator Core

The evaporator core is an aluminum plate/fin type and is located in the heater core and evaporator core housing. A mixture of liquid refrigerant and oil enters the bottom of the evaporator core through the evaporator core inlet tube and continues out of the evaporator core through the evaporator core outlet tube as a vapor. During A/C compressor operation, airflow from the blower motor is cooled and dehumidified as it flows through the evaporator core fins.

Thermostatic Expansion Valve (TXV)

The thermostatic expansion valve (TXV) is located at the evaporator core inlet and outlet tubes at the center rear of the engine compartment. The TXV provides a restriction to the flow of refrigerant from the high-pressure side of the refrigerant system and separates the low-pressure and high-pressure sides of the refrigerant system. Refrigerant entering and exiting the evaporator core passes through the TXV through 2 separate flow paths. An internal temperature sensing bulb senses the temperature of the refrigerant flowing out of the evaporator core and adjusts an internal pin-type valve to meter the refrigerant flow into the evaporator core. The internal pin-type valve decreases the amount of refrigerant entering the evaporator core at lower temperatures and increases

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the amount of refrigerant entering the evaporator core at higher temperatures.

Receiver/Drier

NOTE: Installation of a new receiver/drier desiccant bag is not required when repairing the A/C system, except when there is physical evidence of contamination from a failed A/C compressor or damage to the receiver/drier desiccant bag. Damage to the receiver/drier desiccant bag includes physical damage to the receiver/drier desiccant bag or moisture contamination. Moisture contamination results only from a complete loss of refrigerant and equalization of the refrigerant system pressure with atmospheric pressure for a period longer than one hour. If even a slight amount of positive refrigerant pressure is present in the system before repairs are carried out, the receiver/drier desiccant bag should not be replaced.

The receiver/drier is integral to the A/C condenser. It stores high-pressure liquid refrigerant after it leaves the condenser core. A receiver/drier desiccant bag installed inside the receiver/drier removes any retained moisture from the refrigerant. The receiver/drier desiccant bag is a separate component and can be separately removed and installed with the A/C condenser removed from the vehicle.

Evaporator Temperature Sensor

The evaporator temperature sensor contains a thermistor which receives a reference voltage from the HVAC module. The thermistor then varies the resistance to the reference voltage based on the evaporator fin temperature. The resulting voltage is returned to the HVAC module where it is interpreted as an evaporator fin temperature reading.

The HVAC module maintains evaporator core temperature and prevents icing of the evaporator core. It does this by switching off the A/C request signal to the instrument cluster when the evaporator temperature sensor reading falls below acceptable levels, and by switching on the A/C request when the evaporator temperature rises above acceptable levels.

The evaporator temperature sensor electrical connector is located outside of the heater core and evaporator core housing near the accelerator pedal, with the sensor pigtail leading into the housing to the sensor probe which is inserted between the evaporator core fins.

The evaporator temperature sensor is not available as a separate component. To install a new evaporator temperature sensor, a new heater core and evaporator core housing must be installed.

A/C Pressure Transducer

The A/C pressure transducer monitors the compressor discharge pressure and communicates with the PCM. The PCM will interrupt A/C compressor operation in the event that the A/C pressure transducer indicates high system discharge pressures. It is also used to sense low charge conditions. If the pressure is below a predetermined value for a given ambient temperature, the PCM will not allow the clutch to engage. It is not necessary to recover the refrigerant before removing the A/C pressure transducer.

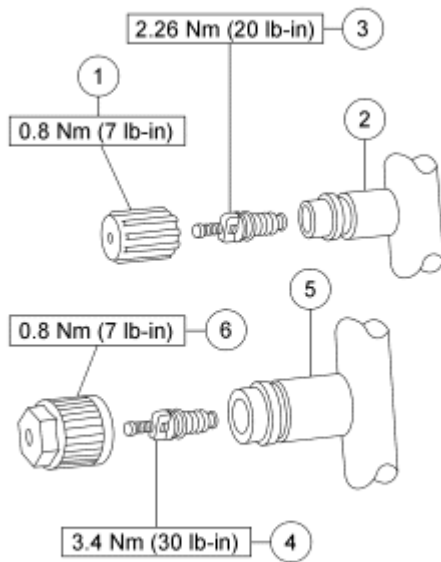
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Service Gauge Port Valves

The high-pressure service gauge port valve is located on the compressor-to-condenser discharge line.

The low-pressure service gauge port valve is located on the compressor suction line.



N0061379

Fig. 1: Exploded View Of Service Gauge Port Valves With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	19D702	Low-pressure service gauge port valve cap
2	-	Low-pressure service gauge port valve
3	19D701	Low-pressure Schrader-type valve
4	19D701	High-pressure Schrader-type valve
5	-	High-pressure service gauge port valve
6	19D702	High-pressure service gauge port valve cap

The fitting is an integral part of the refrigerant line or component.

- Special couplings are required for both the high-side and low-side service gauge ports.
- A very small amount of leakage will always be detectable around the Schrader-type valve with the service gauge port valve cap removed, and is considered normal. A new Schrader-type valve core can be installed if the seal leaks excessively.

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- The service gauge port valve caps are used as primary seals in the refrigerant system to prevent leakage through the Schrader-type valves from reaching the atmosphere. Always install and tighten the A/C service gauge port valve caps to the correct torque after they are removed.

Refrigerant System Dye

A fluorescent refrigerant system dye "wafer" is added to the receiver/drier desiccant bag to assist in refrigerant system leak diagnosis using a Rotunda-approved ultraviolet blacklight. This fluorescent dye "wafer" will dissolve after about 30 minutes of continued A/C operation. It is not necessary to add additional dye to the refrigerant system before diagnosing leaks, even if a significant amount of refrigerant has been removed from the system. Additional refrigerant system dye should only be added if more than 50% of the refrigerant system lubricant capacity has been lost due to a fitting separation or hose rupture. Refer to **CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS** article.

AIR DISTRIBUTION AND FILTERING

Air Distribution

There are 2 sources of air available to the air distribution system:

- Outside air
- Recirculated air

Recirculated air is always used when the climate control assembly is set to the MAX A/C mode on vehicles equipped with electronic manual temperature control (EMTC) and can be selected in any mode except DEFROST. Vehicles equipped with dual-zone electronic automatic temperature control (EATC) use recirculated air only when the RECIRC function is selected in any mode other than DEFROST, or when required by the HVAC module in the AUTOMATIC mode.

RECIRC is a timed function when selected while operating in FLOOR or FLOOR/DEFROST mode, and will automatically default back to fresh air inlet in 5 minutes if operating with A/C ON at an ambient temperature below 10°C (50°F). When operating with A/C OFF, the RECIRC function will always default back to fresh air inlet in 5 minutes regardless of ambient temperature.

Air distribution within the vehicle is determined by the airflow mode selected, or by the dual-zone EATC system in the AUTOMATIC mode. The air distribution mode on vehicles equipped with dual-zone EATC can be overridden by the driver if desired. Airflow mode control doors are used to direct airflow within the heater core and evaporator core housing. Electric mode door actuators are used to position these airflow mode doors.

Vehicles equipped with dual-zone EATC use 2 separate temperature blend doors. This allows individual temperature settings to be selected for the RH and LH air outlets. The airflow mode doors are not partitioned for separate RH and LH airflow positions. Therefore, the airflow mode will always remain similar for the RH and LH sides of the vehicle regardless of the temperatures selected.

Air enters the passenger compartment from the following outlets:

- Instrument panel registers

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- Floor duct outlets
- Windshield defroster
- Side window demisters
- Rear footwell duct
- Center console registers

Passenger compartment air is exhausted from the vehicle through open windows or body air vents.

Cabin Air Filter

A cabin air filter is available as standard equipment on dual-zone EATC vehicles. EMTC vehicles are not equipped from the factory with a cabin air filter, but are equipped with a screen in place of the filter to prevent any large foreign material from entering the heater core and evaporator core housing. The cabin air filter or screen can be accessed by removing the glove compartment and removing the cabin air filter access cover.

A cabin air filter can be installed as a dealer accessory on vehicles not originally equipped with a filter, but only a screen or a filter should be installed, never both.

CONTROL COMPONENTS

Electronic Manual Temperature Control (EMTC)

The electronic manual temperature control (EMTC) components are used to select the following:

- Air inlet source (outside or recirculated)
- Blower motor speed
- Discharge air temperature (temperature blend)
- Discharge air location (defrost, panel, floor)
- A/C compressor operation

Control System Inputs

The EMTC system has 2 system control inputs.

HVAC Module

The HVAC module integrates the temperature control, airflow mode selection, A/C request button, recirculated air request button and rear defog switch into a single unit.

The temperature control selector setting determines air temperature. Movement of the temperature display from COOL (blue) to WARM (red) causes a corresponding movement of the temperature blend door and determines the discharge air temperature that the air distribution system will maintain. The temperature control selector is an integral part of the HVAC module and cannot be installed separately. The temperature selector knob is removable and can be separately replaced.

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The airflow mode setting determines air discharge location. Movement of the airflow mode selector causes a corresponding movement of the airflow mode doors and determines the air discharge location. The airflow mode selector is an integral part of the HVAC module and cannot be installed separately. The airflow mode selector knob is removable and can be separately replaced.

The A/C request button determines A/C compressor operation, except when the airflow mode selector is in the OFF, MAX A/C or DEFROST mode. The A/C request button is an integral part of the HVAC module and cannot be installed separately.

The recirculated air request button can select recirculated air in any mode except DEFROST, and fresh air in any mode except MAX A/C or OFF. The recirculated air request button is an integral part of the HVAC module and cannot be installed separately.

The rear defog button signals activation of the heated backlight. The rear defog button is an integral part of the HVAC module and cannot be installed separately.

Blower Motor Switch

The blower motor switch is mounted in the HVAC module and controls blower motor speed by adding or bypassing resistors in the blower motor resistor in all modes except OFF. The blower motor switch knob is removable and can be separately replaced.

Control System Outputs

The EMTC system has 4 system outputs.

Blower Motor Resistor

The blower motor switch varies the blower motor ground circuit resistance by adding or bypassing series resistance in the blower motor resistor. Increased resistance will lower the blower motor speed, and lowered resistance will increase blower motor speed. When the blower motor is in the HI position, the blower motor resistor is bypassed.

The blower motor resistor is located on the heater core and evaporator core housing.

Temperature Blend Door Actuator

The temperature blend door actuator moves the temperature blend door on command from the HVAC module.

The temperature blend door actuator contains a reversible electric motor and a potentiometer. The potentiometer circuit consists of a 5-volt reference signal connected to one end of a variable resistor, and a signal ground connected to the other. A signal return circuit is connected to a contact wiper, which is driven along the variable resistor by the actuator shaft. The signal returned to the HVAC module from the contact wiper indicates the position of the temperature blend door. The HVAC module drives the actuator motor in whichever direction is necessary to make the actuator contact wiper voltage agree with the expected HVAC module contact wiper voltage value.

Airflow Mode Door Actuator

The airflow mode door actuator uses a cam and lever assembly to position the airflow mode doors on command from the HVAC module. The mode door actuator contains a reversible electric motor and potentiometer. The potentiometer allows the HVAC module to monitor the position of the airflow mode doors. The potentiometer circuit consists of a 5-volt reference signal connected to one end of a variable resistor, and a signal ground connected to the other. A signal return circuit is connected to a contact wiper, which is driven along the variable resistor by the actuator shaft. The signal returned to the HVAC module from the contact wiper indicates the position of the airflow mode doors.

Air Inlet Mode Door Actuator

The air inlet mode door actuator moves the air inlet door between the fresh and RECIRC positions on command from the HVAC module. The air inlet mode door actuator is driven to, and will automatically stop at, the full RECIRC or full fresh air inlet position and does not require a potentiometer circuit to monitor its position.

Electronic Automatic Temperature Control (EATC)

The dual-zone electronic automatic temperature control HVAC module analyzes input from the following major sources:

- Temperature, airflow mode, blower, A/C and RECIRC selection (made by the vehicle occupants)
- In-vehicle temperature sensor
- Ambient temperature sensor
- Solar radiation sensor
- Evaporator temperature sensor
- Vehicle speed
- Engine coolant temperature

Using these inputs, the dual-zone HVAC module determines the correct conditions for the following outputs:

- A/C compressor operation
- Blower speed
- Temperature blend door position
- Airflow mode door position
- Air inlet door position

Control System Inputs

The dual-zone EATC system has 4 control system inputs.

HVAC Module

The HVAC module provides an interface for the vehicle occupants to control the climate control system, or automatically controls the climate control system in the AUTO mode. The HVAC module has a vacuum

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fluorescent display for displaying set temperature, airflow direction, blower speed and DTCs. An on-board diagnostic (OBD) feature is incorporated to supply the technician with DTCs. These DTCs help direct the technician to the inoperative component.

In-Vehicle Temperature Sensor

The in-vehicle temperature sensor contains a thermistor, which measures the in-vehicle air temperature and sends that reading to the HVAC module. An aspirator hose is connected between the heater core and evaporator core housing and the in-vehicle temperature sensor. The aspirator hose uses airflow through the heater core and evaporator core housing to create a venturi-type suction to draw in-vehicle air through the in-vehicle temperature sensor (across the thermistor).

Solar Radiation Sensor (Sunload Sensor)

The solar radiation sensor supplies information to the HVAC module indicating sunload.

Ambient Temperature Sensor

The ambient temperature sensor signal is received by the HVAC module and indicates the outside air temperature. The ambient temperature sensor is located in front of the radiator on the radiator support bolster.

The ambient temperature sensor signal is not the sole factor considered by the HVAC module for the outside air temperature display. Therefore, the ambient temperature PID monitored with a scan tool may not exactly match the HVAC module display. For information regarding the HVAC module outside air temperature display, refer to **CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS** article.

Control System Outputs

The EATC system has 4 control system outputs.

Blower Motor Speed Control

The blower motor speed control controls the blower motor speed by converting low power signals from the HVAC module to a high current, variable ground feed for the blower motor. A delay function is used to provide a gradual increase or decrease in blower motor speed under all conditions.

Temperature Blend Door Actuators

The dual-zone EATC system uses 2 temperature blend door actuators to control 2 separate temperature blend doors. The temperature blend doors independently vary the LH side and RH side temperature settings, as desired. The temperature blend door actuators each contain a reversible electric motor and a potentiometer. The potentiometer circuit consists of a 5-volt reference signal connected to one end of a variable resistor, and a signal ground connected to the other. A signal return circuit is connected to a contact wiper, which is driven along the variable resistor by the actuator shaft. The signal returned to the HVAC module from the contact wiper indicates the position of the temperature blend door. The HVAC module drives the actuator motor in whichever direction is necessary to make the actuator contact wiper voltage agree with the expected HVAC module contact wiper voltage value.

Airflow Mode Door Actuator

The airflow mode door actuator uses a cam and lever assembly to position the airflow mode doors on command from the HVAC module. The mode door actuator contains a reversible electric motor and potentiometer. The potentiometer allows the HVAC module to monitor the position of the airflow mode doors. The potentiometer circuit consists of a 5-volt reference signal connected to one end of a variable resistor, and a signal ground connected to the other. A signal return circuit is connected to a contact wiper, which is driven along the variable resistor by the actuator shaft. The signal returned to the HVAC module from the contact wiper indicates the position of the airflow mode doors.

Air Inlet Mode Door Actuator

The air inlet mode door actuator moves the air inlet door between the fresh and RECIRC positions on command from the HVAC module. The air inlet mode door actuator is driven to, and will automatically stop at, the full RECIRC or full fresh air inlet position and does not require a potentiometer circuit to monitor its position.

HEATING AND VENTILATION

The heating and ventilation system has the following features:

- Controls the temperature and, during A/C operation, reduces the relative humidity of the air inside the vehicle.
- Delivers heated or cooled air to maintain the vehicle interior temperature and comfort level.
- Cooling or heating can be adjusted to maintain the desired temperature.
- Uses a reheat method to provide conditioned air to the passenger compartment.
- All airflow from the blower motor passes through the A/C evaporator core.
- Temperature blending is controlled by the temperature blend door(s), which regulate(s) the amount of air that flows through and around the heater core, where it is then mixed and distributed.

Heater Core

The heater core consists of fins and tubes arranged to extract heat from the engine coolant and transfer the heat to air passing through the heater core.

Blower Motor

The blower motor pulls air from the air inlet and forces it into the heater core and evaporator core housing where it is mixed and distributed.

Heater Core and Evaporator Core Housing

The heater core and evaporator core housing directs airflow from the blower motor through the evaporator core and heater core. All airflow from the blower motor passes through the evaporator core. The airflow is then directed through or around the heater core by the temperature blend door(s). Vehicles equipped with dual-zone electronic automatic temperature control (EATC) use a partitioned heater core and evaporator core housing with 2 electric actuator-positioned temperature blend doors. This allows for separate temperatures to be selected for

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the driver and passenger sides of the passenger compartment. Manual systems use a single electric actuator-positioned temperature blend door to direct airflow through or around the heater core.

REMOVAL AND INSTALLATION

AIR CONDITIONING (A/C) COMPRESSOR

MATERIAL

Item	Specification
PAG Refrigerant Compressor Oil (R-134a Systems) YN-12-D	WSH-M1C231-B

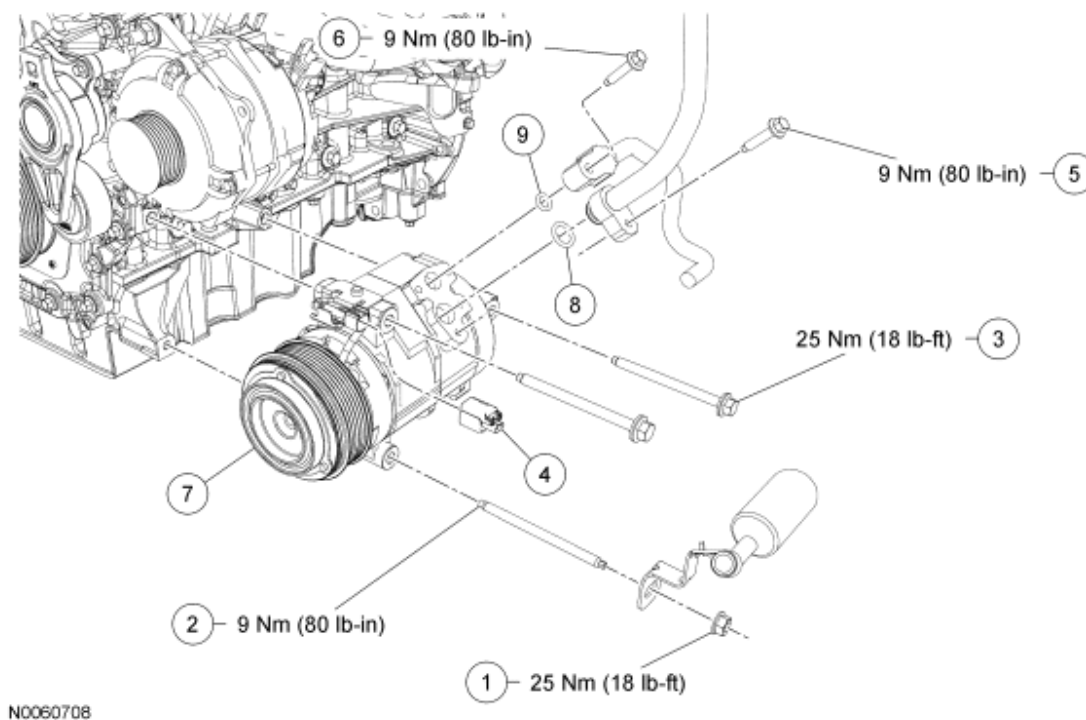


Fig. 2: Exploded View Of Air Conditioning (A/C) Compressor With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W520413	A/C compressor nut
2	W712610	A/C compressor stud
3	W712611	A/C compressor bolt (2 required)
4	12B637	Clutch field coil electrical connector
5	W500215	A/C compressor suction fitting bolt
6	W500215	A/C compressor discharge fitting bolt
7	19703	A/C compressor
8	19E889	Compressor suction fitting O-ring seal

REMOVAL

CAUTION: If installing a new air conditioning (A/C) compressor due to an internal failure of the old unit, you must carry out the following procedures to remove contamination from the A/C system.

- If A/C flushing equipment is available, carry out flushing of the A/C system prior to installing a new A/C compressor. For additional information, refer to CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS article.
- If A/C flushing equipment is not available, carry out filtering of the A/C system after a new A/C compressor has been installed. For additional information, refer to CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS article.
- Install a new thermostatic expansion valve (TXV), as directed by the A/C flushing or filtering procedure.
- Install a new receiver/drier desiccant bag as directed by the A/C flushing or filtering procedure.

NOTE: The O-ring seals used on this vehicle are not interchangeable with all similarly sized O-ring seals used on other Ford products. Use only the O-ring seals specified for this vehicle in the Ford master parts catalog.

NOTE: Installation of a new receiver/drier desiccant bag is not required when repairing the A/C system, except when there is physical evidence of contamination from a failed A/C compressor or damage to the receiver/drier desiccant bag. Damage to the receiver/drier desiccant bag includes physical damage to the receiver/drier desiccant bag or moisture contamination. Moisture contamination results only from a complete loss of refrigerant and equalization of the refrigerant system pressure with atmospheric pressure for a period longer than one hour. If even a slight amount of positive refrigerant pressure is present in the system before repairs are carried out, the receiver/drier desiccant bag should not be replaced.

1. If flushing of the A/C system has not been carried out, recover the refrigerant. For additional information, refer to CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS article.
2. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to JACKING AND LIFTING article.
3. Remove the drive belt from the A/C compressor pulley.
4. Remove the RH fender splash shield. For additional information, refer to FRONT END BODY PANELS article.

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5. Remove the A/C compressor nut and stud.
6. Remove the lower A/C compressor bolt.
7. Disconnect the field coil electrical connector.
8. Remove the upper A/C compressor bolt and reposition the A/C compressor to allow access to the compressor discharge and suction fittings.
9. Remove the A/C compressor suction fitting bolt and disconnect the fitting.
 - Discard the O-ring seal.
10. Remove the A/C compressor discharge fitting bolt and disconnect the fitting.
 - Discard the O-ring seal.
11. Remove the A/C compressor.

INSTALLATION

NOTE: The service A/C compressor for this vehicle is shipped from the factory with PAG oil installed. The refrigerant oil adding procedure specified for this vehicle must be followed to avoid installing excessive PAG oil in the refrigerant system.

1. If filtering of the A/C system is not to be carried out, lubricate the refrigerant system with the correct amount of clean PAG oil. For additional information, refer to **CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS** article.
2. Connect the A/C compressor discharge fitting and install the A/C compressor discharge fitting bolt.
 - Install a new O-ring seal lubricated in clean PAG oil.
 - Tighten to 9 N.m (80 lb-in).
3. Connect the A/C compressor suction fitting and install the A/C compressor suction fitting bolt.
 - Install a new O-ring seal lubricated in clean PAG oil.
 - Tighten to 9 N.m (80 lb-in).
4. Install, but do not tighten the upper A/C compressor bolt.
5. Install, but do not tighten the lower A/C compressor bolt.
6. Install the A/C compressor stud.
 - Tighten to 9 N.m (80 lb-in).
7. Position the compressor-to-condenser discharge line and install, but do not tighten the A/C compressor nut.
8. Tighten the A/C compressor bolts and nut in the sequence shown in illustration.
 - Tighten to 25 N.m (18 lb-ft).

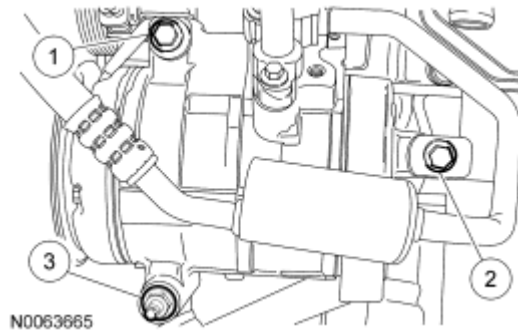


Fig. 3: Tightening A/C Compressor Bolts & Nut In Sequence
Courtesy of FORD MOTOR CO.

9. Connect the field coil electrical connector.
10. Install the RH fender splash shield. For additional information, refer to **FRONT END BODY PANELS** article.
11. Install the drive belt.
12. If filtering of the A/C system is not to be carried out, evacuate, leak test and charge the refrigerant system. For additional information, refer to **CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS** article.

AIR INLET DUCT

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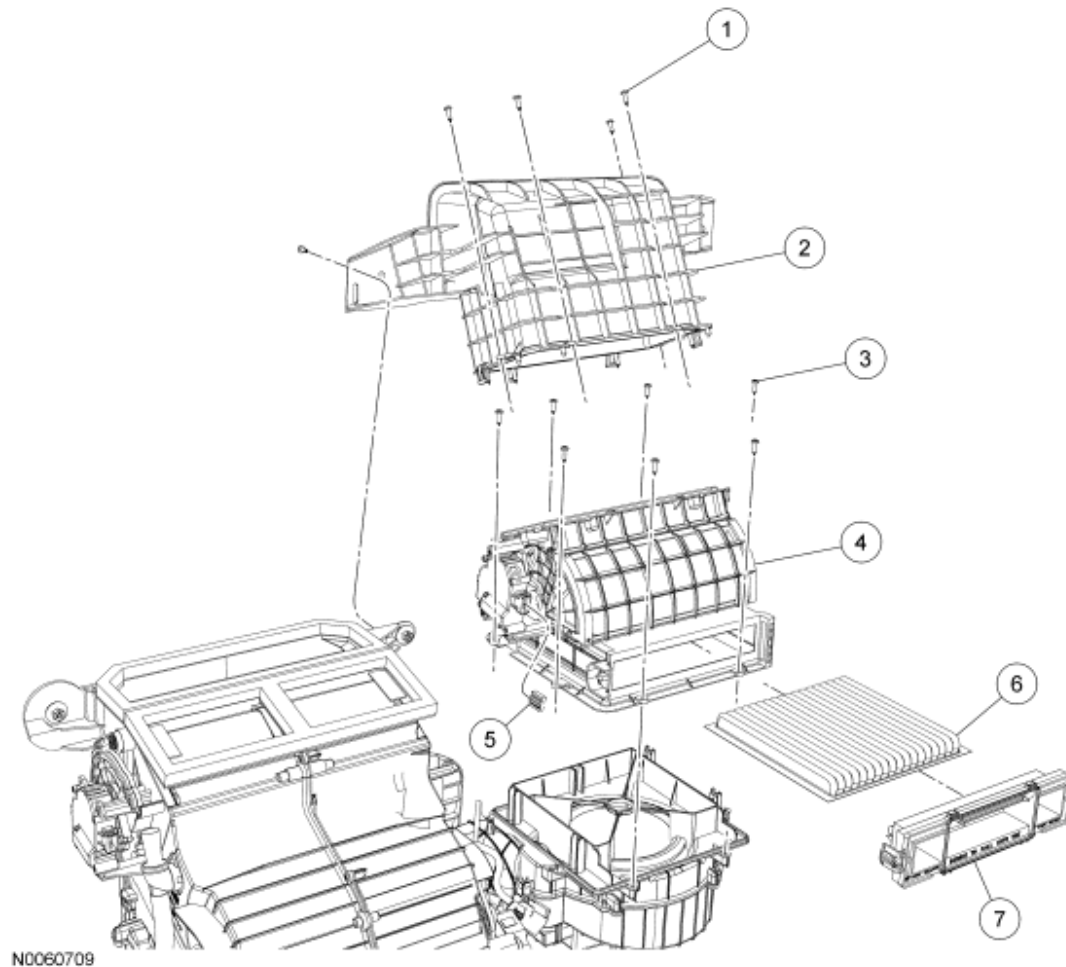


Fig. 4: Exploded View Of Air Inlet Duct
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Fresh air inlet duct screw (5 required)
2	-	Fresh air inlet duct (part of 19B555)
3	-	Air inlet duct screw (7 required)
4	-	Air inlet duct (part of 19B555)
5	19C603	Air inlet mode door actuator electrical connector
6	19N619	Cabin air filter
7	19B735	Cabin air filter cover

REMOVAL AND INSTALLATION

1. Remove the heater core and evaporator core housing. For additional information, refer to **HEATER CORE AND EVAPORATOR CORE HOUSING**.
2. Disconnect the air inlet mode door actuator electrical connector.
3. Remove the cabin air filter cover and the cabin air filter (if equipped) or air inlet screen (if equipped).
4. Remove the 5 fresh air inlet duct screws.
5. Detach the 4 fresh air inlet duct clips and remove the fresh air inlet duct.
6. Remove the 7 air inlet duct screws.
7. Remove the air inlet duct.
8. To install, reverse the removal procedure.

AMBIENT AIR TEMPERATURE SENSOR

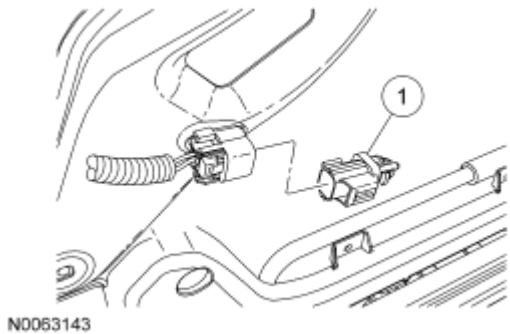


Fig. 5: Identifying Ambient Air Temperature Sensor
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	12A647	Ambient air temperature sensor

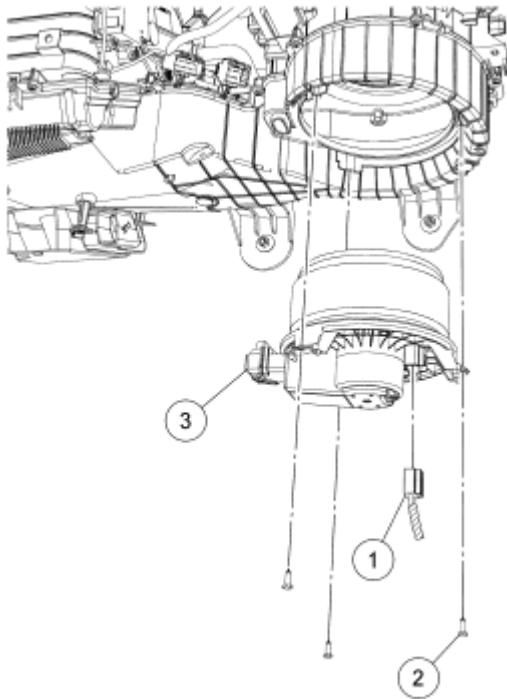
REMOVAL AND INSTALLATION

1. Remove the front bumper cover. For additional information, refer to **BUMPERS** article.
2. Disconnect the ambient air temperature sensor electrical connector and remove the ambient air temperature sensor.
3. To install, reverse the removal procedure.

BLOWER MOTOR

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N0063079

Fig. 6: Exploded View Of Blower Motor
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	19C603	Blower motor electrical connector
2	-	Blower motor screw (3 required)
3	19805	Blower motor

REMOVAL AND INSTALLATION

1. Remove the RH lower instrument panel insulator.
2. Disconnect the blower motor electrical connector.
3. Remove the 3 blower motor screws.
4. Remove the blower motor.
5. To install, reverse the removal procedure.

BLOWER MOTOR RESISTOR

REMOVAL AND INSTALLATION

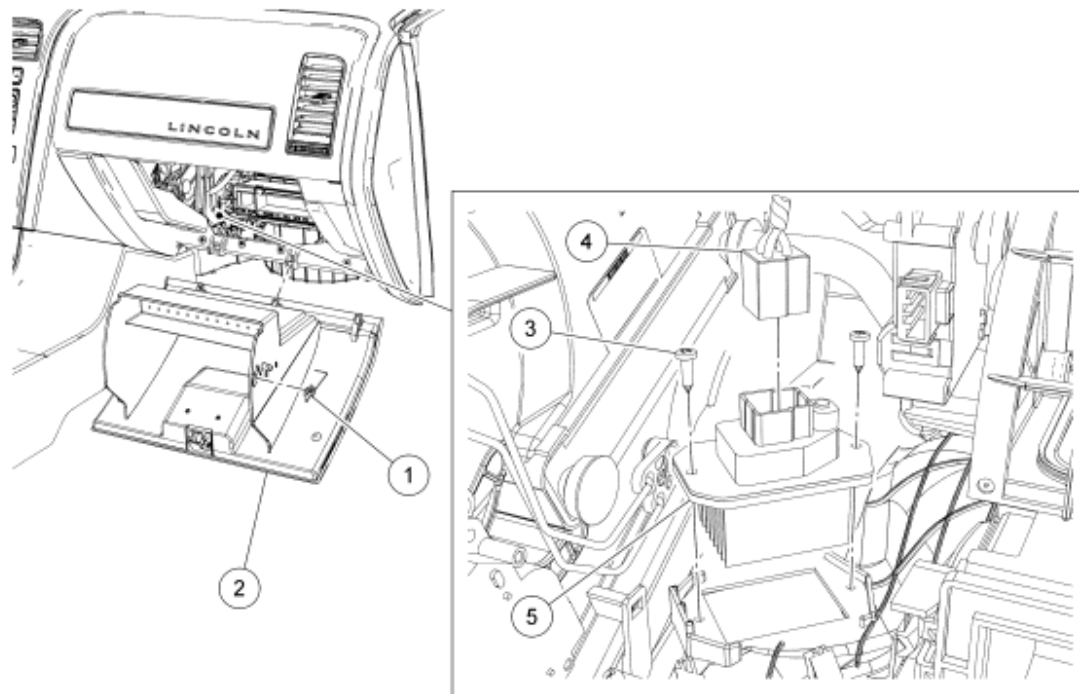
1. For additional information, refer to **BLOWER MOTOR SPEED CONTROL**.

BLOWER MOTOR SPEED CONTROL

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NOTE: Blower motor speed control shown in illustration, blower motor resistor similar.



NO063080

Fig. 7: Identifying Blower Motor Speed Control

Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	7806200	Glove compartment damper
2	7806024	Glove compartment
3	-	Blower motor speed control screw (2 required)
4	19C603	Blower motor speed control electrical connector
5	19E624	Blower motor speed control (dual-zone electronic automatic temperature control [EATC] vehicles)
5	19A706	Blower motor resistor (electronic manual temperature control [EMTC] vehicles)

REMOVAL AND INSTALLATION

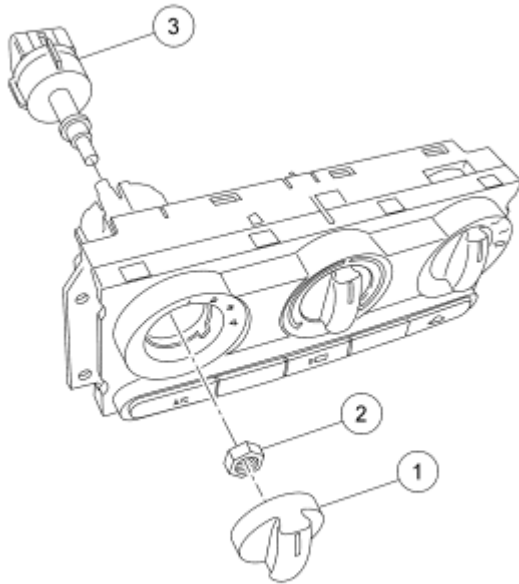
1. Detach the glove compartment damper from the glove compartment.
2. Remove the glove compartment.
3. Disconnect the blower motor speed control electrical connector.
4. Remove the 2 blower motor speed control screws.

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5. Remove the blower motor speed control.
6. To install, reverse the removal procedure.

BLOWER MOTOR SWITCH



N0065314

Fig. 8: Exploded View Of Blower Motor Switch
Courtesy of FORD MOTOR CO.

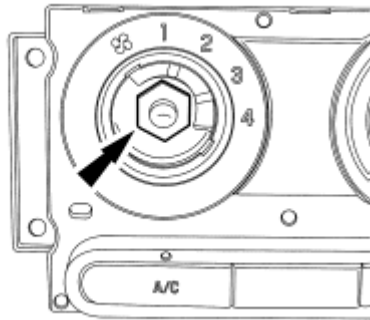
Item	Part Number	Description
1	18519	Blower motor switch knob
2	-	Blower motor switch nut (part of 18578)
3	18578	Blower motor switch

REMOVAL AND INSTALLATION

1. Remove the climate control module. For additional information, refer to **HEATING VENTILATION AIR CONDITIONING (HVAC) MODULE - EDGE.**
2. Pull to remove the blower motor switch knob.
3. Remove the blower motor switch nut.

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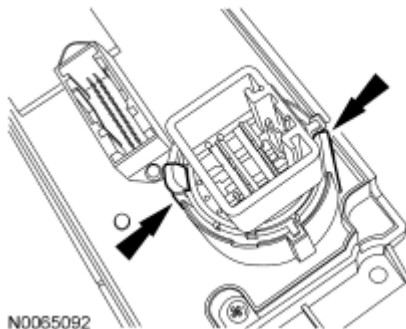
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N0065091

Fig. 9: Identifying Blower Motor Switch Knob
Courtesy of FORD MOTOR CO.

4. Release the 2 retaining tabs with the index finger on each hand, while depressing the blower motor switch shaft with the thumbs to remove the blower motor switch.



N0065092

Fig. 10: Identifying Blower Motor Switch Shaft
Courtesy of FORD MOTOR CO.

5. To install, reverse the removal procedure.

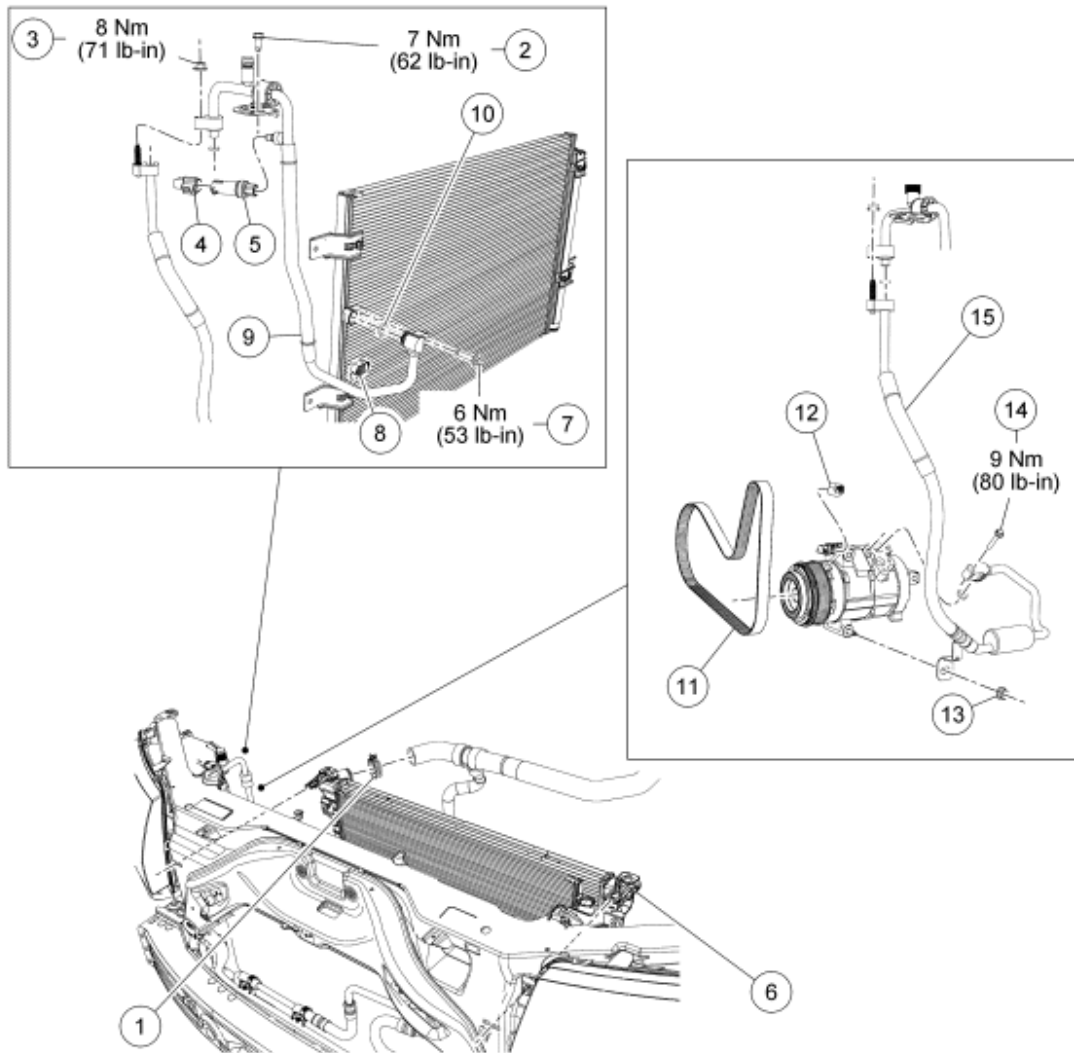
COMPRESSOR TO CONDENSER DISCHARGE LINE

MATERIAL

Item	Specification
PAG Refrigerant Compressor Oil (R-134a Systems) YN-12-D	WSH-M1C231-B

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N0062015

Fig. 11: Identifying Compressor To Condenser Discharge Line With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Upper radiator hose clamp
2	W707400	Compressor-to-condenser discharge line bracket bolt
3	W520413	Compressor-to-condenser discharge line fitting nut
4	12B637	A/C pressure transducer electrical connector
5	19D594	A/C pressure transducer
6	8A193	Radiator bracket
7	W500215	Condenser inlet fitting bolt

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8	W704005	Compressor-to-condenser discharge line-to-radiator clip
9	19972	Compressor-to-condenser discharge line (condenser side)
10	19E889	O-ring seal (3 required)
11	8620	Accessory drive belt
12	12A581	Clutch field coil electrical connector
13	W520413	Compressor-to-condenser discharge line bracket nut
14	W500215	A/C compressor discharge fitting bolt
15	19972	Compressor-to-condenser discharge line (compressor side)

REMOVAL AND INSTALLATION

NOTE: The O-ring seals used on this vehicle are not interchangeable with all similarly sized O-ring seals used on other Ford products. Use only the O-ring seals specified for this vehicle in the Ford master parts catalog.

NOTE: The compressor-to-condenser discharge line consists of 2 separately serviceable lines. If only one line is to be removed, only carry out the steps qualified for that line.

Both compressor-to-condenser discharge lines

1. Recover the refrigerant. For additional information, refer to **CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS** article.
2. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
3. Remove the compressor-to-condenser discharge line fitting nut and disconnect the fitting.
 - Discard the O-ring seal.
 - To install, tighten to 8 N.m (71 lb-in).

Condenser side compressor-to-condenser discharge line

4. Remove the front bumper cover. For additional information, refer to **BUMPERS** article.
5. Drain the engine coolant. For additional information, refer to **ENGINE COOLING** article.
6. Release the upper radiator hose clamp and disconnect the upper radiator hose from the radiator.
7. Remove the engine cooling fan. For additional information, refer to **ENGINE COOLING** article.
8. Remove the compressor-to-condenser discharge line bracket bolt.
 - To install, tighten to 7 N.m (62 lb-in).
9. Disconnect the A/C pressure transducer electrical connector.

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10. Detach the 2 radiator brackets from the core support and position the radiator rearward.
11. Remove the condenser inlet fitting bolt and disconnect the fitting.
 - Discard the O-ring seal.
 - To install, tighten to 6 N.m (53 lb-in).
12. Detach the compressor-to-condenser discharge line-to-radiator clip.
13. Remove the condenser side compressor-to-condenser discharge line.

Compressor side compressor-to-condenser discharge line

14. Remove the drive belt from the A/C compressor pulley.
15. Remove the RH inner fender splash shield.
16. Disconnect the clutch field coil electrical connector.
17. Remove the compressor-to-condenser discharge line bracket nut.
18. Remove the A/C compressor discharge fitting bolt and disconnect the fitting.
 - Discard the O-ring seal.
 - To install, tighten to 9 N.m (80 lb-in).
19. Remove the compressor-to-condenser discharge line.

Both compressor-to-condenser discharge lines

20. To install, reverse the removal procedure.
 - Install new O-ring seals lubricated in clean PAG oil.
 - Lubricate the refrigerant system with the correct amount of clean PAG oil. For additional information, refer to **CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS** article.
21. Evacuate, leak test and charge the refrigerant system. For additional information, refer to **CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS** article.

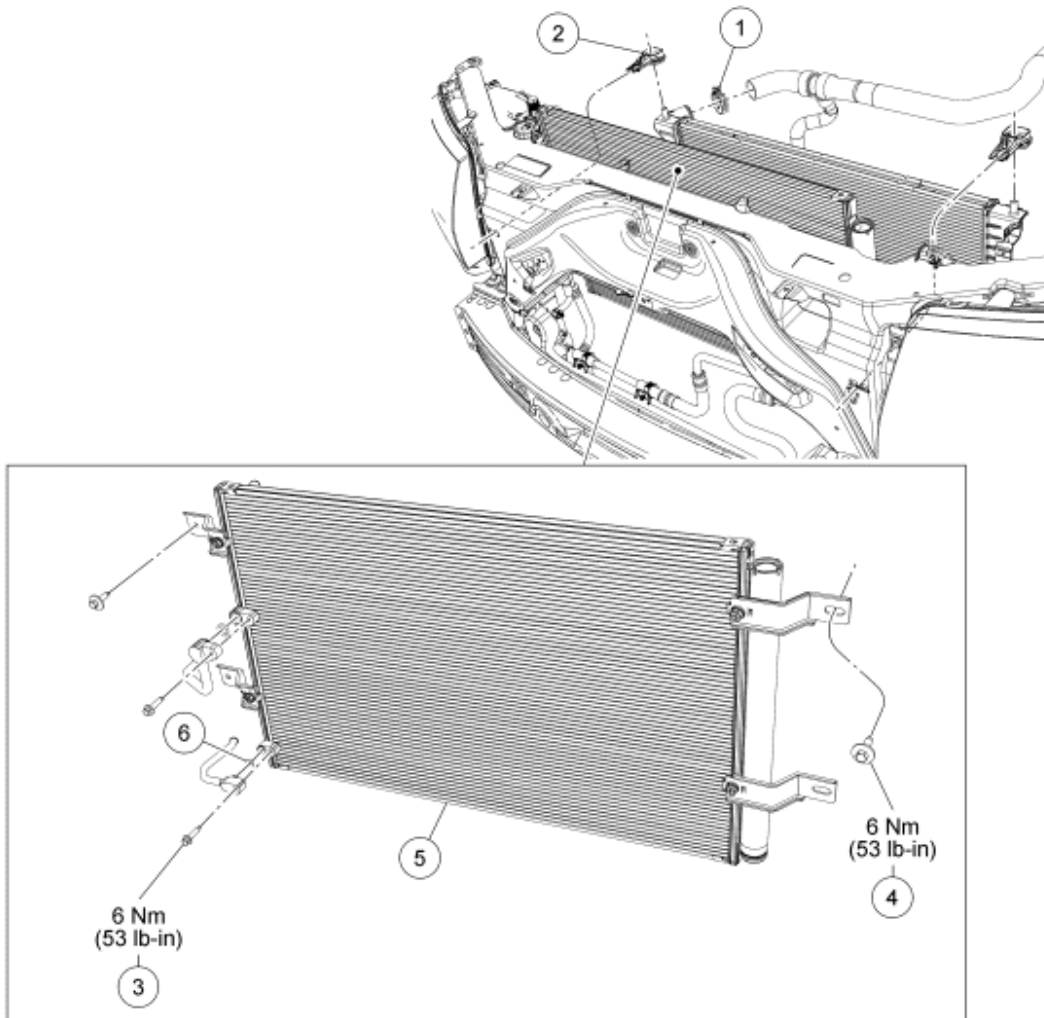
CONDENSER CORE

MATERIAL

Item	Specification
PAG Refrigerant Compressor Oil (R-134a Systems) YN-12-D	WSH-M1C231-B

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N0063081

Fig. 12: Exploded View Of Condenser Core With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Upper radiator hose clamp
2	8A193	Radiator bracket (2 required)
3	W500215	Condenser fitting bolt (2 required)
4	W704652	Condenser bolt (2 required)
5	19710	Condenser core
6	19E889	O-ring seal (2 required)

REMOVAL AND INSTALLATION

NOTE: The A/C condenser for this vehicle is produced from the factory with a

receiver/drier desiccant bag installed in the integrated receiver/drier. This receiver/drier desiccant bag may become contaminated with moisture if the A/C condenser is not installed for an extended period of time. Since it cannot be determined how much time has elapsed between production and installation of the unit as a service part, the receiver/drier desiccant bag must be removed and replaced with the separate receiver/drier desiccant bag, which is supplied together with the A/C condenser service part kit. For additional information, refer to DESICCANT BAG.

NOTE: The O-ring seals used on this vehicle are not interchangeable with all similarly sized O-ring seals used on other Ford products. Use only the O-ring seals specified for this vehicle in the Ford master parts catalog.

1. Recover the refrigerant. For additional information, refer to CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS article.
2. With the vehicle in NEUTRAL position it on a hoist. For additional information, refer to JACKING AND LIFTING article.
3. Drain the engine coolant. For additional information, refer to ENGINE COOLING article.
4. Remove the front bumper cover. For additional information, refer to BUMPERS article.
5. Remove the cooling fan. For additional information, refer to ENGINE COOLING article.
6. Release the upper radiator hose clamp at the radiator and disconnect the hose.
7. Release the 2 radiator bracket clips from the radiator support and position the radiator rearward.
8. Release the 2 condenser lines from the clips at the radiator.
9. Remove the condenser outlet and inlet fitting bolts and disconnect the fittings.
 - Discard the O-ring seals.
 - To install, tighten to 6 N.m (53 lb-in).
10. Remove the 2 condenser bolts.
 - To install, tighten to 6 N.m (53 lb-in).

NOTE: The condenser core must be removed from above the vehicle.

11. Remove the condenser core.
12. To install, reverse the removal procedure.
 - Install new O-ring seals lubricated in clean PAG oil.
 - Lubricate the refrigerant system with the correct amount of clean PAG oil. For additional information, refer to CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS article.
13. Fill the engine cooling system. For additional information, refer to ENGINE COOLING article.
14. Evacuate, leak test and charge the refrigerant system. For additional information, refer to CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS article.

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MATERIAL

Item	Specification
PAG Refrigerant Compressor Oil (R-134a Systems) YN-12-D	WSH-M1C231-B

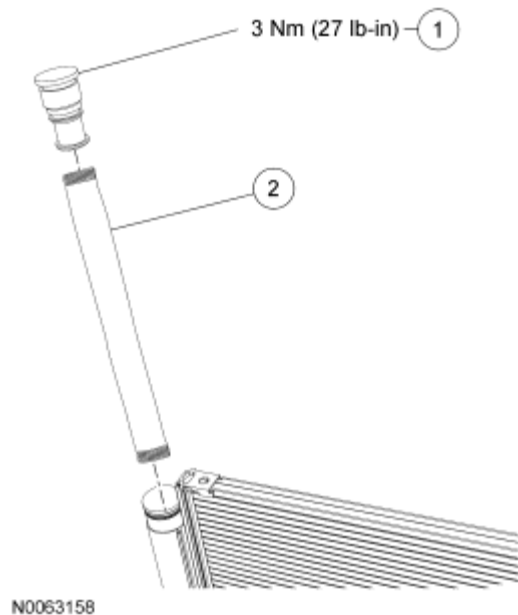


Fig. 13: Exploded View Of Desiccant Bag With Torque Specification
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Receiver/drier plug (part of 19E989 kit)
2	19C836	Receiver/drier desiccant bag

REMOVAL AND INSTALLATION

NOTE: Installation of a new receiver/drier desiccant bag is not required when repairing the A/C system, except when there is physical evidence of contamination from a failed A/C compressor or damage to the receiver/drier desiccant bag. Damage to the receiver/drier desiccant bag includes physical damage to the receiver/drier desiccant bag or moisture contamination. Moisture contamination results only from a complete loss of refrigerant and equalization of the refrigerant system pressure with atmospheric pressure for a period longer than one hour. If even a slight amount of positive refrigerant pressure is present in the system before repairs are carried out, the receiver/drier desiccant bag should not be replaced.

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1. Remove the A/C condenser core. For additional information, refer to **CONDENSER CORE**.
2. Remove the receiver/drier plug.
 - To install, tighten to 3 N.m (27 lb-in).
3. Using a suitable tool. Grasp and remove the receiver/drier desiccant bag.
4. To install, reverse the removal procedure.
 - Lubricate the O-ring seal on the new receiver/drier plug with clean PAG oil.

EVAPORATOR OUTLET LINE

MATERIAL

Item	Specification
PAG Refrigerant Compressor Oil (R-134a Systems) YN-12-D	WSH-M1C231-B

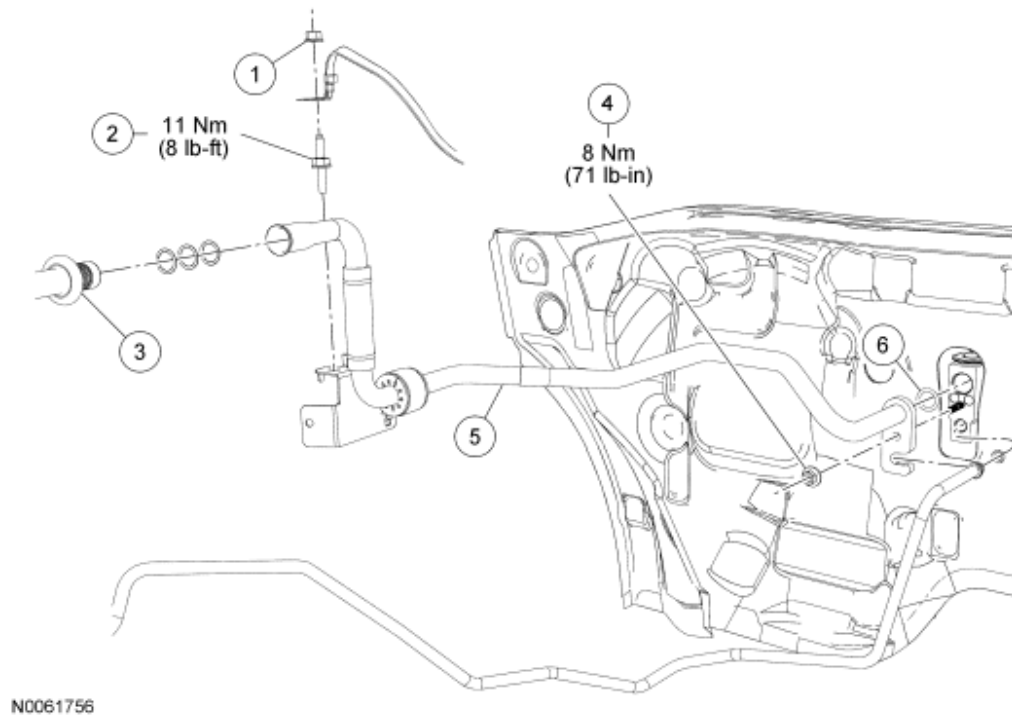


Fig. 14: Exploded View Of Evaporator Outlet Line With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W520413	Ground strap nut
2	N806925	Evaporator outlet line bracket stud bolt
3	-	Evaporator outlet line spring lock coupling (part of 19835)
4	W707142	Thermostatic expansion valve (TXV)

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		fitting nut
5	19835	Evaporator outlet line
6	19E889	O-ring seal (5 required)

REMOVAL AND INSTALLATION

NOTE: The O-ring seals used on this vehicle are not interchangeable with all similarly sized O-ring seals used on other Ford products. Use only the O-ring seals specified for this vehicle in the Ford master parts catalog.

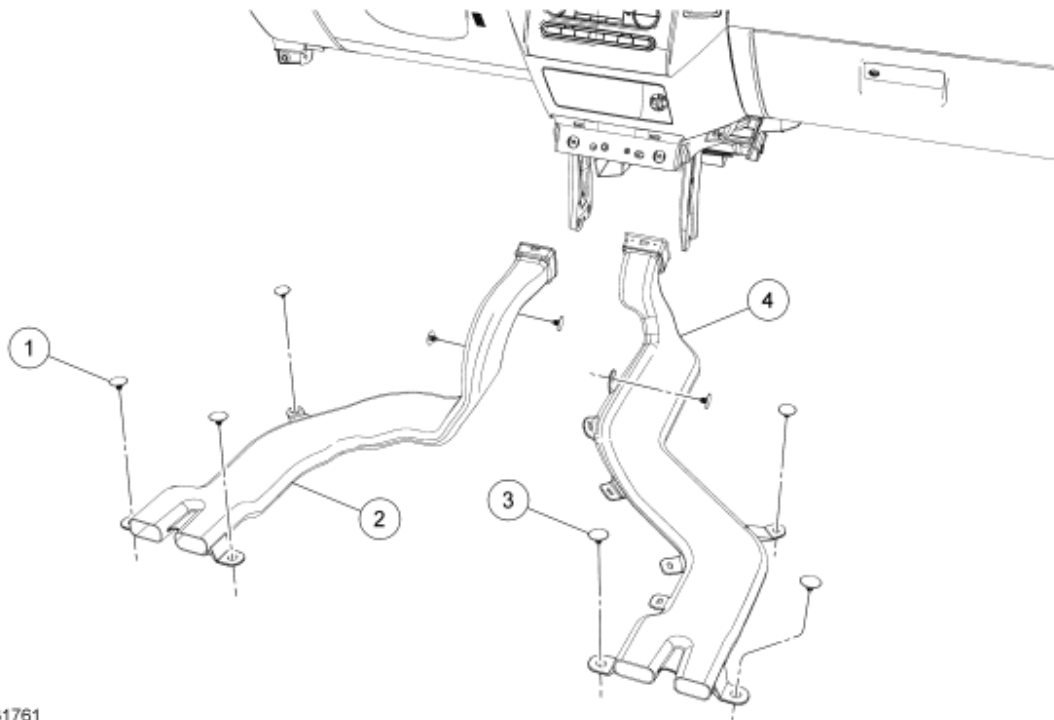
NOTE: Installation of a new receiver/drier desiccant bag is not required when repairing the A/C system, except when there is physical evidence of contamination from a failed A/C compressor or damage to the receiver/drier desiccant bag. Damage to the receiver/drier desiccant bag includes physical damage to the receiver/drier desiccant bag or moisture contamination. Moisture contamination results only from a complete loss of refrigerant and equalization of the refrigerant system pressure with atmospheric pressure for a period longer than one hour. If even a slight amount of positive refrigerant pressure is present in the system before repairs are carried out, the receiver/drier desiccant bag should not be replaced.

1. Recover the refrigerant. For additional information, refer to **CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS** article.
2. Remove the ground strap nut at the RH shock tower.
3. Remove the evaporator outlet line bracket stud bolt.
 - To install, tighten to 11 N.m (8 lb-ft).
4. Disconnect the evaporator outlet line spring lock coupling.
 - Discard the O-ring seals.
5. Remove the thermostatic expansion valve (TXV) fitting nut and disconnect the fitting.
 - Discard the O-ring seals.
 - To install, tighten to 8 N.m (71 lb-in).
6. Remove the evaporator outlet line.
7. To install, reverse the removal procedure.
 - Install new O-ring seals lubricated in clean PAG oil.
 - Lubricate the refrigerant system with the correct amount of clean PAG oil. For additional information, refer to **CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS** article.
8. Evacuate, leak test and charge the refrigerant system. For additional information, refer to **CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS** article.

FOOTWELL DUCT - REAR

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N0061761

Fig. 15: Exploded View Of Rear Footwell Duct
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	N806113	LH rear footwell duct pin-type retainer (5 required)
2	18C420	LH rear footwell duct
3	N806113	RH rear footwell duct pin-type retainer (4 required)
4	18C420	RH rear footwell duct

REMOVAL AND INSTALLATION

LH rear footwell duct only

1. Remove the LH front seat. For additional information, refer to **SEATING** article.
2. Remove the floor console. For additional information, refer to **INSTRUMENT PANEL AND CONSOLE** article.
3. Remove the LH front door sill plate.
4. Remove the driver floor carpet pin-type retainers and position the carpet away from the LH rear footwell duct.
5. Remove the 5 LH rear footwell duct pin-type retainers.
6. Remove the 7 floor console support bracket bolts.

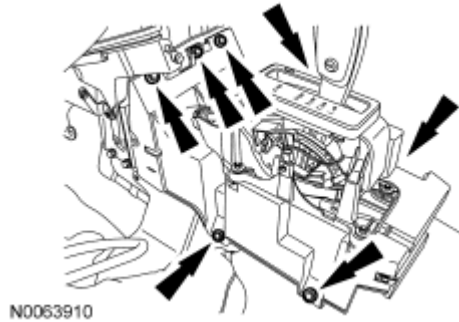


Fig. 16: Identifying Floor Console Support Bracket Bolts
Courtesy of FORD MOTOR CO.

NOTE: The floor console support bracket will have to be repositioned to remove the LH rear footwell duct.

7. Remove the LH rear footwell duct.

RH rear footwell duct only

8. Remove the RH floor console side finish panel.
9. Remove the RH front seat. For additional information, refer to **SEATING** article.
10. Remove the RH front door sill plate.
11. Remove the passenger front floor carpet pin-type retainers and position the carpet away from the RH rear footwell duct.
12. Remove the 5 RH rear footwell duct pin-type retainers.
13. Remove the RH rear footwell duct.

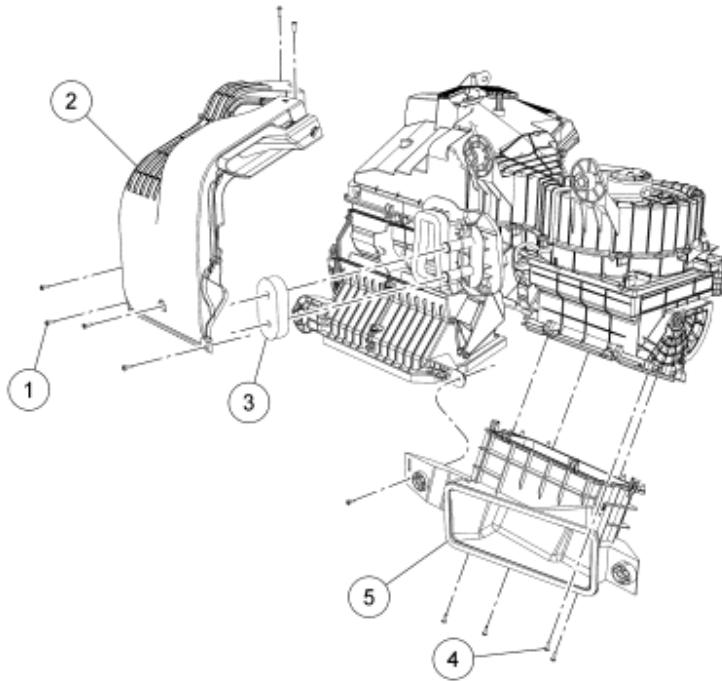
Both rear footwell ducts

14. To install, reverse the removal procedure.

HEATER CORE

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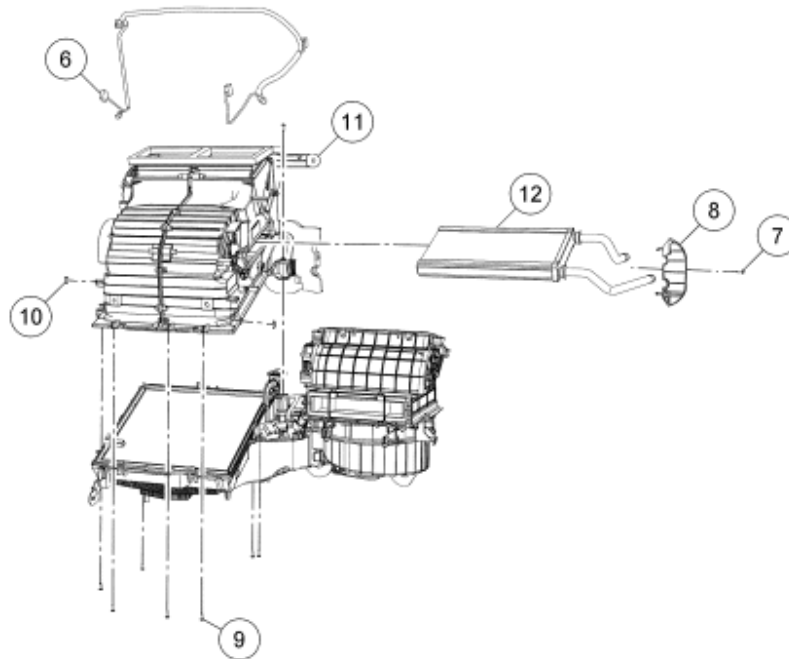
N0061758

Fig. 17: Exploded View Of Heater Core (1 Of 2)
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Floor duct screw (6 required)
2	-	Floor duct (part of 19B555)
3	-	Dash panel seal (part of 18529 kit)
4	-	Fresh air inlet duct screw (5 required)
5	-	Fresh air inlet duct (part of 19B555)

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N0061759

Fig. 18: Exploded View Of Heater Core (2 Of 2)
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
6	19C603	Plenum chamber wire harness
7	-	Heater core tube bracket screw
8	-	Heater core tube bracket (part of 19B555)
9	-	Plenum chamber screw (8 required)
10	-	Plenum chamber clip (2 required)
11	-	Plenum chamber
12	18476	Heater core

REMOVAL AND INSTALLATION

NOTE: If a heater core leak is suspected, the heater core must be pressure leak tested before it is removed from the vehicle. For additional information, refer to **CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS** article.

1. Remove the heater core and evaporator core housing. For additional information, refer to **HEATER CORE AND EVAPORATOR CORE HOUSING**.
2. Remove the 6 floor duct screws and the floor duct.
3. Remove the heater core tube dash panel seal.
4. Remove the heater tube bracket screw and the heater tube bracket.

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5. Remove the 5 fresh air inlet duct screws and the fresh air inlet duct.
6. Disconnect the wire harness from the plenum chamber.
7. Remove the 7 lower facing plenum chamber screws.
8. Orient the heater core and evaporator core housing with the plenum chamber upright.
9. Remove the upper facing plenum chamber screw.
10. Remove the 2 plenum chamber clips and remove the plenum chamber being careful not to allow the evaporator core to become dislodged from the installed position.

CAUTION: Do not handle the heater core by the inlet and/or outlet tube to remove. Handling the heater core by the tubes may damage the joints and lead to failure of the heater core.

11. Remove the heater core in the following sequence.
 1. Grasp the heater core by the core-side of the heater tube connections and partially remove it from the plenum chamber.
 2. Grasp the heater core by the top of the core and remove it from the plenum chamber.

NOTE: It is not necessary to carry out this step if the evaporator core has not become dislodged from the installed position during this procedure.

12. If the evaporator core has been moved at any point during heater core removal, remove the evaporator core, verify that the drain seal is installed in the correct position and install the evaporator core in the correct position.

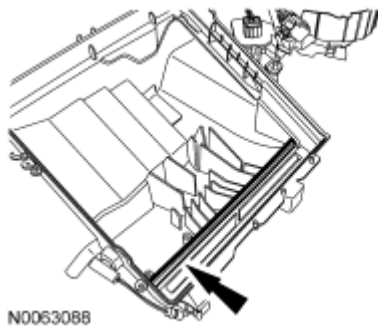


Fig. 19: Identifying Heater Core
Courtesy of FORD MOTOR CO.

13. To install, reverse the removal procedure.

HEATER CORE AND EVAPORATOR CORE HOUSING

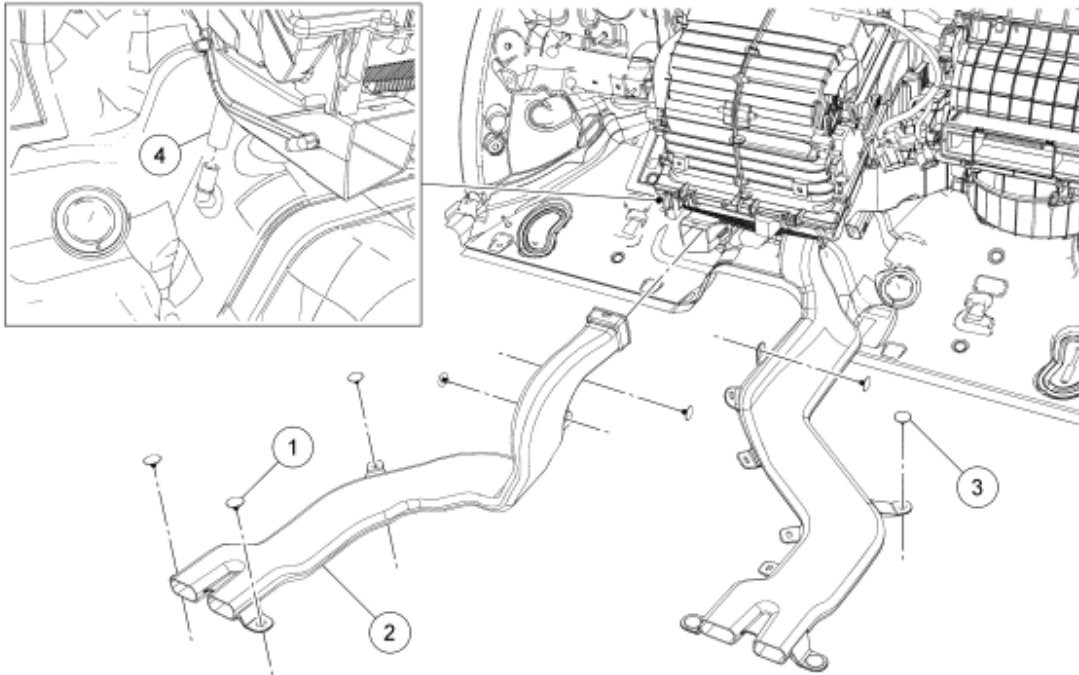
MATERIAL

Item	Specification
PAG Refrigerant Compressor Oil (R-134a Systems)	WSH-M1C231-B

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YN-12-D



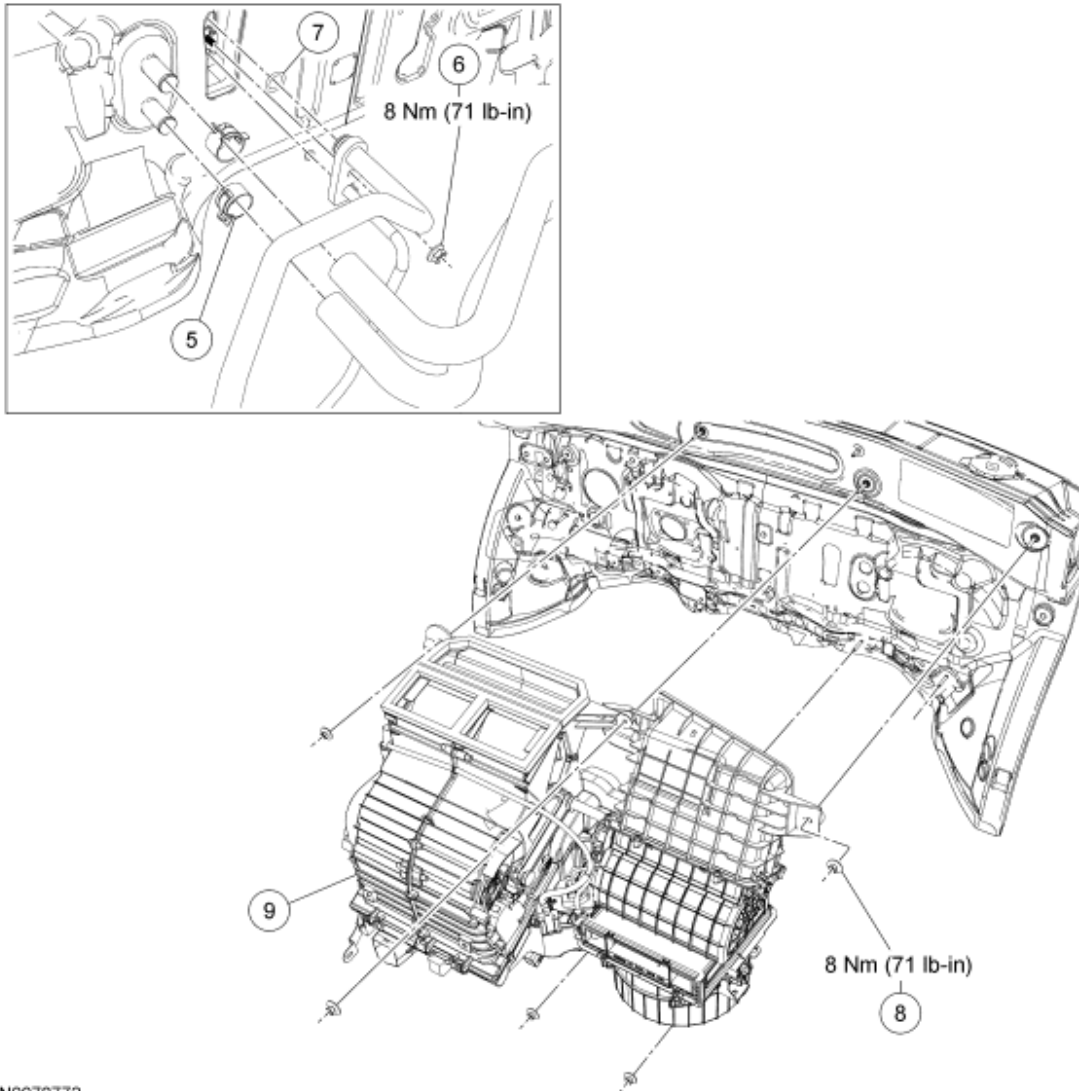
N0063082

Fig. 20: Exploded View Of Heater Core & Evaporator Core Housing (1 Of 2)
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	N806113	LH rear footwell duct pin-type retainer (5 required)
2	18C420	LH rear footwell duct
3	N806113	RH rear footwell duct front pin-type retainer (2 required)
4	19728	Evaporator drain tube

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N0078772

Fig. 21: Identifying Heater Core And Evaporator Core Housing With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
5	W525859	Heater hose clamp (2 required)
6	W707142	Thermostatic expansion valve (TXV) fitting nut
7	19E889	O-ring seal (2 required)
8	W709450	Heater core and evaporator core housing nut (5 required)
9	19B555	Heater core and evaporator core housing

REMOVAL AND INSTALLATION

NOTE: Installation of a new receiver/drier desiccant bag is not required when repairing the A/C system, except when there is physical evidence of contamination from a failed A/C compressor or damage to the receiver/drier desiccant bag. Damage to the receiver/drier desiccant bag includes physical damage to the receiver/drier desiccant bag or moisture contamination. Moisture contamination results only from a complete loss of refrigerant and equalization of the refrigerant system pressure with atmospheric pressure for a period longer than one hour. If even a slight amount of positive refrigerant pressure is present in the system before repairs are carried out, the receiver/drier desiccant bag should not be replaced.

NOTE: The O-ring seals used on this vehicle are not interchangeable with all similarly sized O-ring seals used on other Ford products. Use only the O-ring seals specified for this vehicle in the Ford master parts catalog.

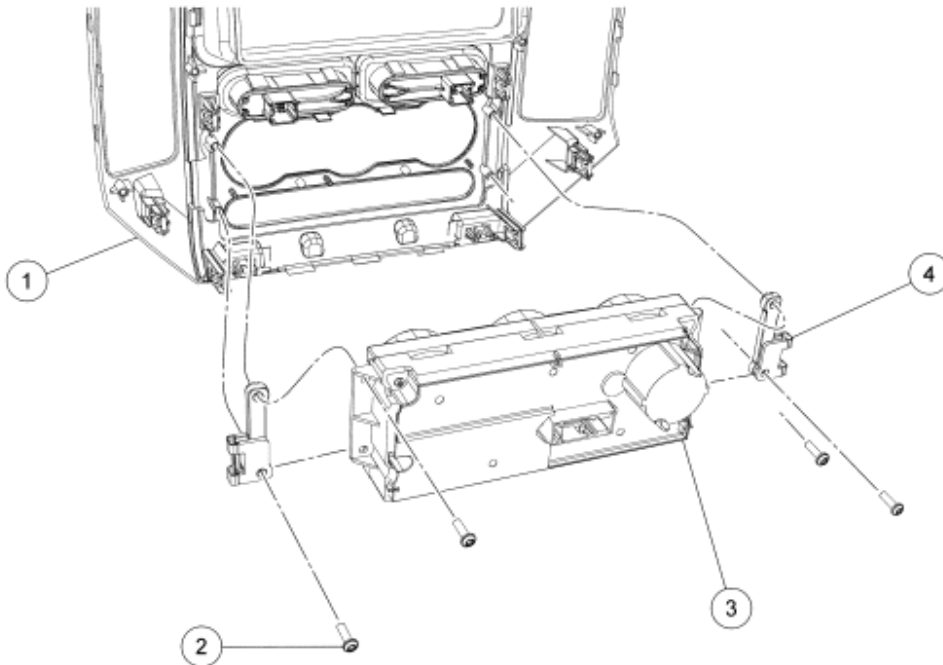
1. Recover the refrigerant. For additional information, refer to **CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS** article.
2. Drain the engine coolant. For additional information, refer to **ENGINE COOLING** article.
3. Remove the instrument panel. For additional information, refer to **INSTRUMENT PANEL AND CONSOLE** article.
4. Remove the 3 driver side carpet pin-type retainers and position the driver side front carpet aside.
5. Remove the 4 LH rear footwell duct pin-type retainers and the LH rear footwell duct.
6. Remove the 2 passenger front carpet pin-type retainers and position the passenger front carpet aside enough to access the 2 front pin-type retainers on the RH rear footwell duct.
7. Remove the 2 front pin-type retainers on the RH rear footwell duct and detach the RH rear footwell duct from the heater core and evaporator core housing.
8. Release the 2 heater hose clamps at the heater core and disconnect the heater hoses.
9. Remove the thermostatic expansion valve (TXV) fitting nut and detach the A/C lines from the TXV.
 - Discard the O-ring seals.
 - To install, tighten to 8 N.m (71 lb-in).

CAUTION: The retaining clips on the plastic evaporator drain tube elbow will break if the elbow is removed from inside the vehicle. If the evaporator drain tube elbow must be removed, the retaining clips must be detached from below the vehicle. If the plastic evaporator drain tube elbow does not need to be removed, only detach the evaporator drain tube from the elbow.

10. Disconnect the evaporator drain tube from the elbow at the floorpan.
11. Remove the 5 heater core and evaporator core housing nuts.
 - To install, tighten to 8 N.m (71 lb-in).
12. Remove the heater core and evaporator core housing.

13. To install, reverse the removal procedure.
 - Install new O-ring seals lubricated in clean PAG oil.
 - Lubricate the refrigerant system with the correct amount of clean PAG oil. For additional information, refer to **CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS** article.
14. Fill the engine coolant level. For additional information, refer to **ENGINE COOLING** article.
15. Evacuate, leak test and charge the refrigerant system. For additional information, refer to **CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS** article.

HEATING VENTILATION AIR CONDITIONING (HVAC) MODULE - EDGE



N0075784

Fig. 22: Exploded View Of Climate Control Module - Edge
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	04302	Upper center instrument panel finish panel
2	W505945	HVAC module screw (4 required)
3	19980	HVAC module
4	78042K50	HVAC module spacer (2 required)

REMOVAL AND INSTALLATION

1. Using a suitable tool, gently pry out the upper center instrument panel finish panel to disengage the clips

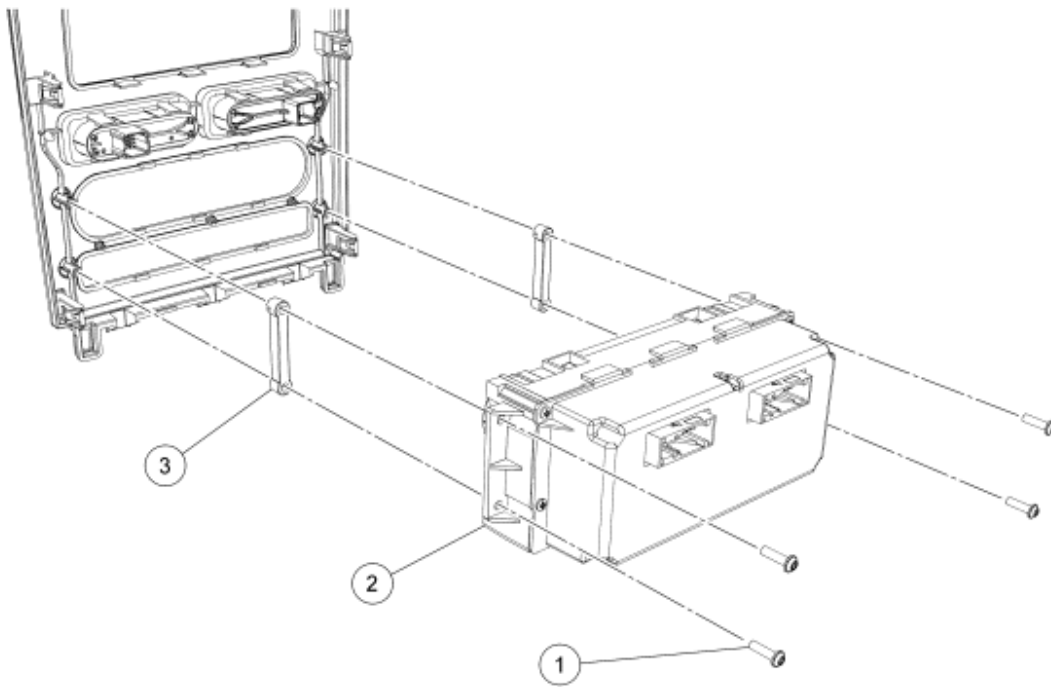
and remove the panel.

- Disconnect the electrical connectors.
2. Remove the 4 HVAC module screws.
 3. Remove the HVAC module.

NOTE: Be sure the HVAC module spacers are correctly positioned before installing the HVAC module.

4. To install, reverse the removal procedure.

HEATING VENTILATION AIR CONDITIONING (HVAC) MODULE - MKX



N0061586

Fig. 23: Exploded View Of Climate Control Module - MKX
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W505945	HVAC module screw (4 required)
2	19980	HVAC module
3	78042K50	HVAC module spacer (2 required)

REMOVAL AND INSTALLATION

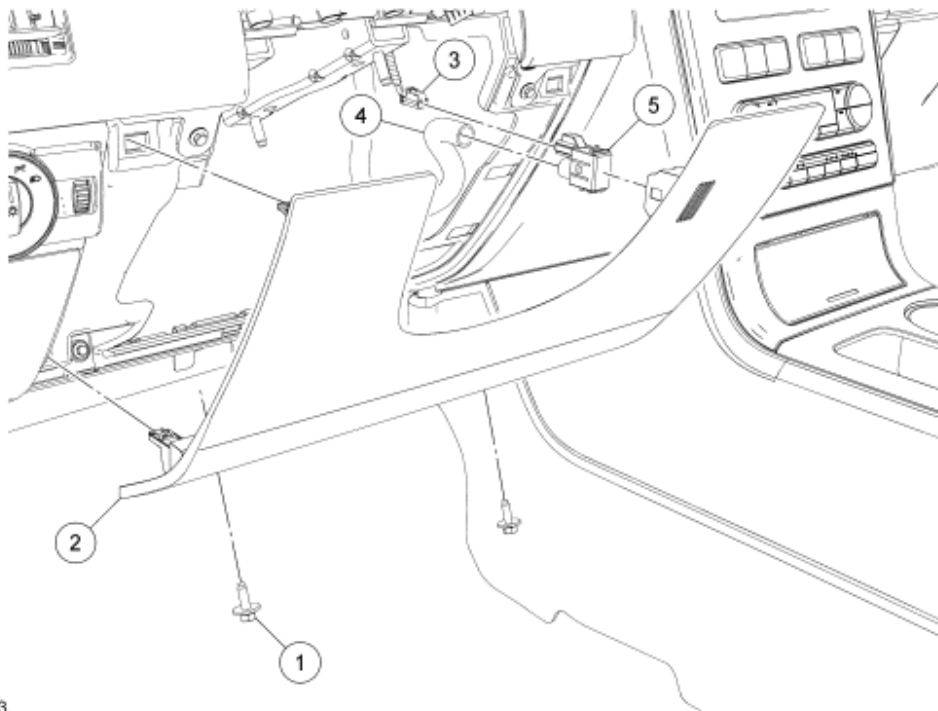
1. Remove the center instrument panel finish panel. For additional information, refer to **INSTRUMENT PANEL AND CONSOLE** article.

2. Remove the 4 HVAC module screws.
3. Remove the HVAC module.

NOTE: Be sure the HVAC module spacers are correctly positioned before installing the HVAC module.

4. To install, reverse the removal procedure.

IN-VEHICLE TEMPERATURE SENSOR



N0063083

Fig. 24: Exploded View Of In-Vehicle Temperature Sensor
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W708268	Lower steering column opening cover bolt (2 required)
2	7804459	Lower steering column opening cover
3	14401	In-vehicle temperature sensor electrical connector
4	-	In-vehicle temperature hose and venturi assembly (part of 19C734 kit)
5	19C734	In-vehicle temperature sensor

REMOVAL AND INSTALLATION

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1. Remove the 2 lower steering column opening cover bolts.
2. Detach the lower steering column opening cover.
3. Disconnect the in-vehicle temperature sensor aspirator hose.
4. Disconnect the in-vehicle temperature sensor electrical connector.
5. Release the 2 in-vehicle temperature sensor clips and remove the in-vehicle temperature sensor.

NOTE: The in-vehicle temperature sensor hose and venturi assembly can be accessed from below the driver side of the instrument panel.

6. Remove the in-vehicle temperature sensor hose and venturi assembly.
7. To install, reverse the removal procedure.

MODE DOOR ACTUATOR - AIR INLET DOOR

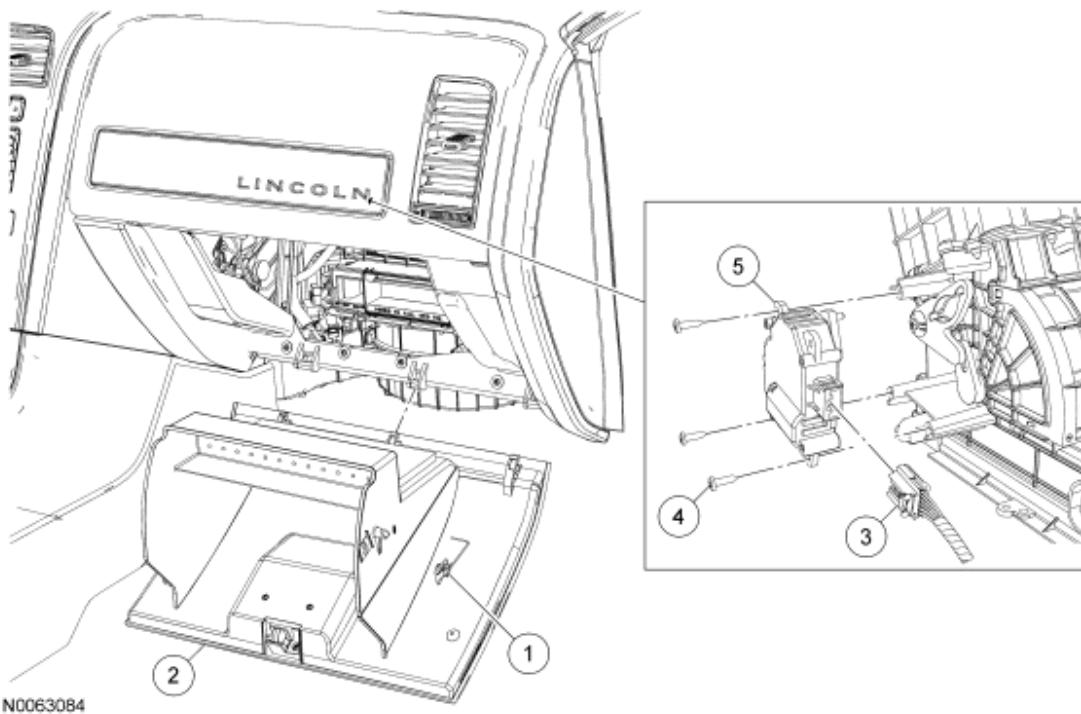


Fig. 25: Air Inlet Mode Door Actuator
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	7806200	Glove compartment damper
2	7806024	Glove compartment
3	19C603	Air inlet mode door actuator electrical connector
4	-	Air inlet mode door actuator screw (2

		required)
5	19E616	Air inlet mode door actuator

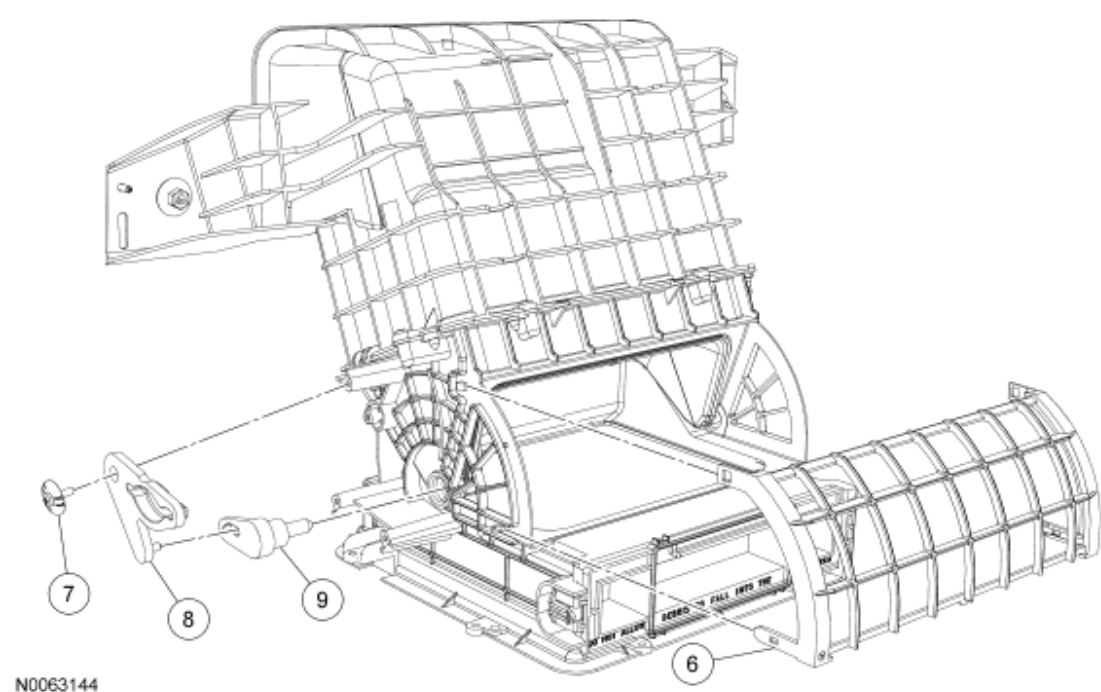


Fig. 26: Air Inlet Mode Door Actuator Linkage
 Courtesy of FORD MOTOR CO.

Item	Part Number	Description
6	-	Air inlet duct grille (part of 19B555)
7	-	Air inlet mode door intermediate lever screw (part of 19E616 kit)
8	-	Air inlet mode door intermediate lever (part of 19E616 kit)
9	-	Air inlet mode door lever (part of 19E616 kit)

REMOVAL AND INSTALLATION

NOTE: It is not required to install the new air inlet door linkage parts included in the air inlet mode door actuator service part kit if only the mode door actuator requires replacement and the linkage parts are not broken or binding.

Air inlet mode door actuator and linkage

1. Disconnect the glove compartment damper from the glove compartment.

2. Remove the glove compartment.
3. Remove the 3 air inlet mode door actuator screws.
4. Disconnect the air inlet mode door actuator electrical connector.
5. Remove the air inlet mode door actuator.
6. Manually operate and inspect the air inlet mode door and air inlet mode door linkage for incorrect placement or damage.

Air inlet mode door linkage only

7. Remove the air inlet mode door intermediate lever screw.
8. Remove the air inlet mode door intermediate lever.
9. Working through the air inlet duct grille opening, depress the clip and detach the air inlet mode door lever from the air inlet mode door.

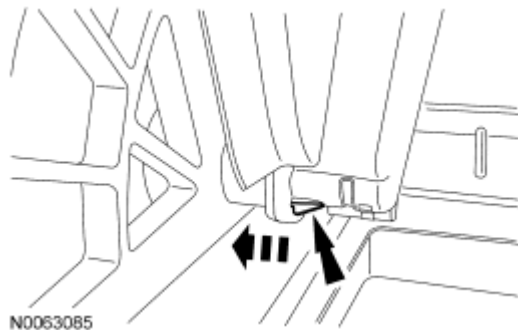


Fig. 27: Depressing Clip & Detaching Air Inlet Mode Door Lever From Air Inlet Mode Door
Courtesy of FORD MOTOR CO.

Air inlet mode door actuator and linkage

NOTE: The air inlet mode door must be fully seated in the correct position during installation. If the air inlet mode door is not fully seated, excessive noise will result when the climate control system is operated.

10. To install, reverse the removal procedure.

MODE DOOR ACTUATOR - DEFROST/PANEL/FLOOR DOOR

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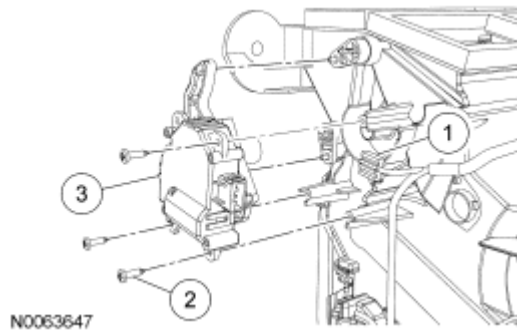


Fig. 28: Exploded View Of Mode Door Actuator - Defrost/Panel/Floor Door
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	19C603	Defrost/panel/floor mode door actuator electrical connector
2	-	Defrost/panel/floor mode door actuator screw (3 required)
3	19C772	Defrost/panel/floor mode door actuator

REMOVAL AND INSTALLATION

NOTE: The defrost/panel/floor mode door actuator can be accessed from below the driver side of the instrument panel.

NOTE: It is not required to install the new defrost/panel/floor mode door linkage parts included in the defrost/panel/floor mode door actuator service part kit if only the mode door actuator requires replacement and the linkage parts are not broken or binding.

Mode door actuator and linkage

1. Remove the 3 defrost/panel/floor mode door actuator screws.
2. Disconnect the defrost/panel/floor mode door actuator electrical connector.
3. Remove the defrost/panel/floor mode door actuator.
4. Manually operate, and inspect the defrost/panel/floor mode door linkage for incorrect placement or damage.

Mode door actuator linkage only

5. Remove the mode door linkage.
 1. Depress the tab and remove the panel door lever.
 2. Remove the floor/defrost door intermediate lever screw and floor/defrost door intermediate lever.
 3. Pull straight outward to remove the floor/defrost door lever.

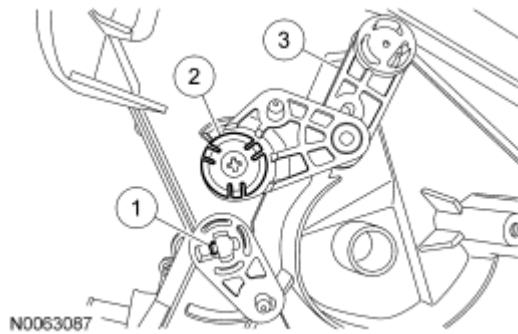


Fig. 29: Identifying Floor/Defrost Door Intermediate Lever
Courtesy of FORD MOTOR CO.

Mode door actuator and linkage

NOTE: When installing the mode door actuator, make sure that the panel door lever and floor/defrost intermediate lever are correctly aligned with the mode door actuator cam.

6. To install, reverse the removal procedure.

REGISTER - CENTER, EDGE

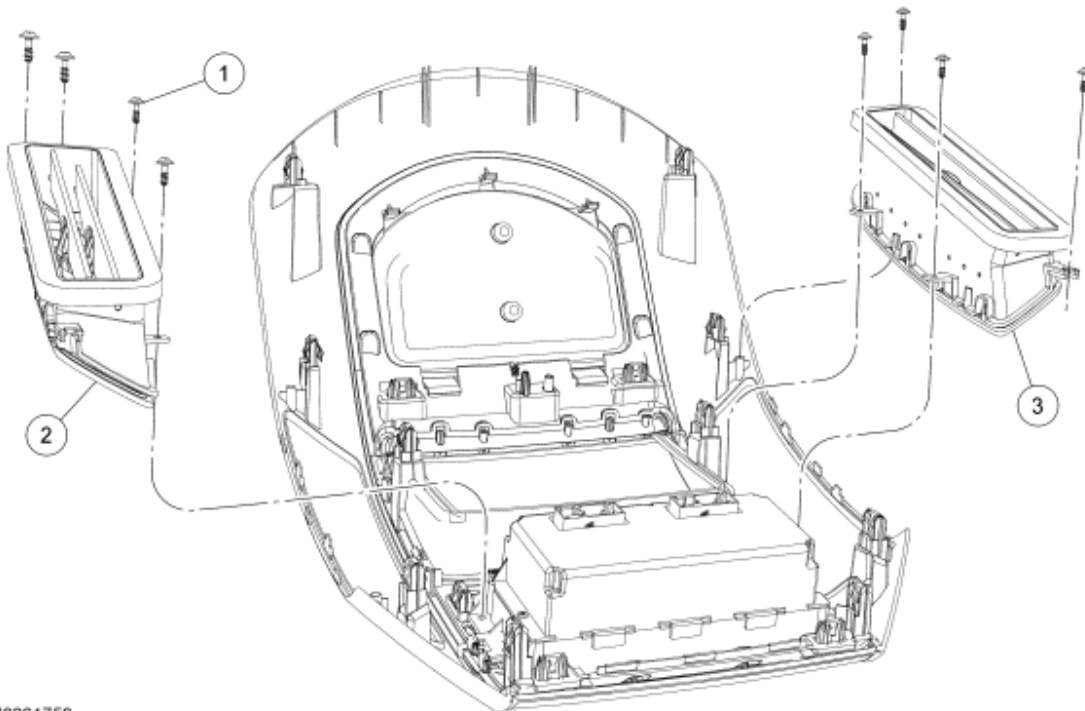


Fig. 30: Exploded View Of Register - Center, Edge
Courtesy of FORD MOTOR CO.

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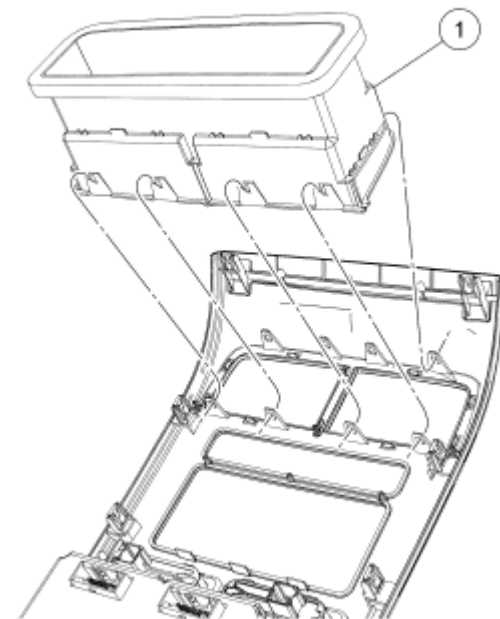
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Item	Part Number	Description
1	-	Center register screw (8 required)
2	19893	RH center register
3	19893	LH center register

REMOVAL AND INSTALLATION

1. Remove the center instrument panel finish panel. For additional information, refer to **INSTRUMENT PANEL AND CONSOLE** article.
2. Remove the 8 center register screws.
3. Remove the center registers.
4. To install, reverse the removal procedure.

REGISTER - CENTER, MKX



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Fig. 31: Exploded View Of Register - Center, MKX
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	19893	Center register

REMOVAL AND INSTALLATION

1. Remove the center instrument panel finish panel. For additional information, refer to **INSTRUMENT**

PANEL AND CONSOLE article.

- 2. Release the 8 center register clips and remove the center register assembly.
- 3. To install, reverse the removal procedure.

REGISTER - DRIVER AND PASSENGER SIDE, EDGE

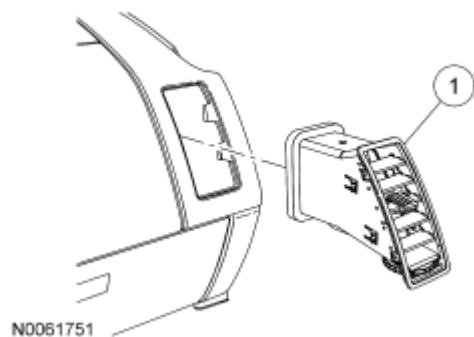


Fig. 32: Exploded View Of Register - Driver and Passenger Side, Edge
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	19893	Register

REMOVAL AND INSTALLATION

NOTE: Passenger side register shown in illustration, driver side similar.

CAUTION: Do not pry on the outer register trim bezel. The clips on the outer register trim bezel break if pried on to remove the register.

- 1. Working through the register vein openings, insert a suitable hook shaped pick behind each of the 4 register clips while gently pulling the register rearward with the hook to disengage the clip.
- 2. Remove the driver or passenger side instrument panel register.
- 3. To install, reverse the removal procedure.

REGISTER - DRIVER SIDE, MKX

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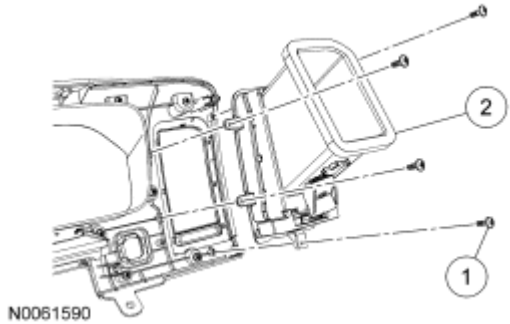


Fig. 33: Exploded View Of Register - Driver Side, MKX
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W701679	Driver side register screw (4 required)
2	19893	Driver side register

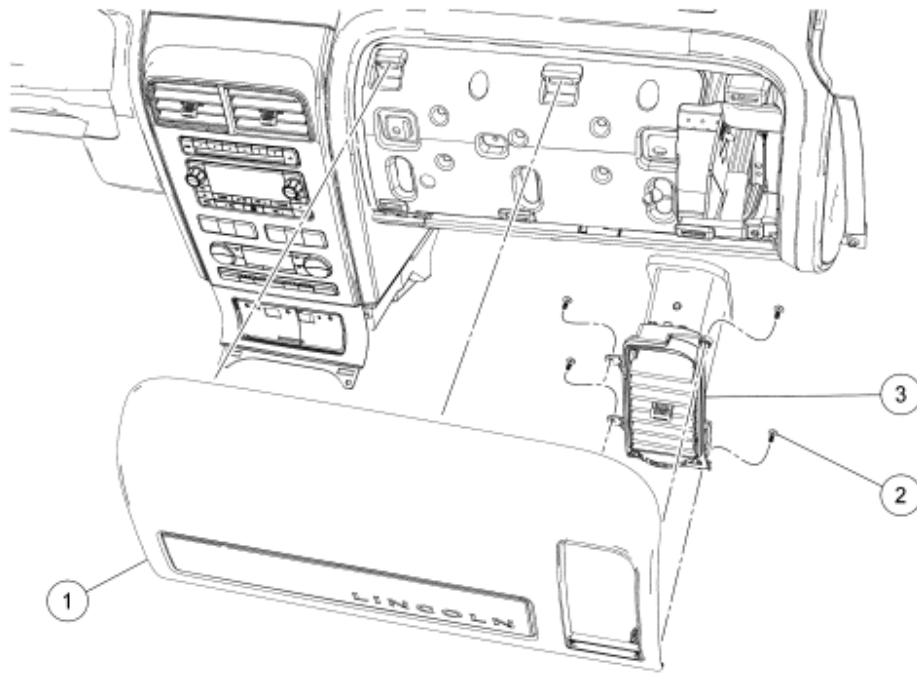
REMOVAL AND INSTALLATION

1. Remove the instrument cluster finish panel. For additional information, refer to **INSTRUMENT PANEL AND CONSOLE** article.
2. Remove the 4 driver side register screws and the driver side register.
3. To install, reverse the removal procedure.

REGISTER - PASSENGER SIDE, MKX

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Fig. 34: Exploded View Of Register - Passenger Side, MKX
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	7804338	RH instrument panel finish panel
2	W701679	Passenger side register screw (4 required)
3	19893	Passenger side register

REMOVAL AND INSTALLATION

1. Disengage the 6 clips and remove the RH instrument panel finish panel.
2. Remove the 4 passenger side register screws and the passenger side register.
3. To install, reverse the removal procedure.

TEMPERATURE BLEND DOOR ACTUATOR - LH

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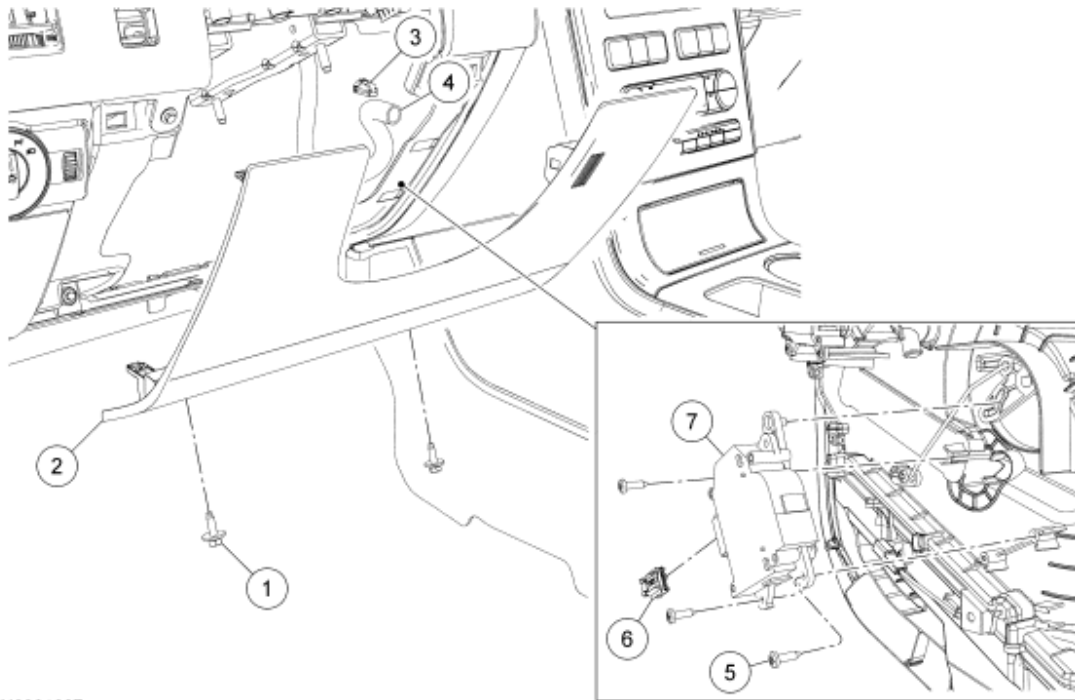


Fig. 35: Exploded View Of Temperature Blend Door Actuator - LH
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W708268	Lower steering column opening cover bolt (2 required)
2	7804459	Lower steering column opening cover
3	14401	In-vehicle temperature sensor electrical connector
4	19C739	In-vehicle temperature sensor aspirator hose
5	-	LH temperature blend door actuator screw (3 required)
6	19C603	LH temperature blend door actuator electrical connector
7	19E616	LH temperature blend door actuator

REMOVAL AND INSTALLATION

NOTE: It is not required to install the new LH temperature blend door linkage parts included in the LH temperature blend door actuator service part kit if only the temperature blend door actuator requires replacement and the linkage parts are not broken or binding.

LH temperature blend door actuator and linkage

1. Remove the 2 lower steering column opening cover bolts.
2. Detach the lower steering column opening cover.
3. Disconnect the in-vehicle temperature sensor aspirator hose and electrical connector.
4. Position the instrument panel wire harness away from the LH temperature blend door actuator.
5. Remove the 3 LH temperature blend door actuator screws.
6. Disconnect the LH temperature blend door actuator electrical connector.
7. Remove the LH temperature blend door actuator.
8. Manually operate, and inspect the LH temperature blend door linkage for incorrect placement, binding or damage.

LH temperature blend door linkage only

9. Detach the 2 retaining tabs and remove the LH temperature blend door linkage.

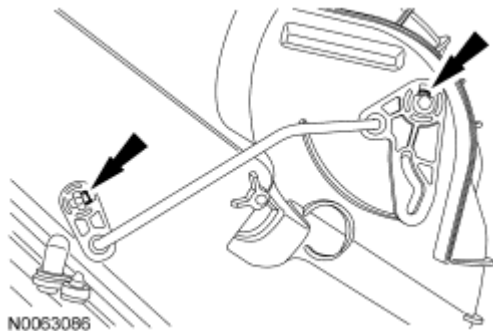


Fig. 36: Locating Retaining Tabs
Courtesy of FORD MOTOR CO.

LH temperature blend door actuator and linkage

10. To install, reverse the removal procedure.

TEMPERATURE BLEND DOOR ACTUATOR - RH, DUAL-ZONE EATC

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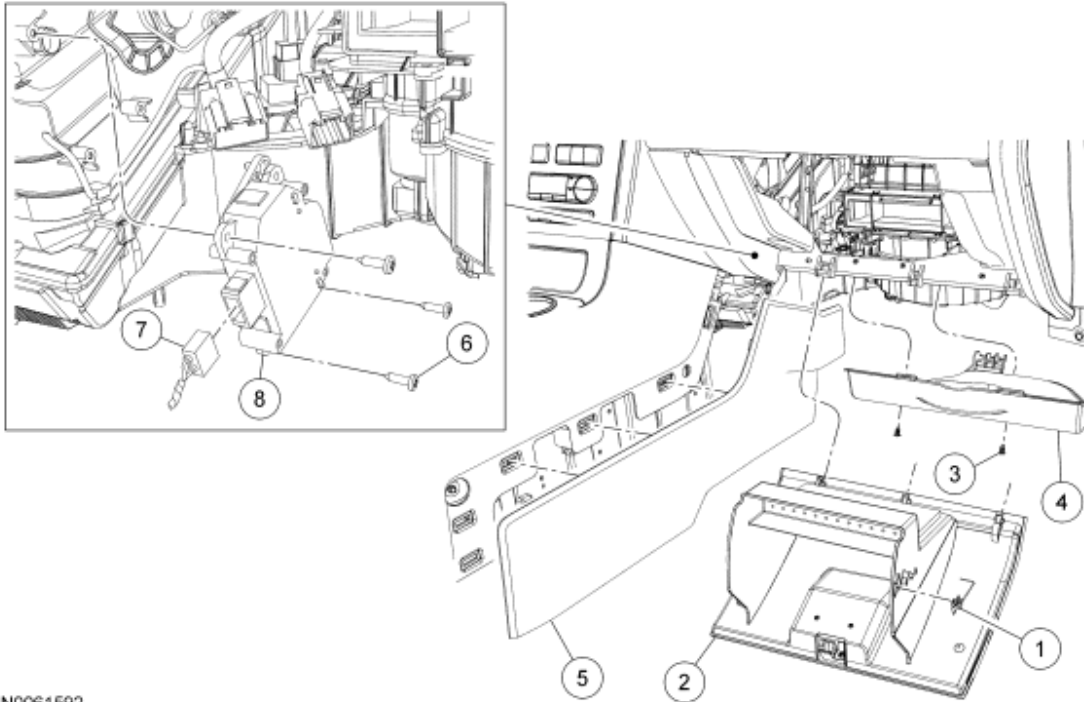


Fig. 37: Exploded View Of Temperature Blend Door Actuator - RH, Dual-Zone EATC
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	7806200	Glove compartment damper
2	7806024	Glove compartment
3	-	RH lower instrument panel insulator pin-type retainer (2 required)
4	-	RH lower instrument panel insulator
5	7804608	RH lower floor console trim panel
6	-	RH temperature blend door actuator screw (3 required)
7	19C603	RH temperature blend door actuator electrical connector
8	19E616	RH temperature blend door actuator

REMOVAL AND INSTALLATION

NOTE: It is not required to install the new RH temperature blend door linkage parts included in the RH temperature blend door actuator service part kit if only the temperature blend door actuator requires replacement and the linkage parts are not broken or binding.

RH temperature blend door actuator and linkage

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1. Disconnect the glove compartment damper from the glove compartment.
2. Remove the glove compartment.
3. Remove the RH lower instrument panel insulator.
4. Remove the RH lower floor console trim panel.
5. Remove the 3 RH temperature blend door actuator screws.
6. Disconnect the RH temperature blend door actuator electrical connector.
7. Remove the RH temperature blend door actuator.
8. Manually operate, and inspect the RH temperature blend door linkage for incorrect placement, binding or damage.

RH temperature blend door actuator linkage only

9. Detach the 2 retaining tabs and remove the RH temperature blend door linkage.

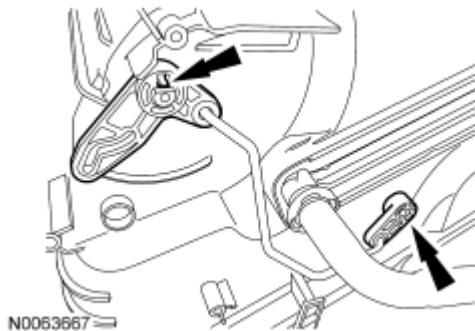


Fig. 38: Locating Retaining Tabs
Courtesy of FORD MOTOR CO.

All vehicles

10. To install, reverse the removal procedure.

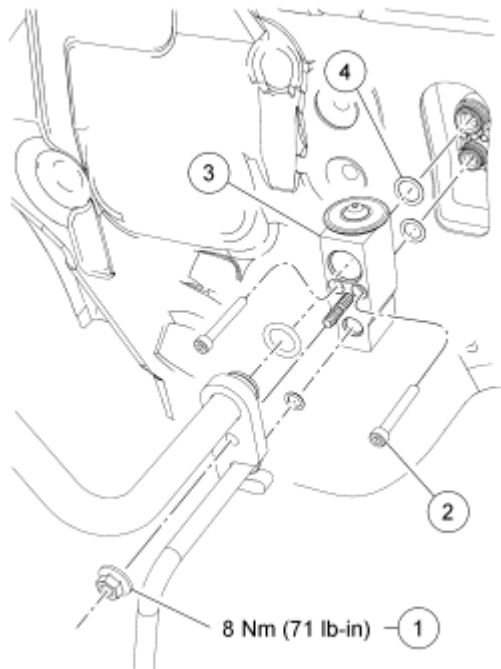
THERMOSTATIC EXPANSION VALVE

MATERIAL

Item	Specification
PAG Refrigerant Compressor Oil (R-134a Systems) YN-12-D	WSH-M1C231-B

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Fig. 39:
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W707142	Thermostatic expansion valve (TXV) fitting nut
2	-	TXV bolt (2 required) (part of 19849 service kit)
3	19849	TXV
4	19E889	O-ring seal (4 required)

REMOVAL AND INSTALLATION

NOTE: Installation of a new receiver/drier desiccant bag is not required when repairing the A/C system, except when there is physical evidence of contamination from a failed A/C compressor or damage to the receiver/drier desiccant bag. Damage to the receiver/drier desiccant bag includes physical damage to the receiver/drier desiccant bag or moisture contamination. Moisture contamination results only from a complete loss of refrigerant and equalization of the refrigerant system pressure with atmospheric pressure for a period longer than one hour. If even a slight amount of positive refrigerant pressure is present in the system before repairs are carried out, the receiver/drier desiccant bag should not be replaced.

NOTE: The O-ring seals used on this vehicle are not interchangeable with all similarly

sized O-ring seals used on other Ford products. Use only the O-ring seals specified for this vehicle in the Ford master parts catalog.

1. Recover the refrigerant. For additional information, refer to **CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS** article.
2. Remove the thermostatic expansion valve (TXV) fitting nut and disconnect the fitting.
 - Discard the O-ring seals.
 - To install, tighten to 8 N.m (71 lb-in).
3. Remove the 2 TXV bolts.
4. Remove the TXV.
 - Discard the O-ring seals.
5. To install, reverse the removal procedure.
 - Install new O-ring seals lubricated in clean PAG oil.
 - Lubricate the refrigerant system with the correct amount of clean PAG oil. For additional information, refer to **CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS** article.
6. Evacuate, leak test and charge the refrigerant system. For additional information, refer to **CLIMATE CONTROL SYSTEM - GENERAL INFORMATION AND DIAGNOSTICS** article.